
0.5um Analog CMOS 2P3M 5V Design Tutorial

Version 1.4

National Semiconductor Public Laboratory (NSPL)

- Electronics and Telecommunications Research Institute (ETRI)
- Seoul National University (SNU)
- Daegu Gyeongbuk Institute of Science & Technology (DGIST)



서울대학교
SEOUL NATIONAL UNIVERSITY



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1. Overview

1. Overview

1.1 Overview

The purpose of this paper is 0.5um Analog CMOS 2P3M 5V Design tutorial.

1.2 Tools and Software Version

- Virtuoso 6.1.7 version
- Calibre 2019.4_36.18
- Quantus 212

1.3 File Configuration

File : NSPL_05_CMOS_PDK.tar.gz

- NSPL_05_CMOS_PDK
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1.tf : tech file
 - display.drf : display file
 - Calibre
 - ANT
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_V1.1_ANT.cal : ANT rule file
 - ant_header.cal : with script file
 - DRC
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_V1.1_DRC.cal : DRC rule file
 - drc_header.cal : with script file
 - LVL
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_V1.1_LVL.cal : LVL rule file
 - lvl_header.cal : with script file
 - LVS
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_V1.1_LVS.cal : LVS rule file
 - lvs_header.cal : with script file
 - PEX
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_V1.1_PEX.cal : PEX rule file
 - pex_header.cal : with script file
 - Assuura
 - DRC
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_V1.1_DRC.rul : DRC rule file
 - LVS
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_V1.1_LVS_compare.rul : LVS compare rule file
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_V1.1_LVS_extract.rul : LVS extract rule file
 - QRC
 - techRuleSets : QRC rule set file
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1_pcell : Pcell library
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1_std_lay : std lay library
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1_std_sch : std sch library
 - ETRI_NSPL_GPIO_240925 : GPIO library
 - MPW_PAD_28pin : 28-pin pad library

1.4 Documents

- 0.5um Analog CMOS 2P3M 5V Design Rule_Electrical target V1.1 : Design rule
- 0.5um Analog CMOS 2P3M 5V Design Guide V1.0 : Design Guide
- 0.5um Analog CMOS 2P3M 5V Design Guide V1.4 : Design Tutorial
- ETRI_5V_GPIO_230912 : GPIO

1.5 Document history

Rev	Date	From	Description
1.0	2023.09.15	ETRI	Added Items 1. Overview 2. PDK Library 설정 3. 회로 설계 및 검증(Pre-simulation) 4. Layout 5. DRC 6. LVS 7. PEX
1.1	2023.10.10	ETRI	8. 회로 설계 및 검증(Post-simulation)
1.2	2023.10.24	ETRI	7. PEX – Calibrereview 방식 8. 회로 설계 및 검증(Post-simulation) – Calibre 방식
1.3	2023.12.29	ETRI	5.2 Assura DRC 검토 6.2 Assura LVS 검토
1.4	2024.10.31	ETRI	PDK 업데이트에 따른 관련 내용 수정 및 추가 - Pcell, Sch 통합 → Pcell - Assura-QRC 3. 회로 설계 및 검증(Pre-simulation) 4. Layout 5. DRC 6. LVS 7. PEX 8. 회로 설계 및 검증(Post-simulation) 9. 28-pin Package Pads 및 GPIO 추가

2. PKD Library 설정

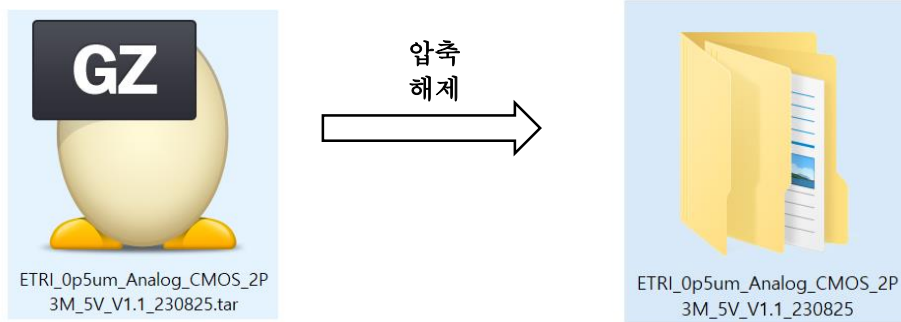
2. PDK Library 설정

- 목적 : 회로 설계 및 검증에 앞서 제공받은 PDK를 설정

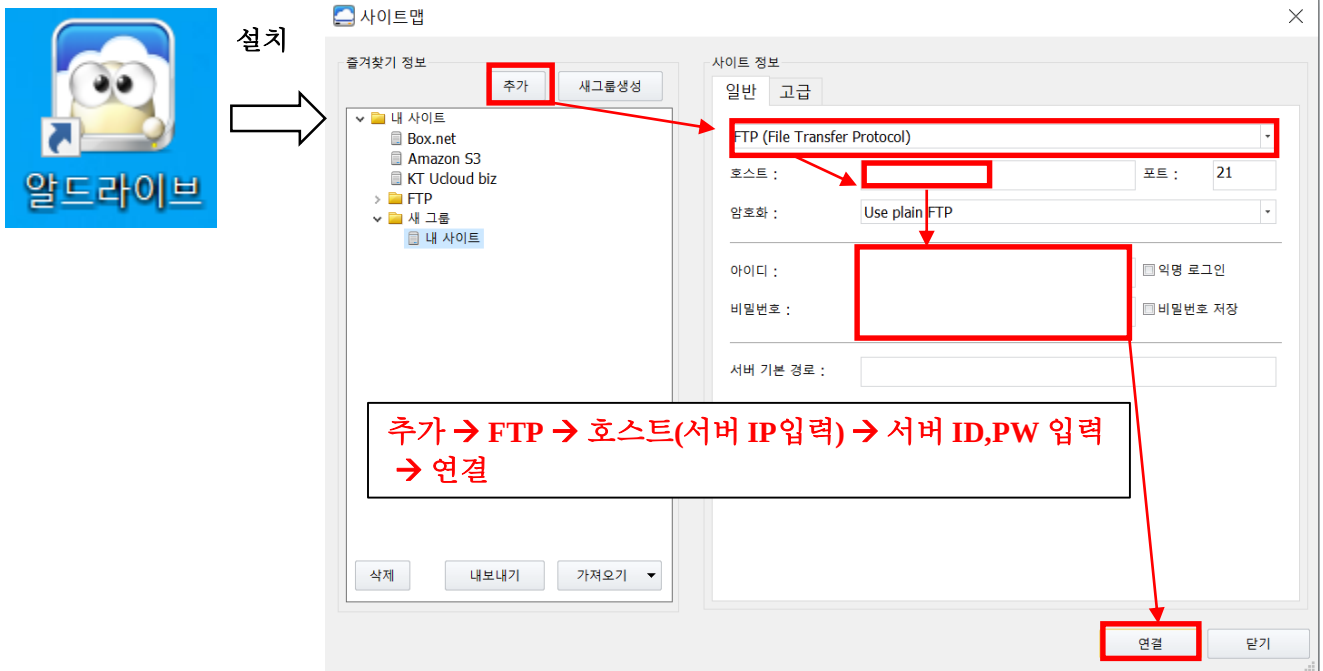
1) PDK Library 전송 (서버 기반의 워크스테이션 접속일 경우)

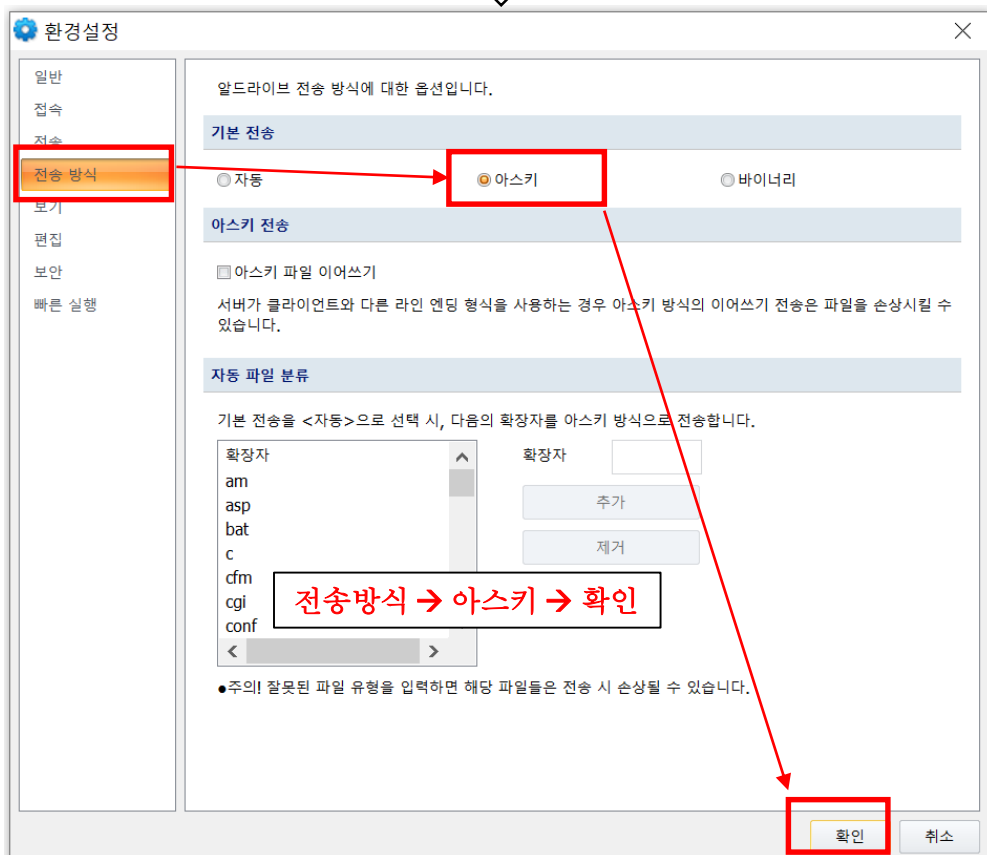
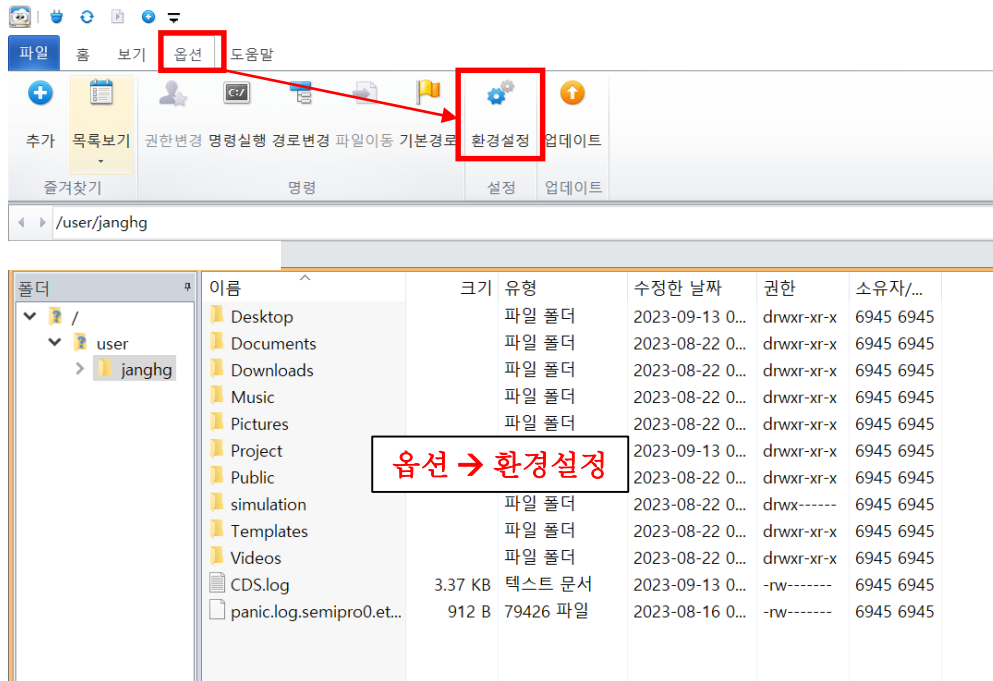
- 제공된 PDK를 FTP(File Transfer Protocol)을 이용하여 워크스테이션에 전송

① 제공된 PDK 파일 압축 해제



② FTP 프로그램 설치 및 설정 (예시로 알드라이브 사용)





③ 폴더 설정 및 파일 전송

마우스 우클릭 → 새폴더만들기 → 폴더명 기입(예시 Project)

폴더	이름	크기	유형	수정한 날짜	권한	소유자/...
/	Desktop		파일 폴더	2023-09-13 0...	drwxr-xr-x	6945 6945
	Documents		파일 폴더	2023-08-22 0...	drwxr-xr-x	6945 6945
	Downloads		파일 폴더	2023-08-22 0...	drwxr-xr-x	6945 6945
	Music		파일 폴더	2023-08-22 0...	drwxr-xr-x	6945 6945
	Pictures		파일 폴더	2023-08-22 0...	drwxr-xr-x	6945 6945
	Project		파일 폴더	2023-09-13 0...	drwxr-xr-x	6945 6945
	Public		파일 폴더	2023-08-22 0...	drwxr-xr-x	6945 6945
user	Videos		파일 폴더	2023-08-22 0...	drwxr-xr-x	6945 6945
	CDS.log	3.37 KB	텍스트 문서	2023-09-13 0...	-rw-----	6945 6945
panic.log.semipro0.et...	912 B	79426 파일	2023-08-16 0...	-rw-----	6945 6945	



MOS_PDK_240925 > NSPL_05CMOS_PDK_240925 > NSPL_05CMOS_PDK_240206

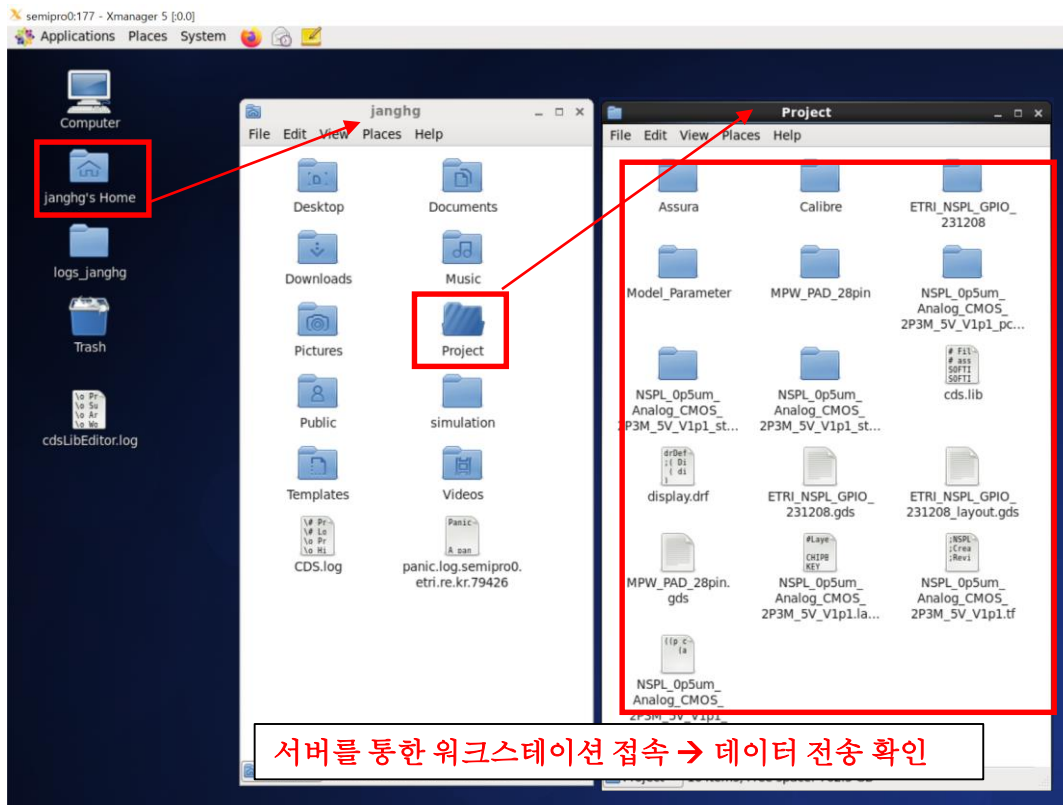
모두선택

붙여넣기

이름	수정한 날짜	유형	크기
Assura	2024-09-27 오전 9:46	파일 폴더	
Calibre	2024-09-27 오전 9:46	파일 폴더	
ETRI_NSPL_GPIO_2312...	2024-09-27 오전 9:46	파일 폴더	
Model_Parameter	2024-09-27 오전 9:46	파일 폴더	
MPW_PAD_28pin	2024-09-27 오전 9:46	파일 폴더	
NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p...	2024-09-27 오전 9:46	파일 폴더	
NSPL_0p5um_An...			
NSPL_0p5um_An...			
cds.lib			
display			
ETRI_NSPL_GPIO...			
ETRI_NSPL_GPIO...			
MPW_PAD_28pin			
NSPL_0p5um_An...			
NSPL_0p5um_An...			
NSPL_0p5um_An...			

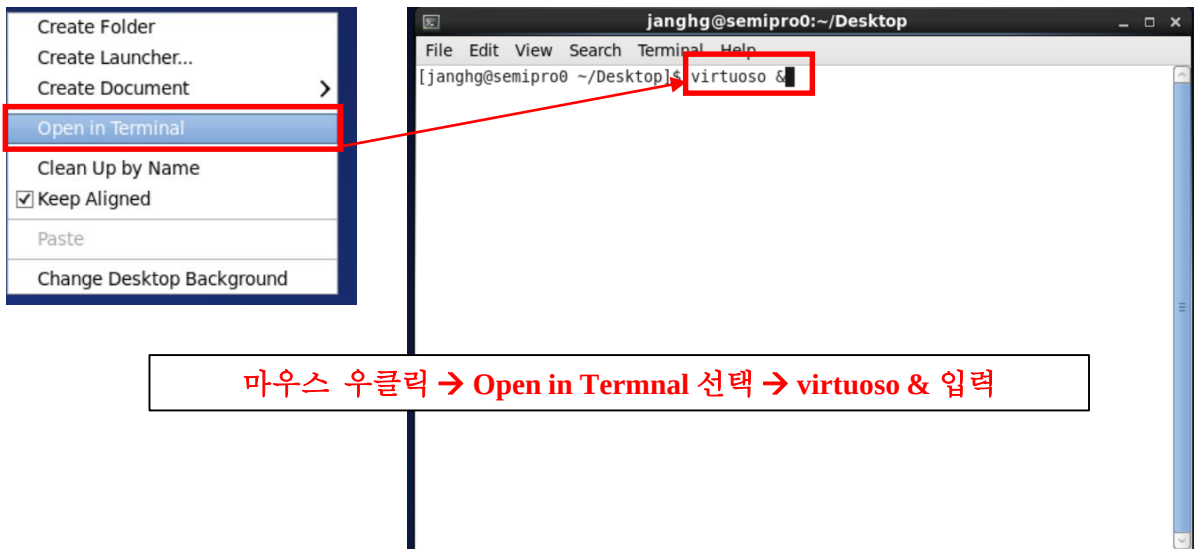
폴더	이름	크기	유형	수정한 날짜	권한	소유자/...
/	Calibre		파일 폴더	2023-09-13 0...	drwxr-xr-x	6945 6945
	ETRI_0p5um_Analog_...		파일 폴더	2023-09-13 0...	drwxr-xr-x	6945 6945
	ETRI_0p5um_Analog_...		파일 폴더	2023-09-13 0...	drwxr-xr-x	6945 6945
	ETRI_0p5um_Analog_...		파일 폴더	2023-09-13 0...	drwxr-xr-x	6945 6945
	05mos_TT.scs	5.47 KB	SCS 파일	2023-09-13 0...	-rw-r--r--	6945 6945
	display.drf	45.5 KB	DRF 파일	2023-09-13 0...	-rw-r--r--	6945 6945
	ETRI_0p5um_Analog_...	3.93 KB	LAYERMAP 파...	2023-09-13 0...	-rw-r--r--	6945 6945
user	ETRI_0p5um_Analog_...	28.4 KB	TF 파일	2023-09-13 0...	-rw-r--r--	6945 6945
	ETRI_0p5um_Analog_...	28.4 KB	TF 파일	2023-09-13 0...	-rw-r--r--	6945 6945

④ 전송된 파일 확인

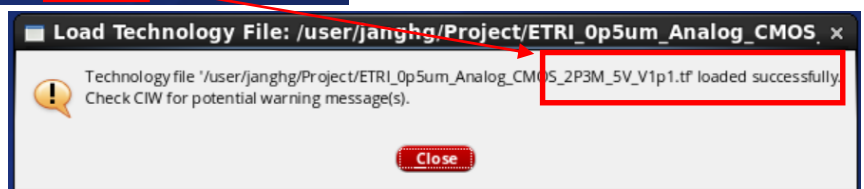
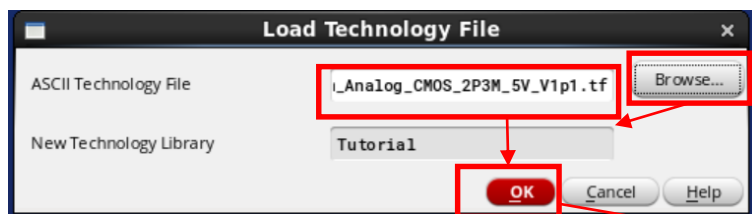
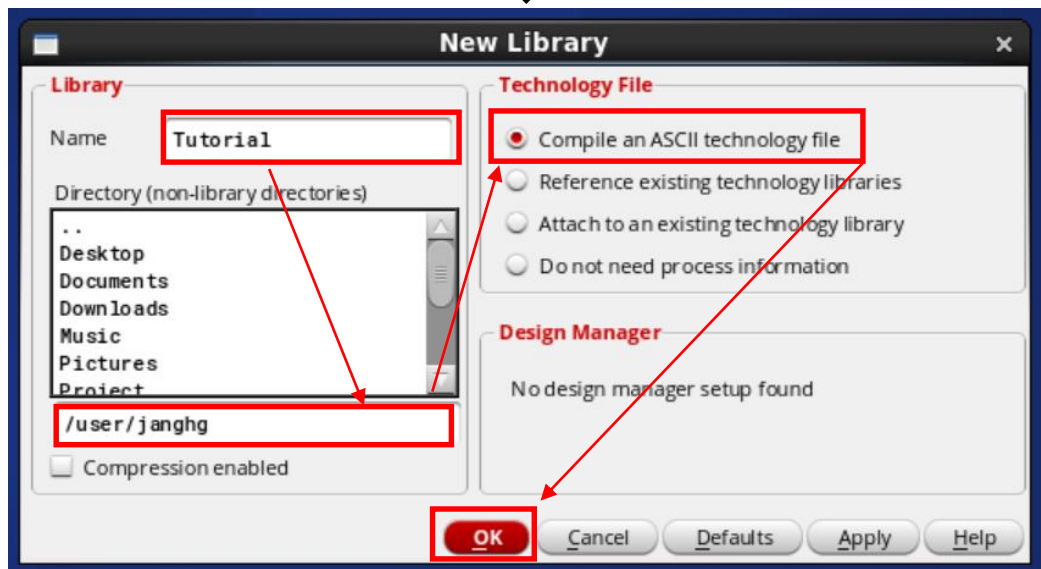
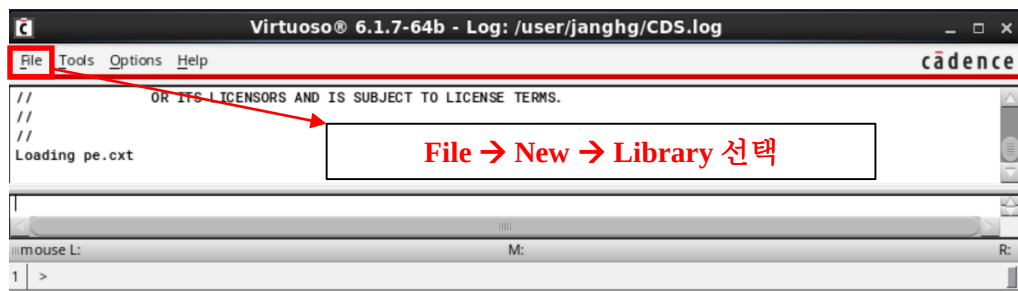


2) Library 생성 및 추가

① Cadence 실행

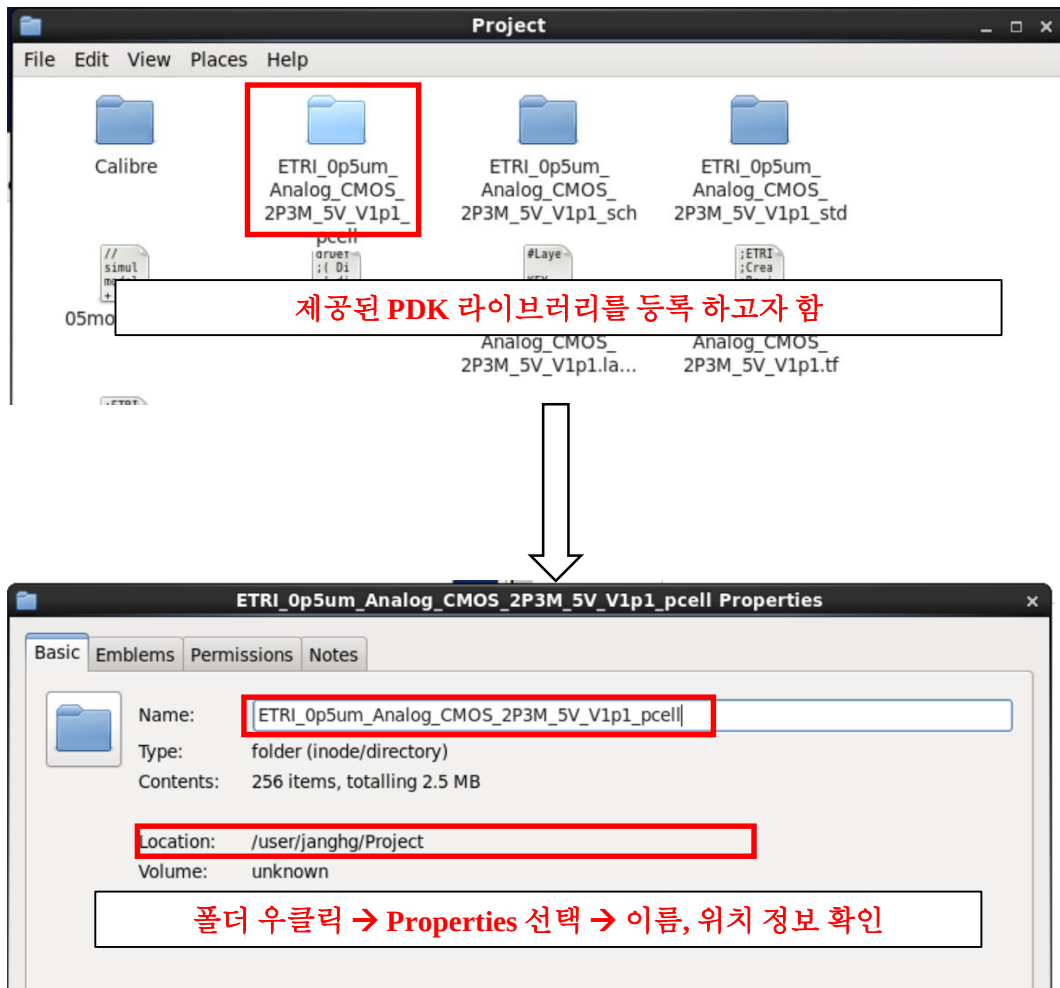


② 라이브러리 생성



Browse 선택 → Project 폴더에 있는 xxx. tf 파일 선택 → OK
→ 문제없을 시 성공 메시지 확인 가능

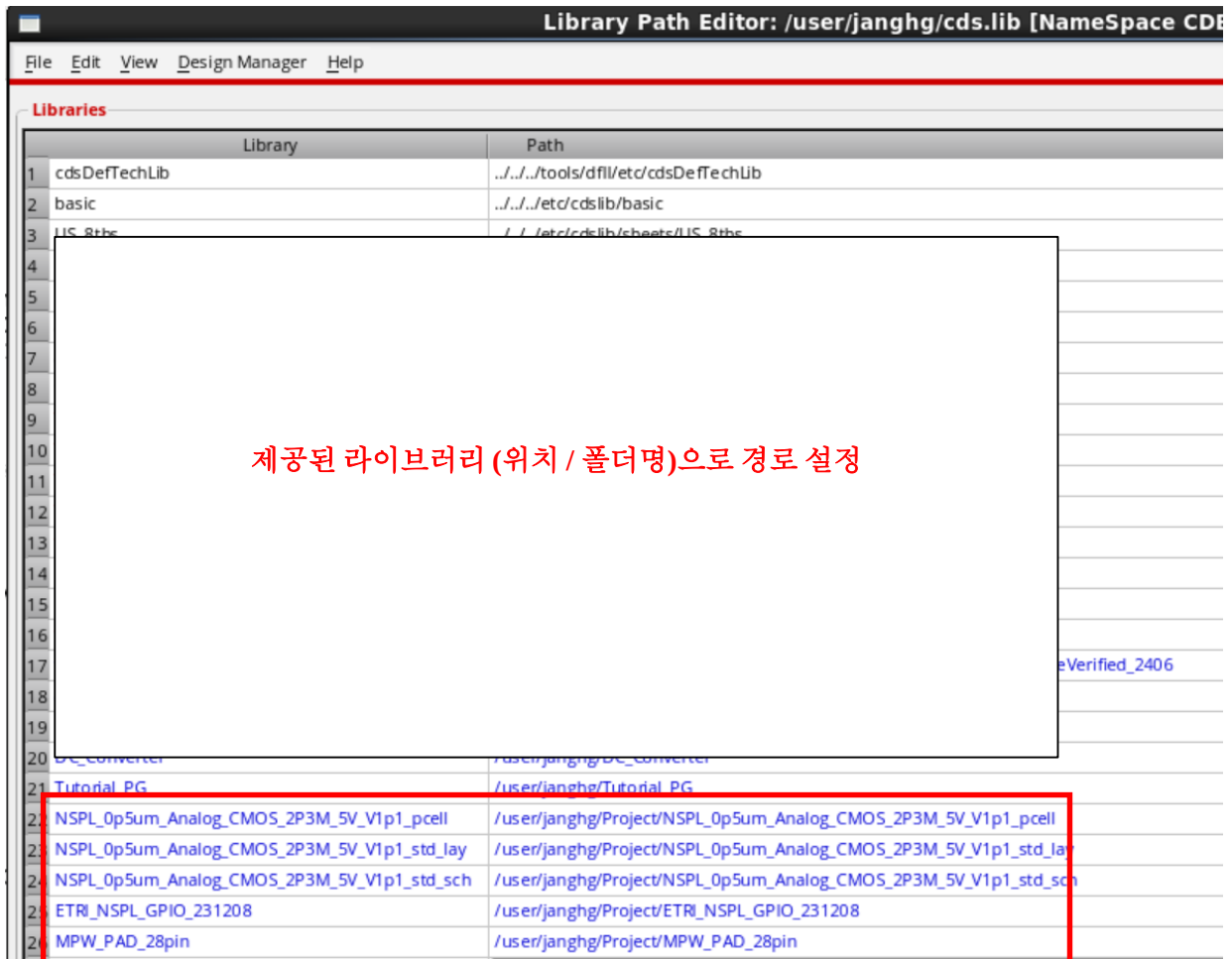
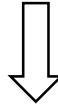
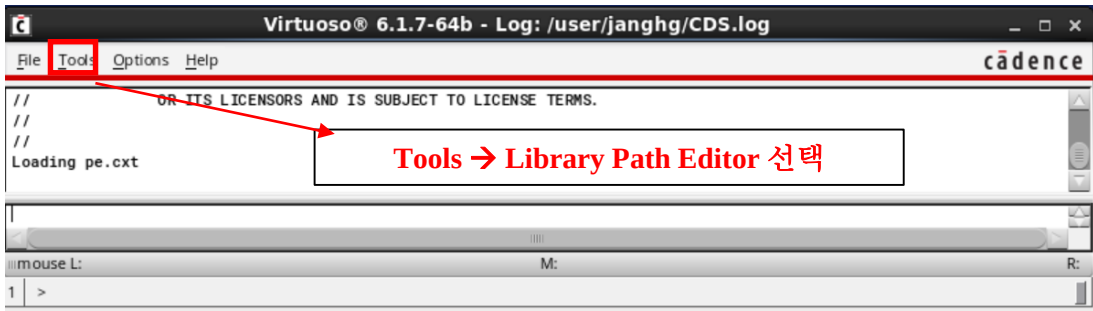
③ 제공된 PDK 라이브러리 추가



- 이름 : ETRI_Op5um_Analog_CMOS_2p3M_5V_V1p1_pcell
- 위치 : /user/janghg/Project

→ 라이브러리 추가에 따른 경로 설정에 필요함

→ 라이브러리 경로 설정시 위치/이름 삽입 필요! (뒷장 계속)



폴더명

위치/폴더명

- 만일, 라이브러리 추가 오류 발생시 빨간색 글씨로 바뀜
- 파란색일 경우 정상적인 등록

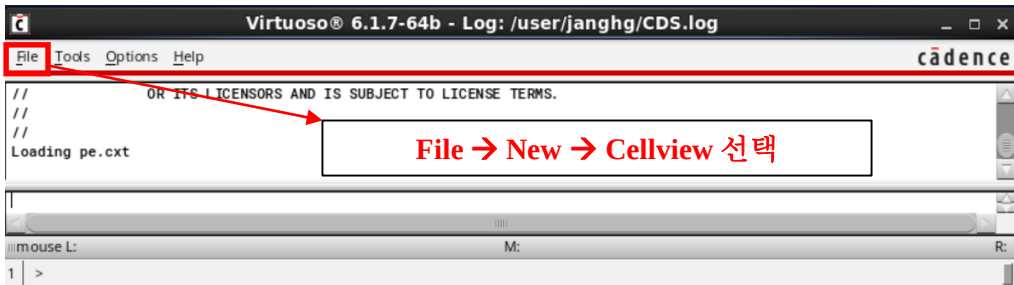
3. 회로 설계 및 검증 (Pre-simulation)

3. 회로 설계 및 검증(Pre-simulation)

- 목적 : 회로 설계 및 시뮬레이션을 통한 회로 검증
- 예: 인버터 회로 설계 및 검증

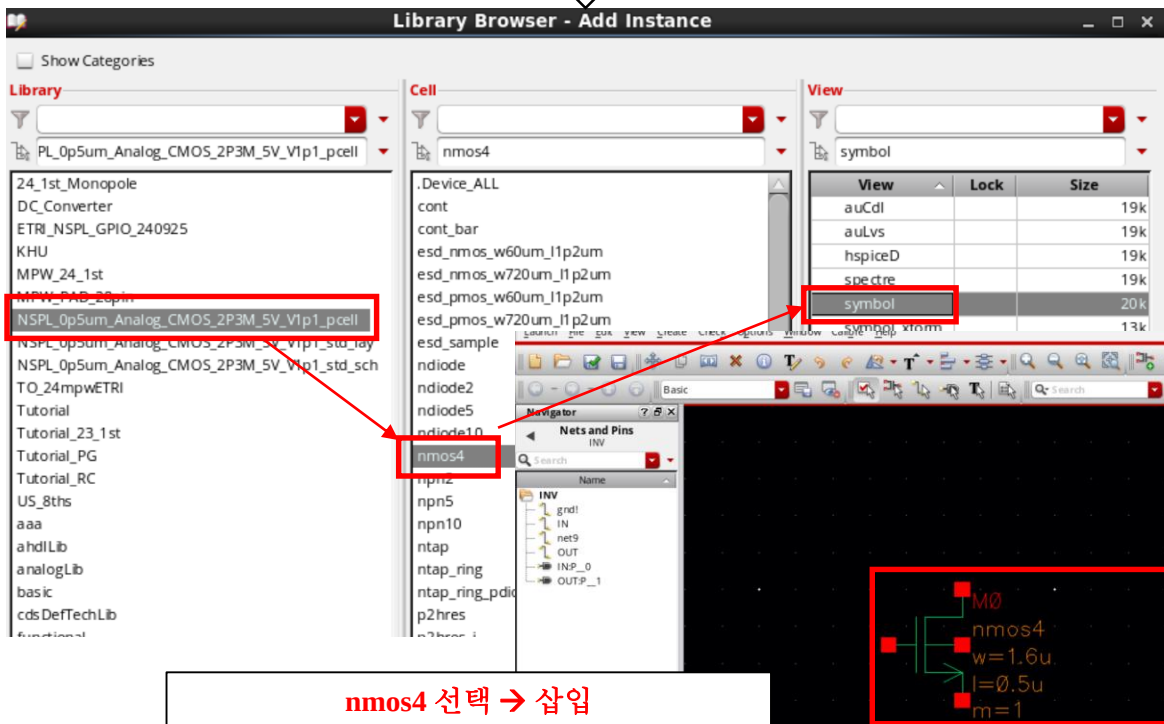
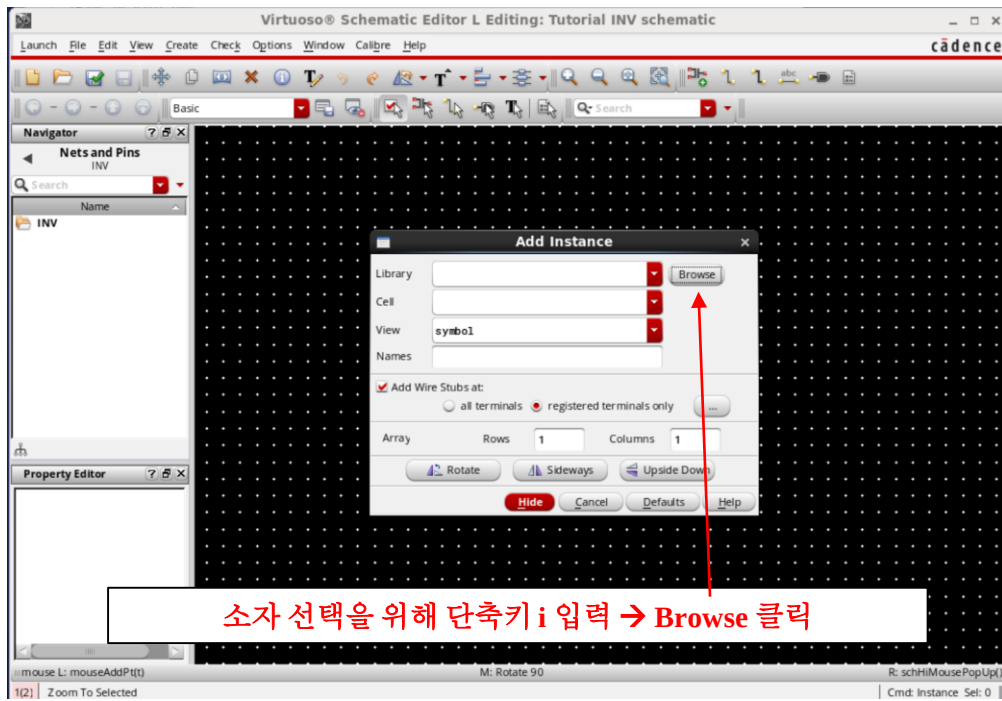
1) 인버터 설계

① Schematic 설계 프로그램 실행



Cell 명 → Type 설정 → 실행 가능한 시뮬레이터 선정 → OK 클릭

② 소자 설정



소자별 parameter 입력을 위해 소자 클릭 후 단축키 Q

- Model name은 시뮬레이션을 위해 모델파라미터 이름 입력 (pmos4, nmos4)
- Width, Length 값 기입

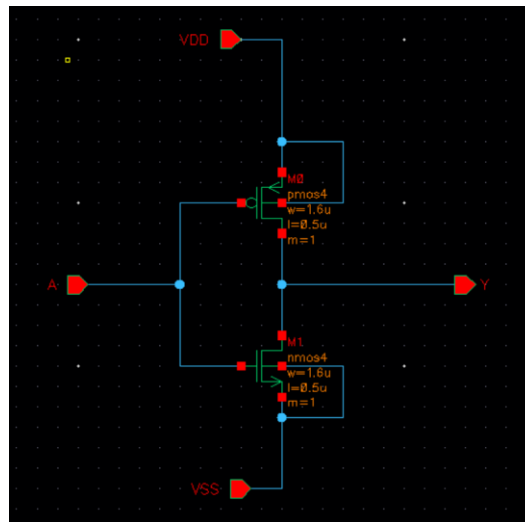
③ Pin 설정



포트 설정(단축키 P), 핀명과 방향(Input, Output) 설정

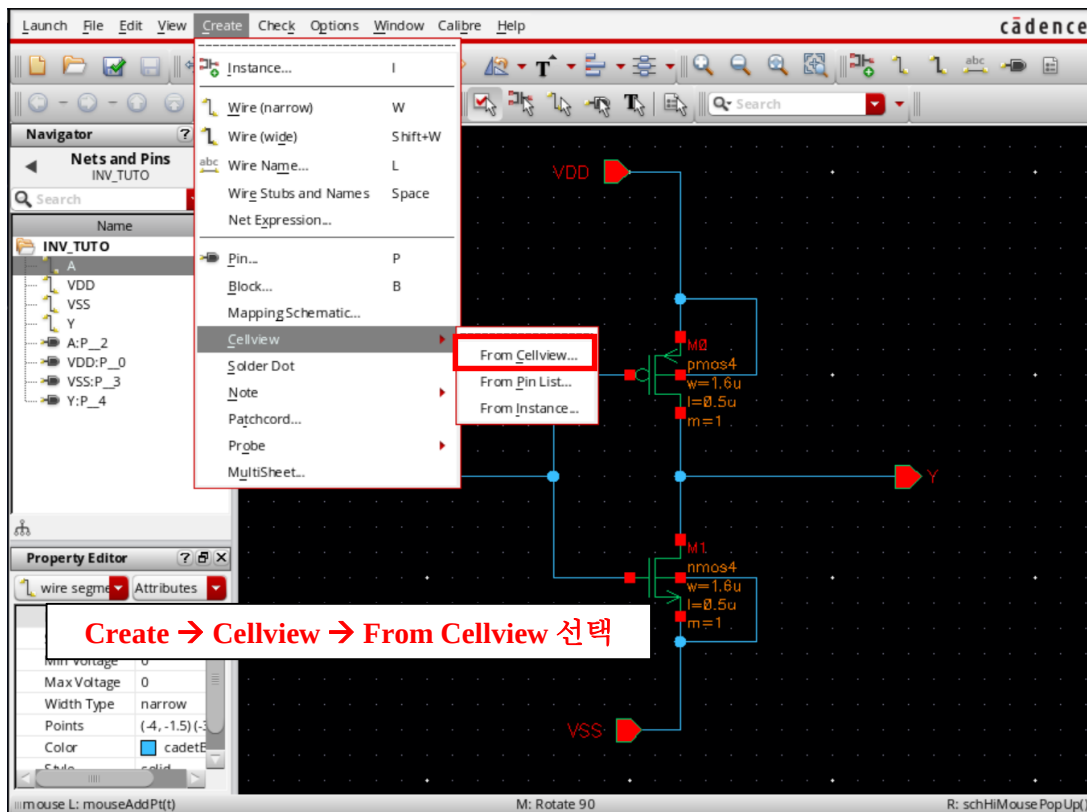
A, Y, VDD, VSS 포트 삽입

④ 와이어 연결

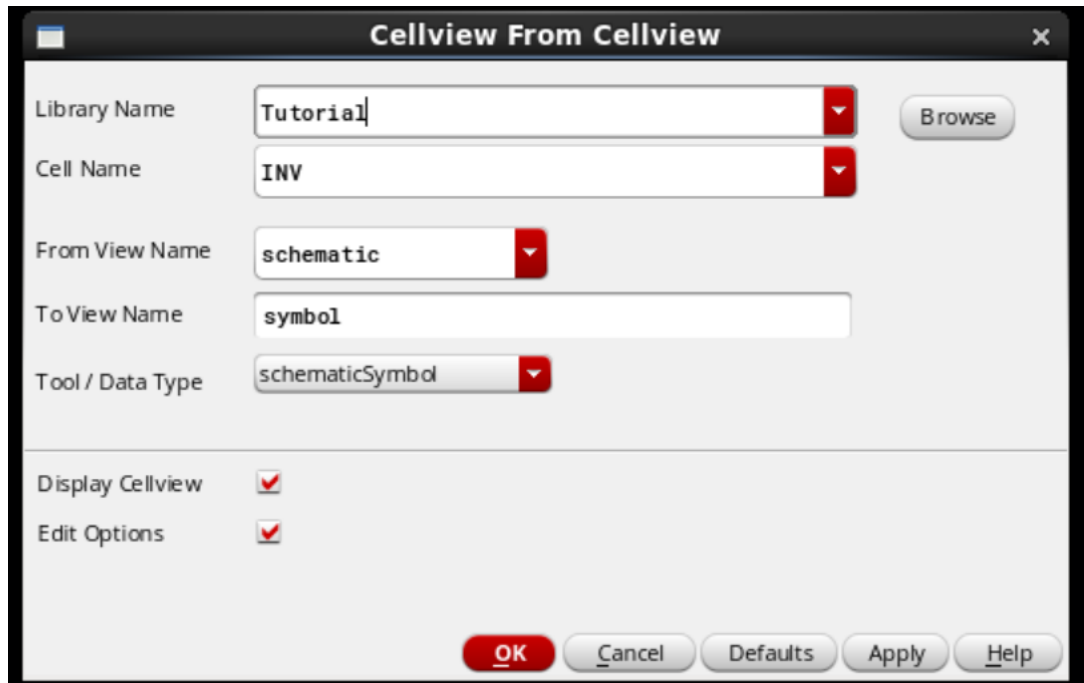


와이어 연결(단축키 w)

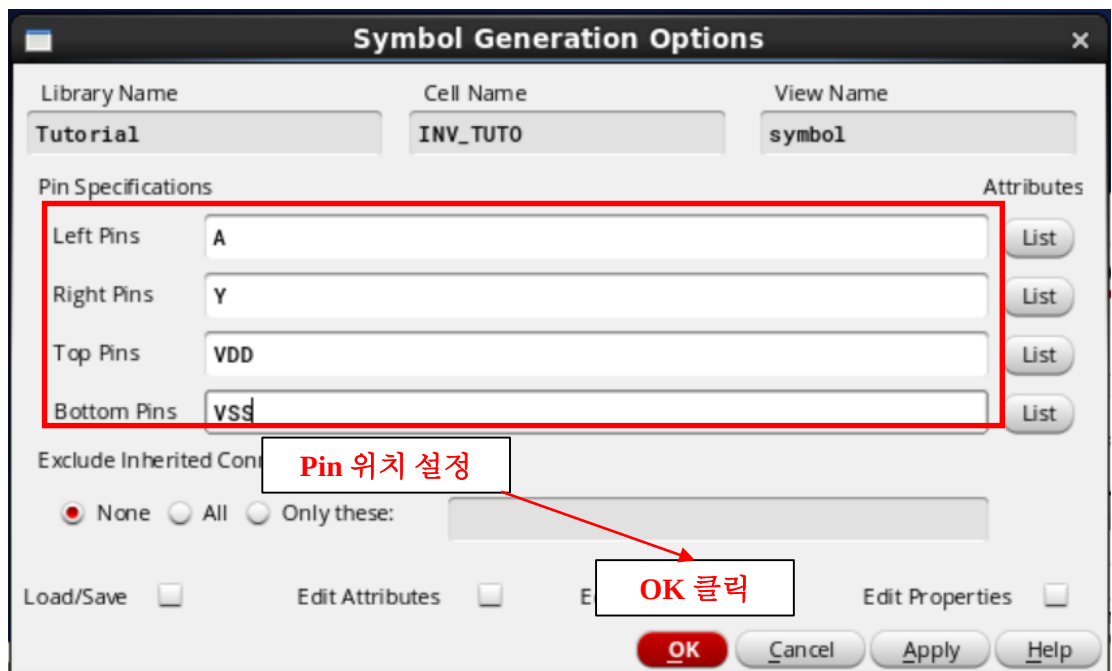
⑤ 심볼 형성

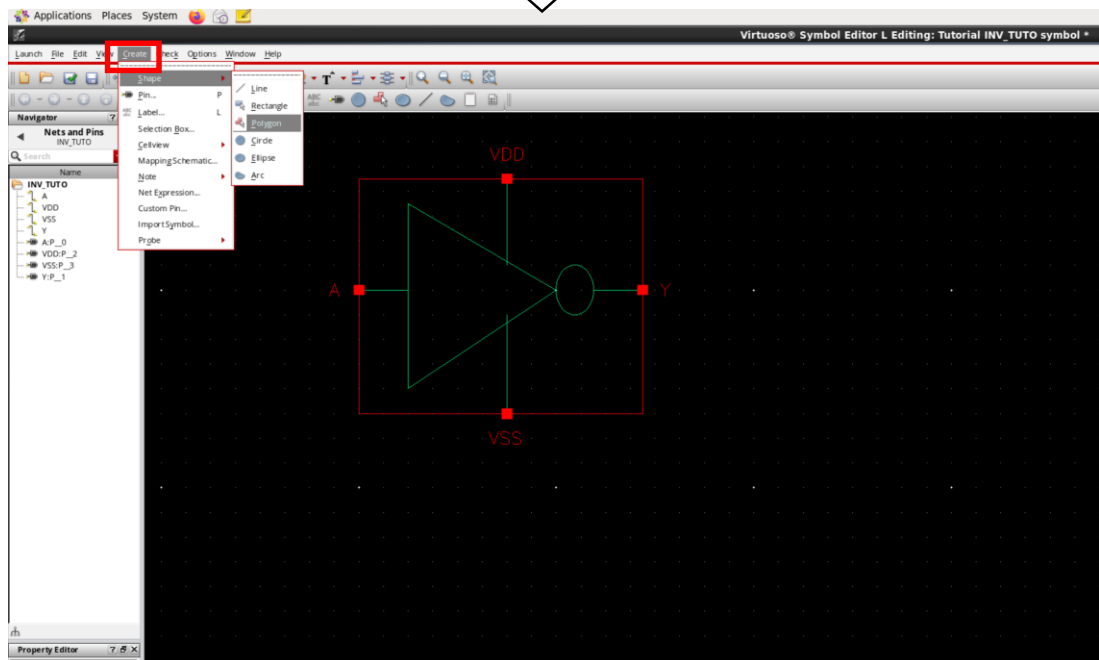
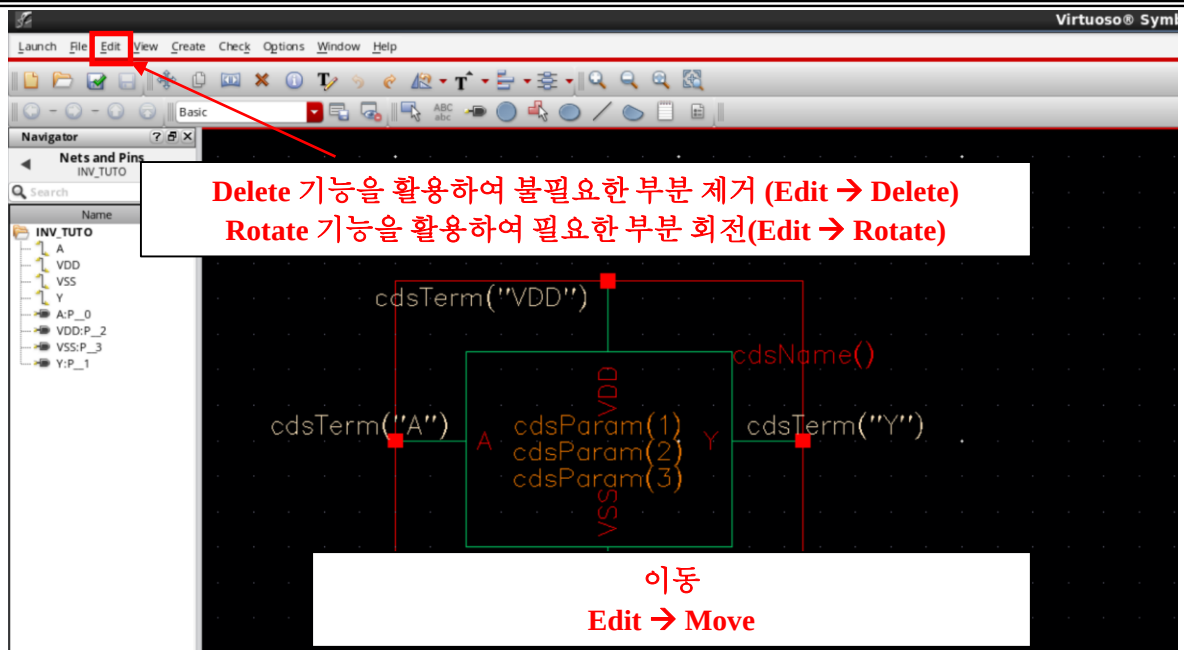


Create → Cellview → From Cellview 선택



OK 선택

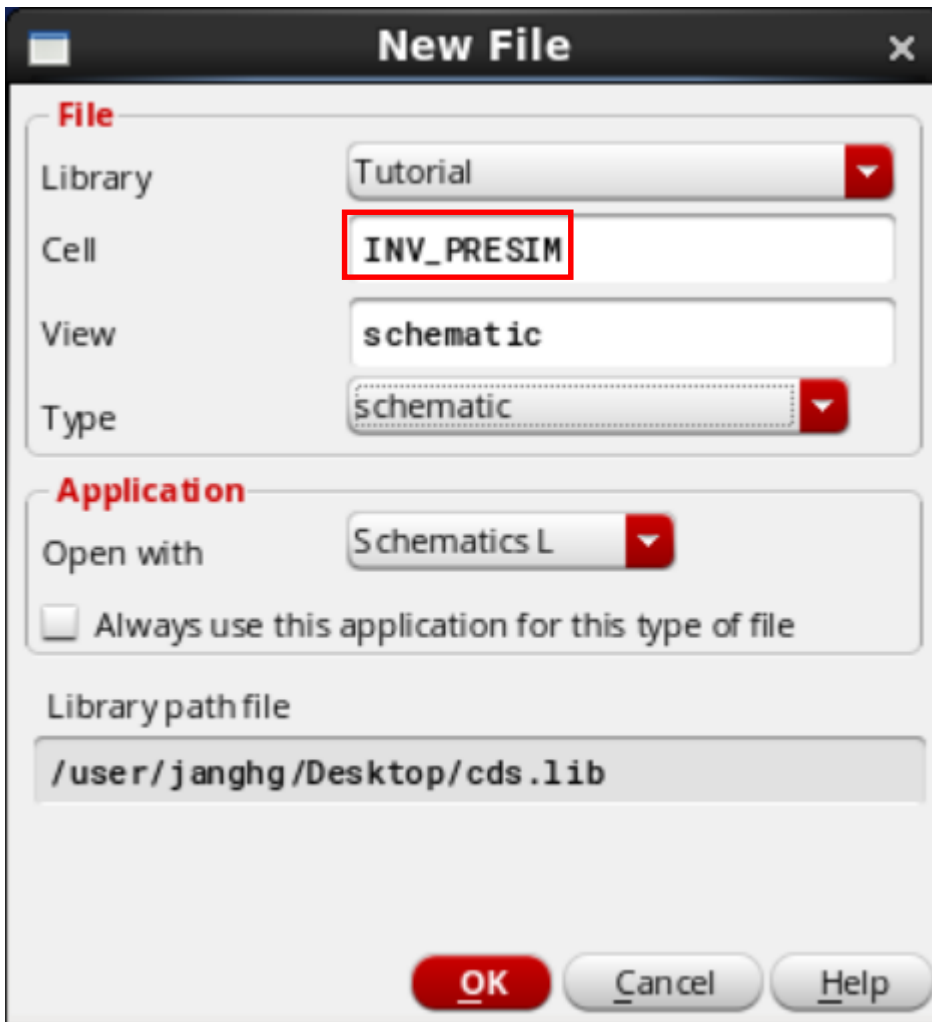
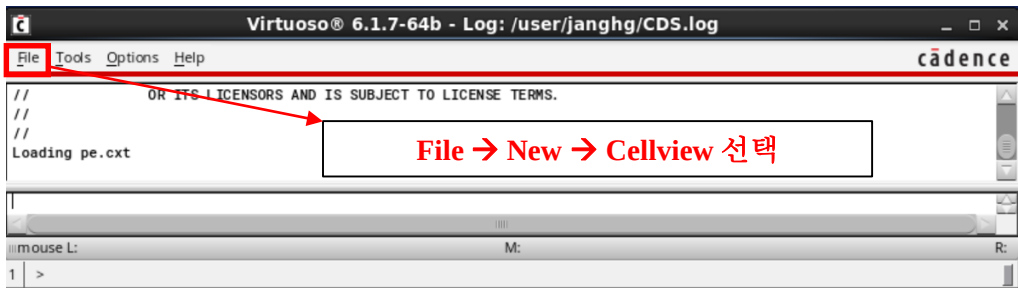




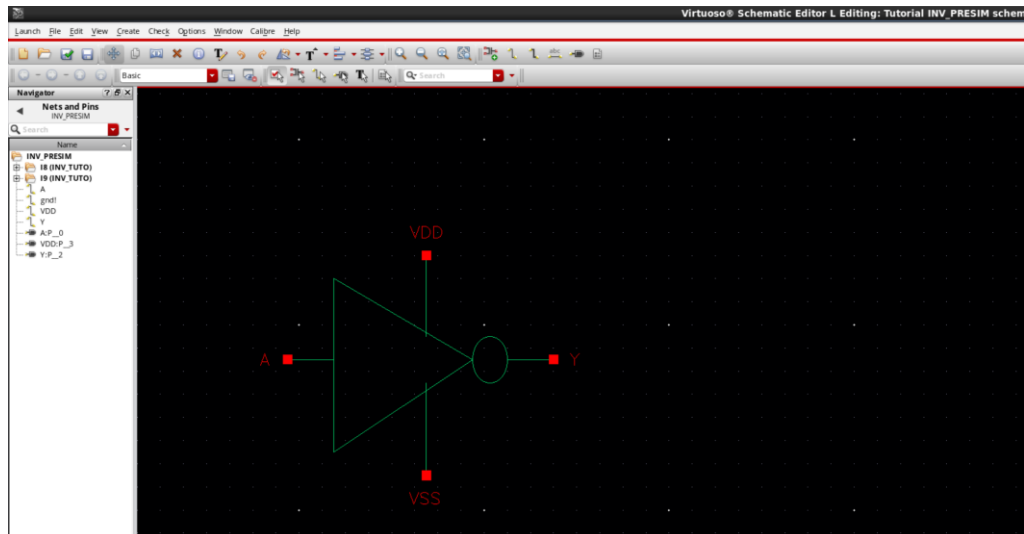
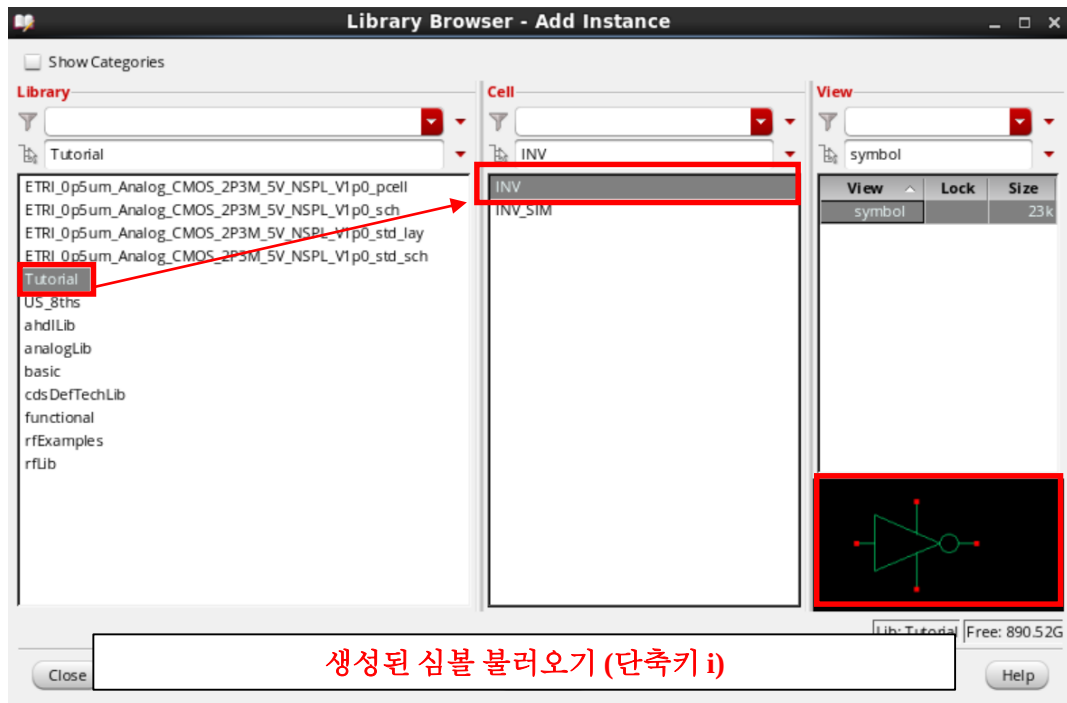
2) 시뮬레이션을 위한 회로 설계

① Schematic 설계 프로그램 실행

새로운 Cell 생성

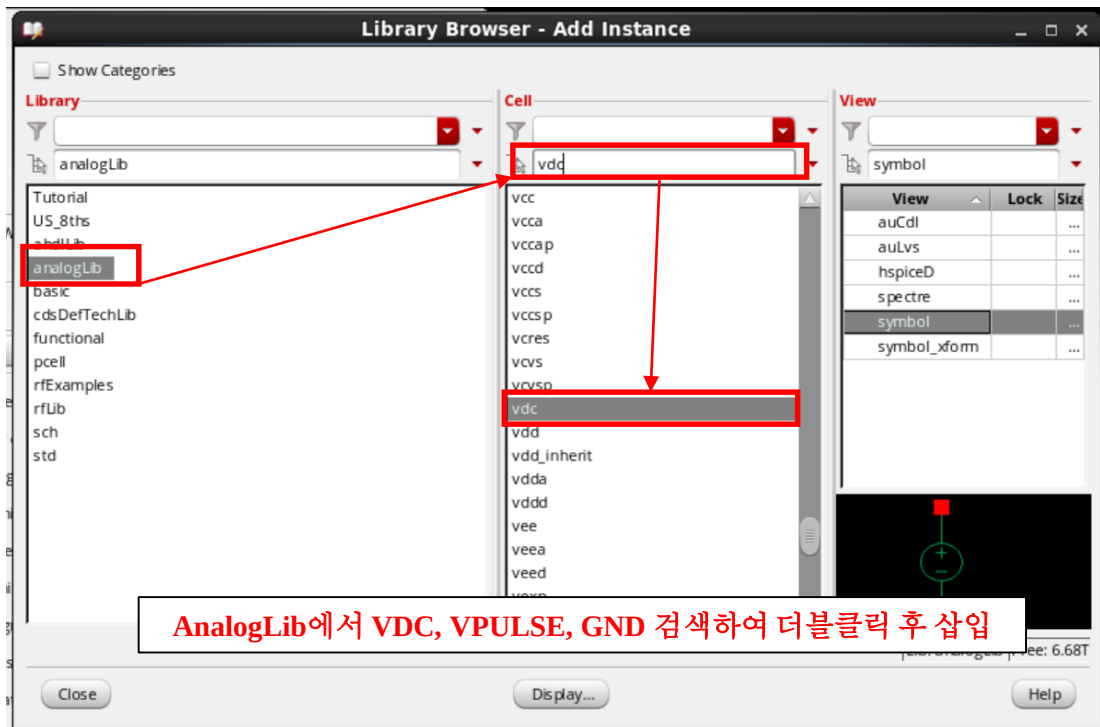


② 인버터 심볼 불러오기

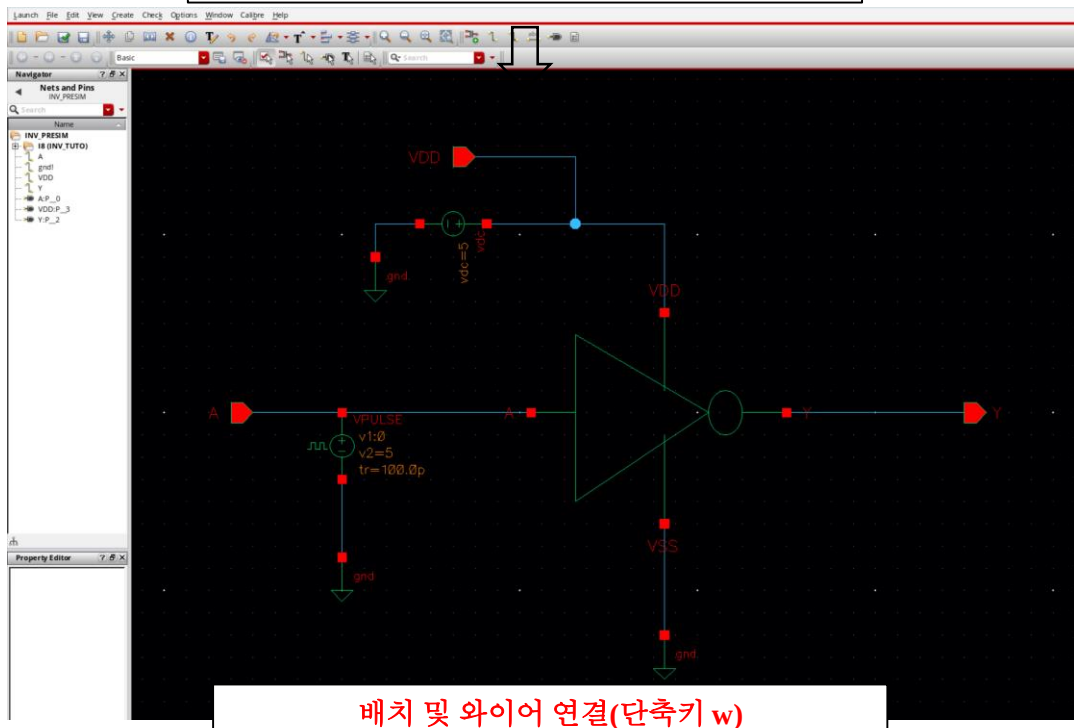


심볼 배치

③ 소자 배치 및 와이어 연결



IN, OUT 포트 생성 (단축키 P)



④ 인가값 설정

Edit Object Properties

Apply To:

Show: ☐ system ☒ user ☒ CDF

Property	Value	Display
Library Name	analogLib	off
Cell Name	vdc	off
View Name	symbol	off
Instance Name	v0	off

User Property	Master Value	Local Value	Display
lvignore	TRUE		off

CDF Parameter	Value	Display
Noise file name		off
Number of noise/freq pairs	0	off
DC voltage	5 V	off

Edit Object Properties

Apply To:

Show: ☐ system ☒ user ☒ CDF

Property	Value	Display
Library Name	analogLib	off
Cell Name	vpulse	off
View Name	symbol	off
Instance Name	VPULSE	off

User Property	Master Value	Local Value	Display
lvignore	TRUE		off

CDF Parameter	Value	Display
Frequency name for 1/period		off
Noise file name		off
Number of noise/freq pairs	0	off
DC voltage		off
AC magnitude		off
AC phase		off
XF magnitude		off
PAC magnitude		off
PAC phase		off
Voltage 1	0 V	off
Voltage 2	5 V	off
Period	10n s	off
Delay time	500.0p s	off
Rise time	100.0p s	off
Fall time	100.0p s	off
Pulse width	5n s	off
Temperature coefficient 1		off
Temperature coefficient 2		off
Nominal temperature		off
Type of rising & falling edge		off

시뮬레이션을 위해 VDC 및 VPULSE 값 입력

⑤ 저장 및 검증



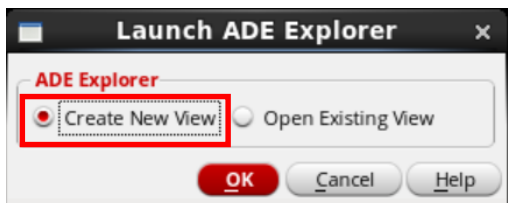
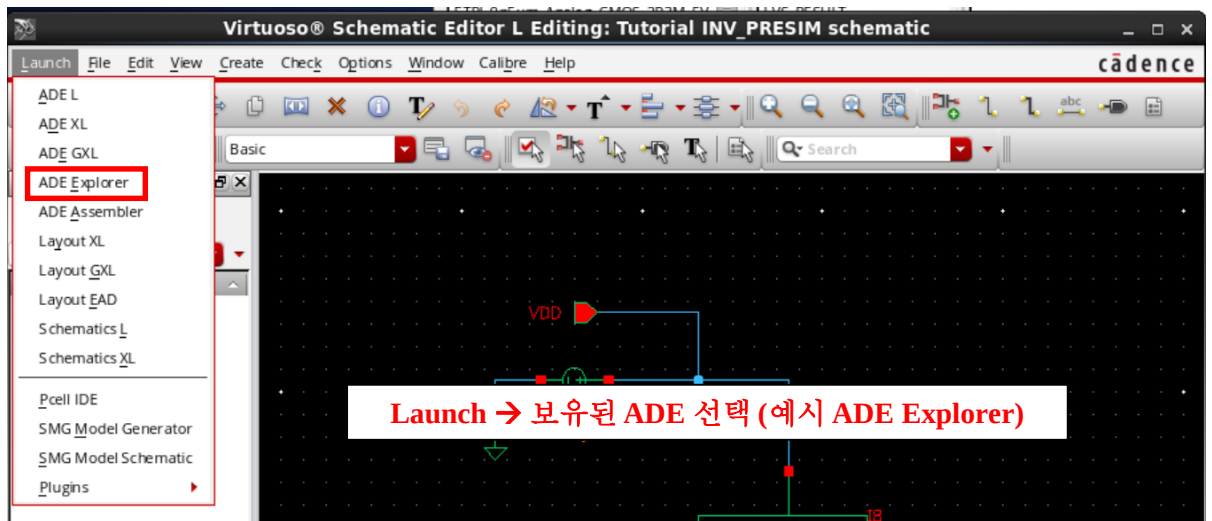
Check & save 버튼 클릭



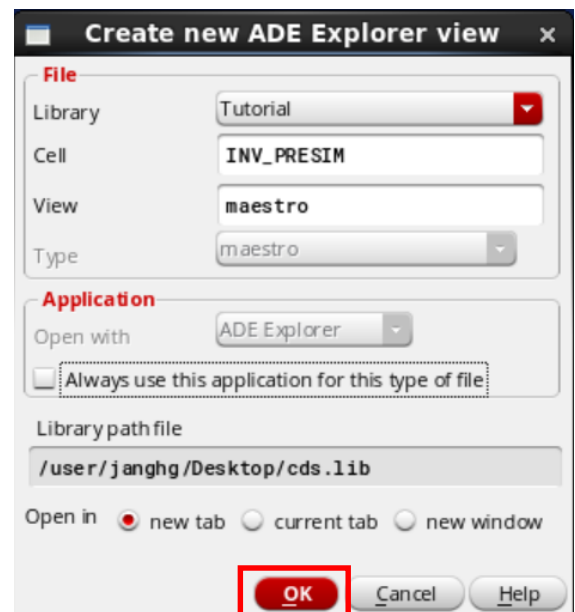
에러 및 경고 메시지 확인, 에러 0까지 수정 및 검증

3) 인버터 시뮬레이션

① Schematic 시뮬레이션 프로그램 실행

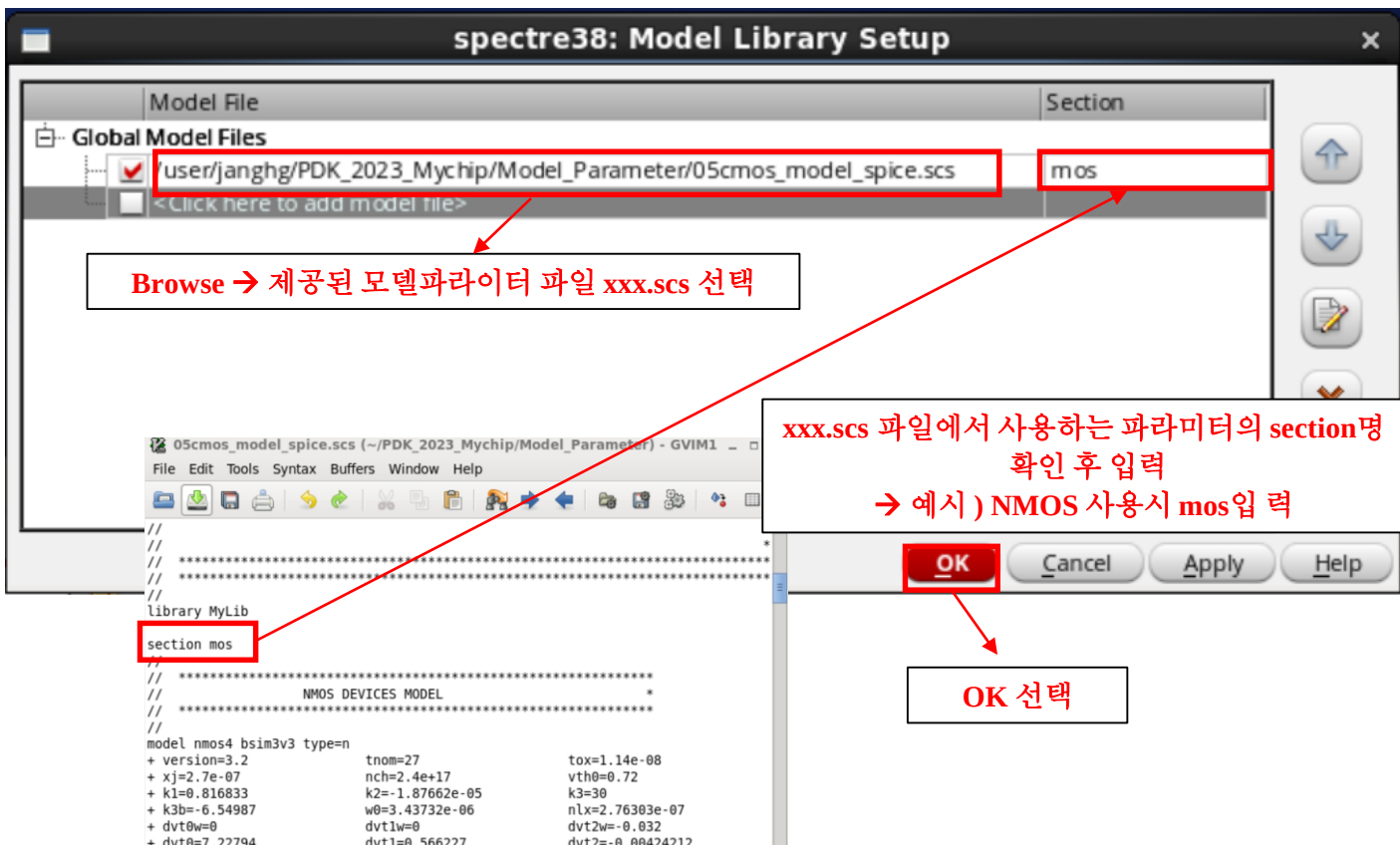
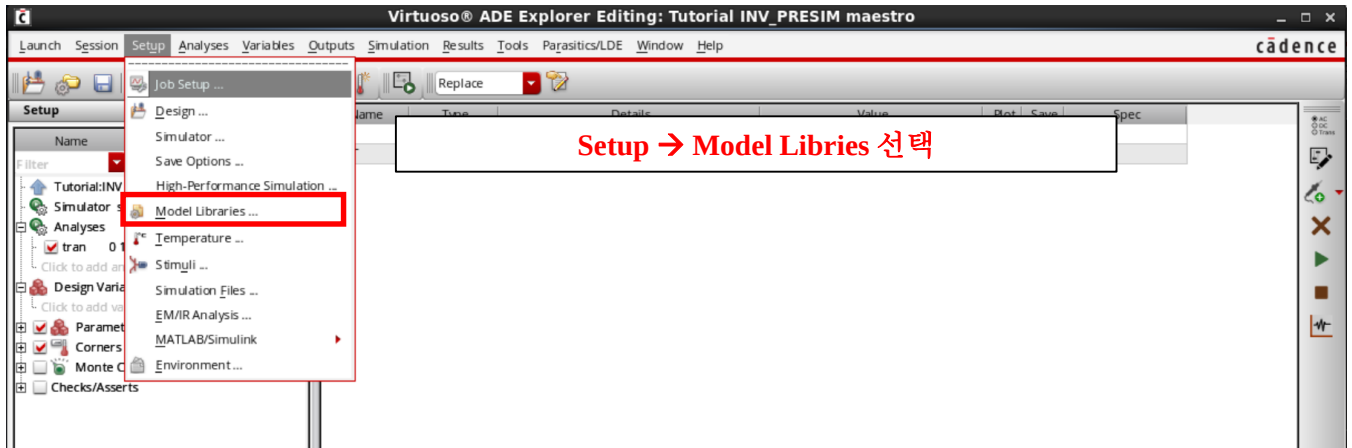


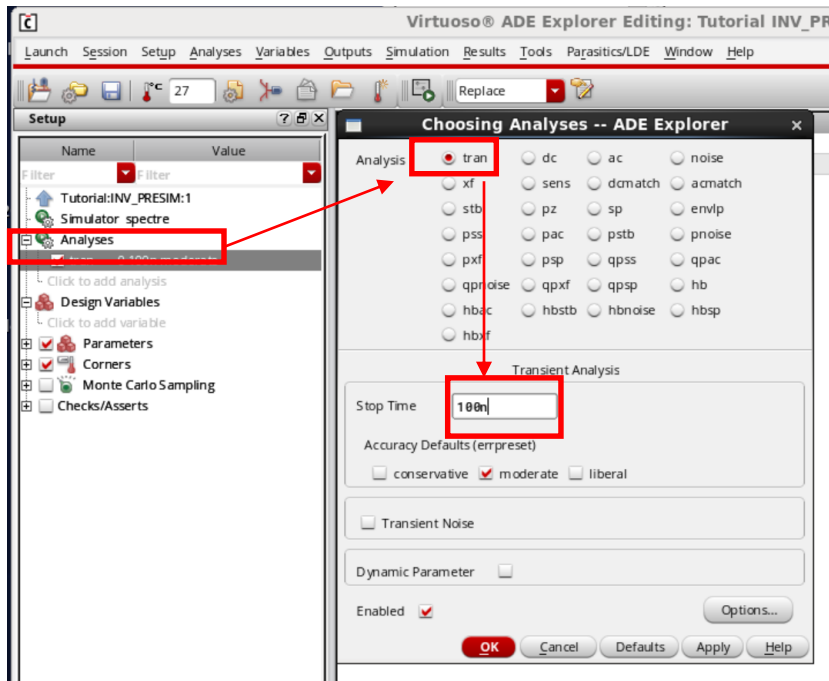
Create New View 선택
(시뮬레이션 완료 저장 후에는 Open Existing View 선택)



오픈 방식 설정 후 OK 클릭

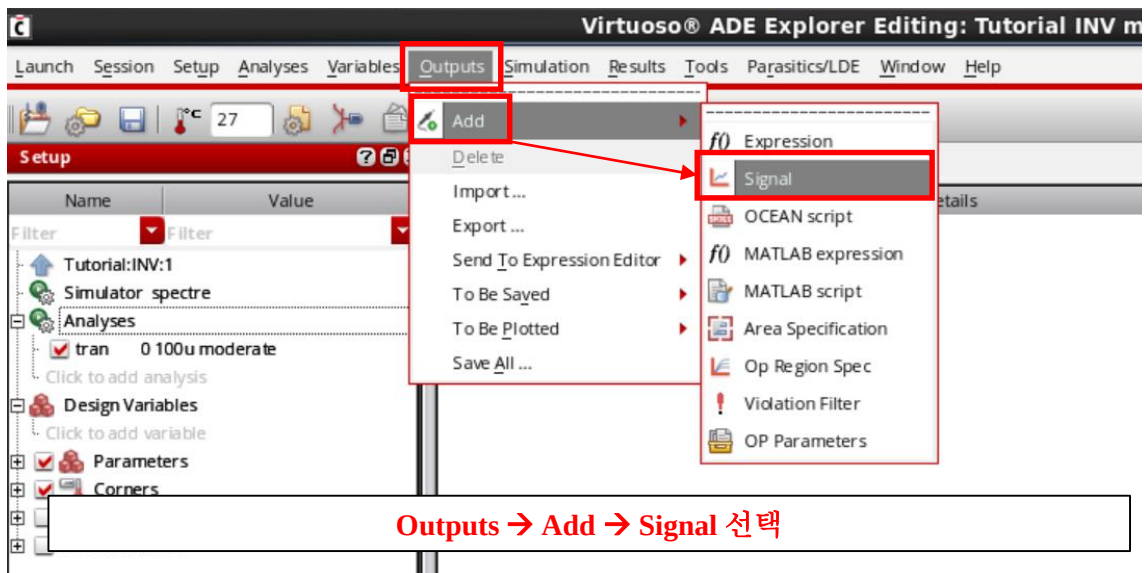
② 회로 검증 시뮬레이션 환경 설정





시뮬레이션 설정 : 분석 항목 선택 및 조건 기입 (예시에서는 Tran. 100ns 로 기입)

③ 시뮬레이션 포트 설정

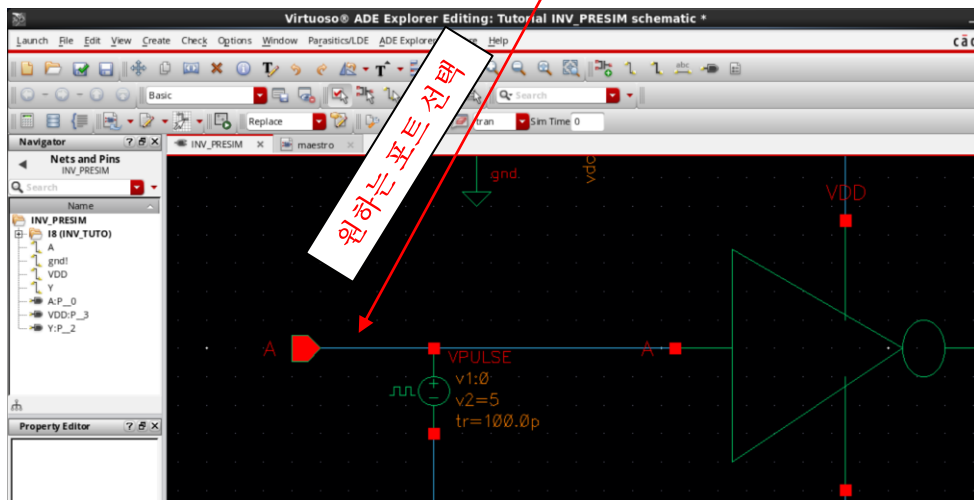


Outputs → Add → Signal 선택

Browser 선택

Name	Type	Details	Value	Plot	Save
VIN	signal	/A		☒	☐
VOUT	signal	/Y		☒	☒

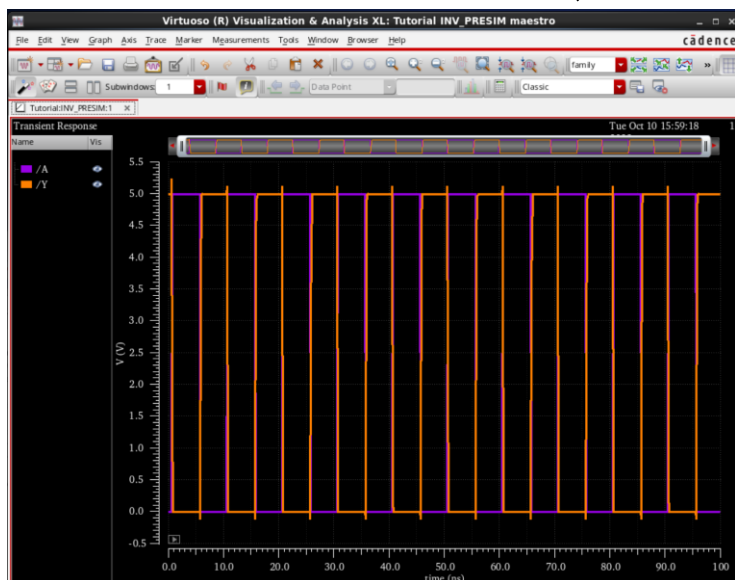
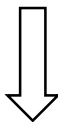
출력명 설정



④ 시뮬레이션 실행



저장 및 시뮬레이션 실행



시뮬레이션 결과 출력

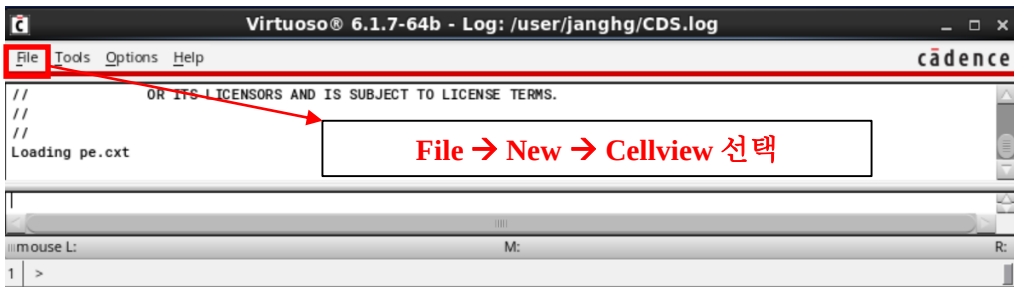
4. Layout

4. Layout

- 목적 : 반도체 회로 제작을 위한 레이아웃 설계
- 예: 설계된 인버터 회로 적용

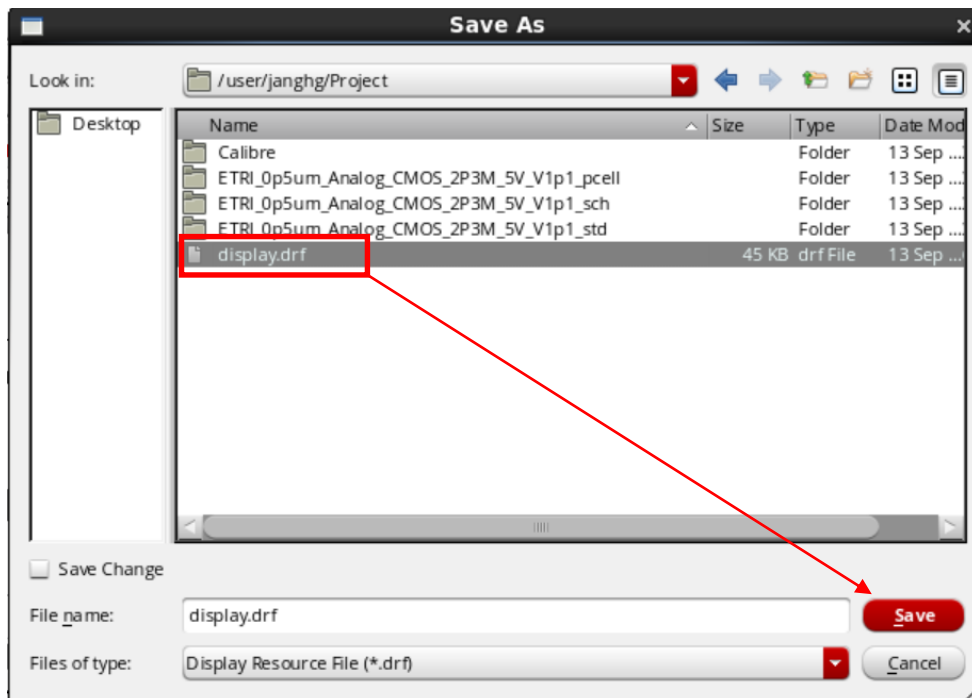
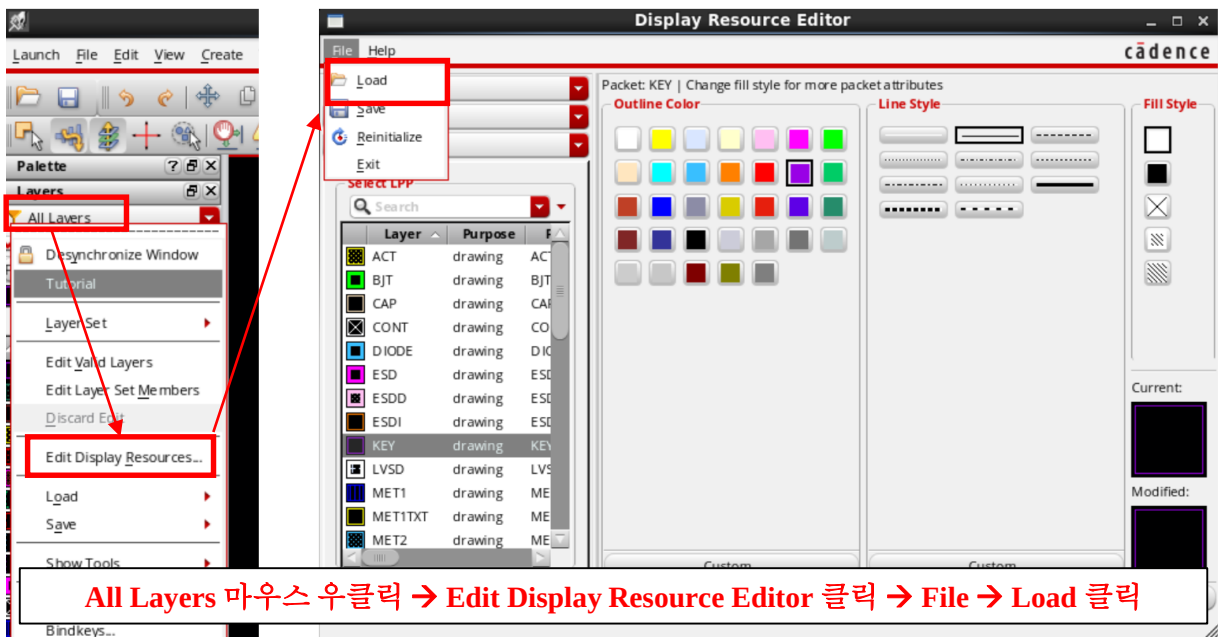
1) 인버터 회로의 레이아웃 설계

① 레이아웃 디자인 설계 프로그램 실행



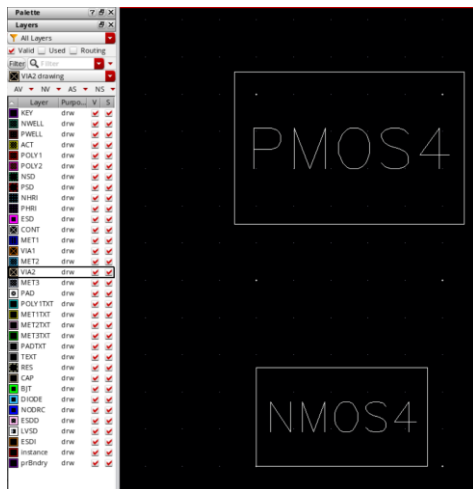
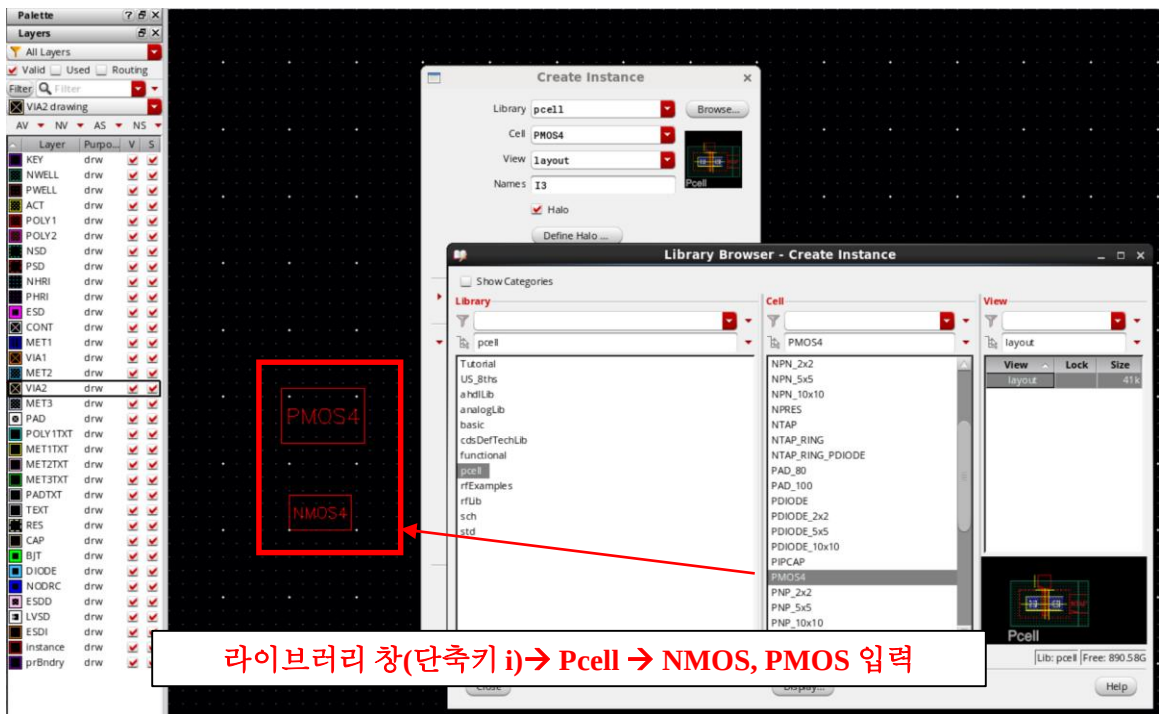
Cell 명 → Type 설정 → 실행 가능한 시뮬레이터 선정 → OK 클릭

② Display resource 파일 불러오기



제공된 xxx.drf 선택 → 저장

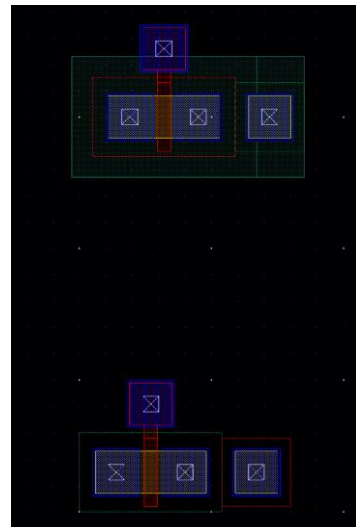
③ Pcell 불러오기



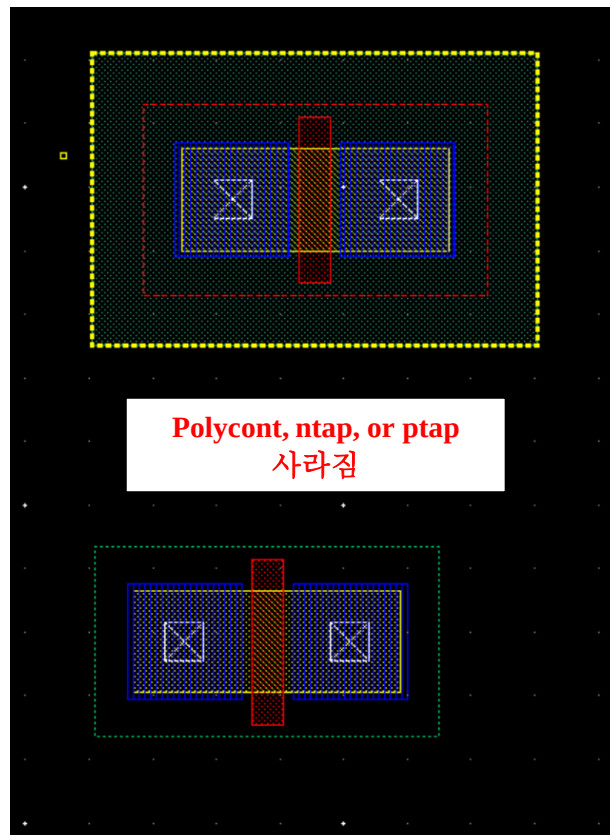
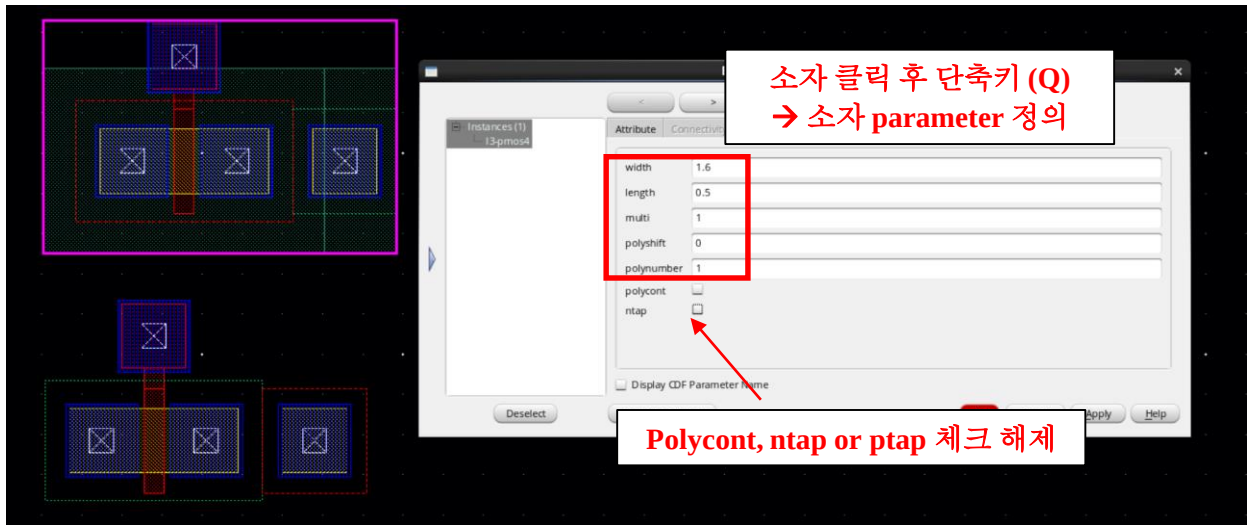
Shift + F



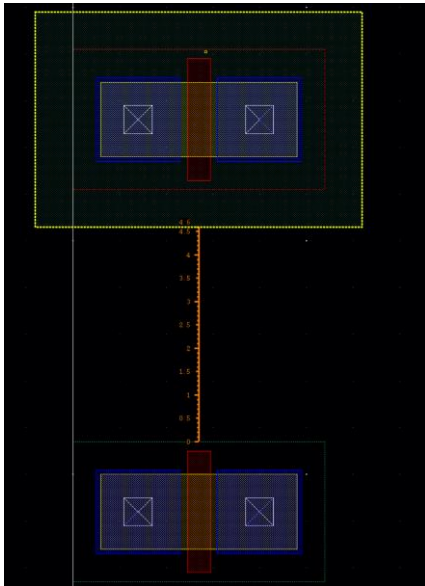
Ctrl + F



④ Pcell 파라미터 설정

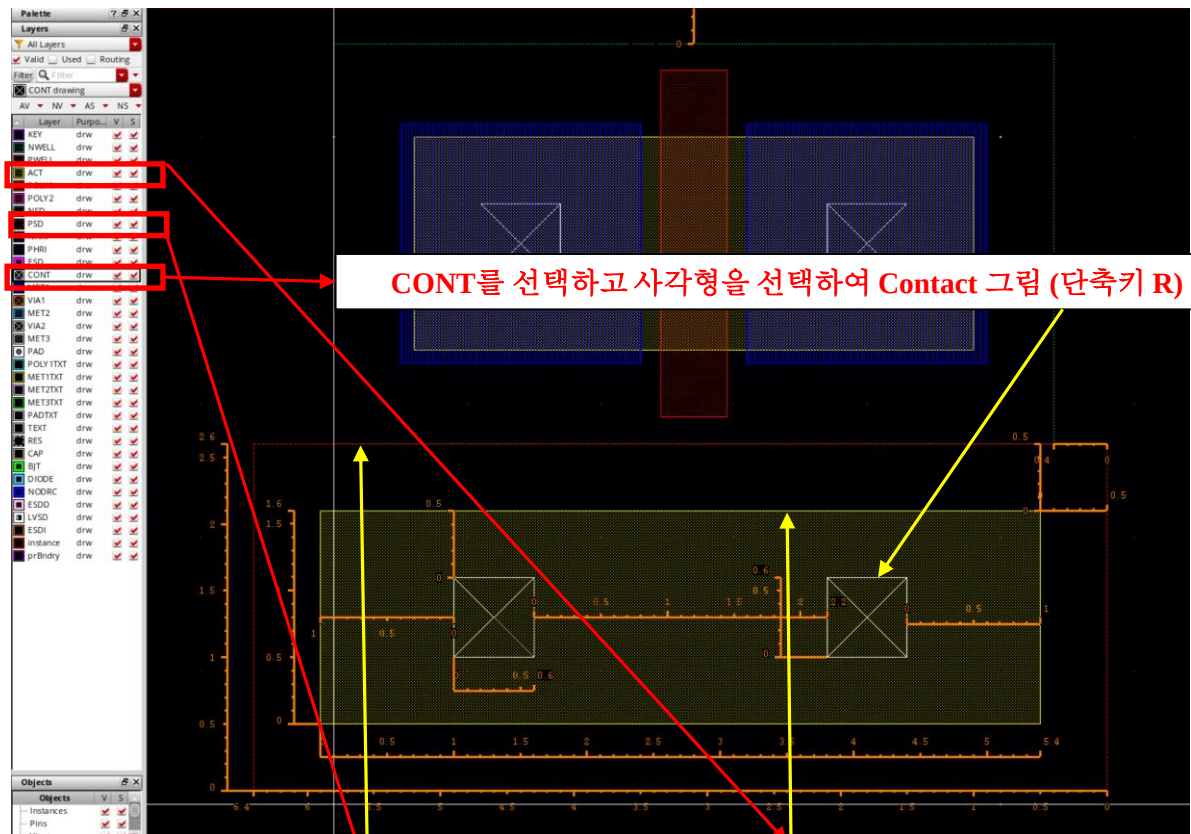


⑤ Layout 진행



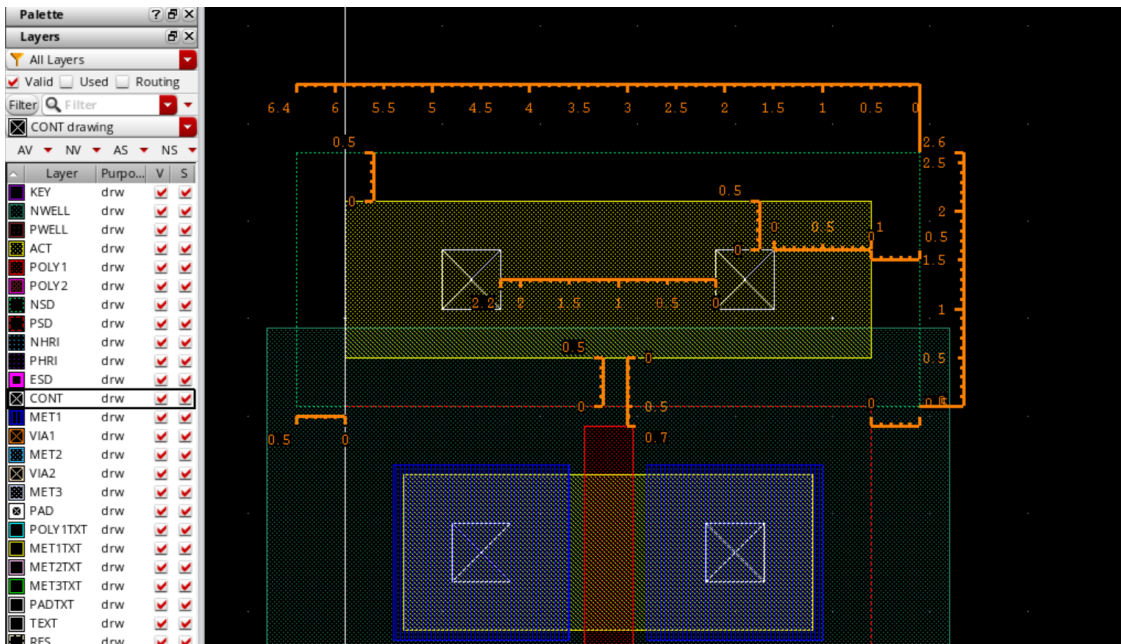
Ruler를 이용하여 사이즈를 확인함(단축키 K)
모든 Ruler를 사라지게 하는 단축키 → Shift + K

PMOS, NMOS 적절한 위치 배치

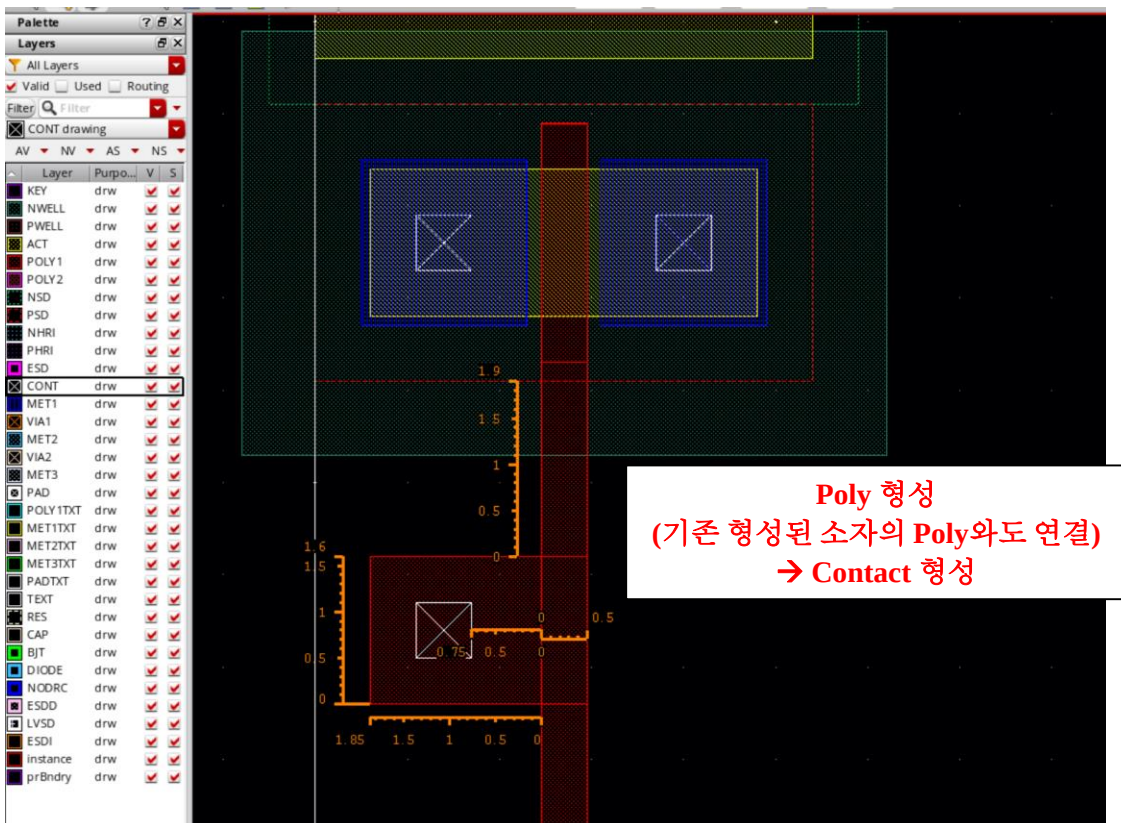


PSD 선택 → 단축키 R → PSD 그림

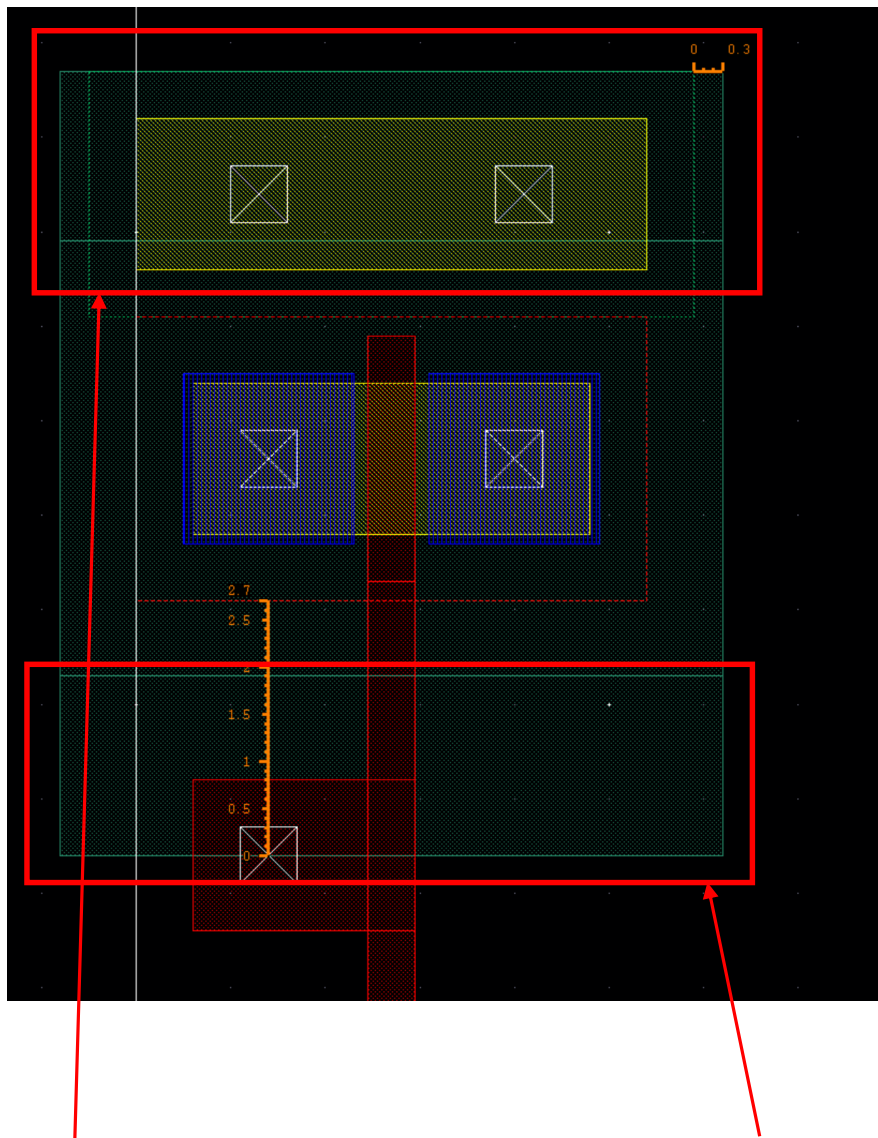
ACT 선택 → 단축키 R → ACT 그림



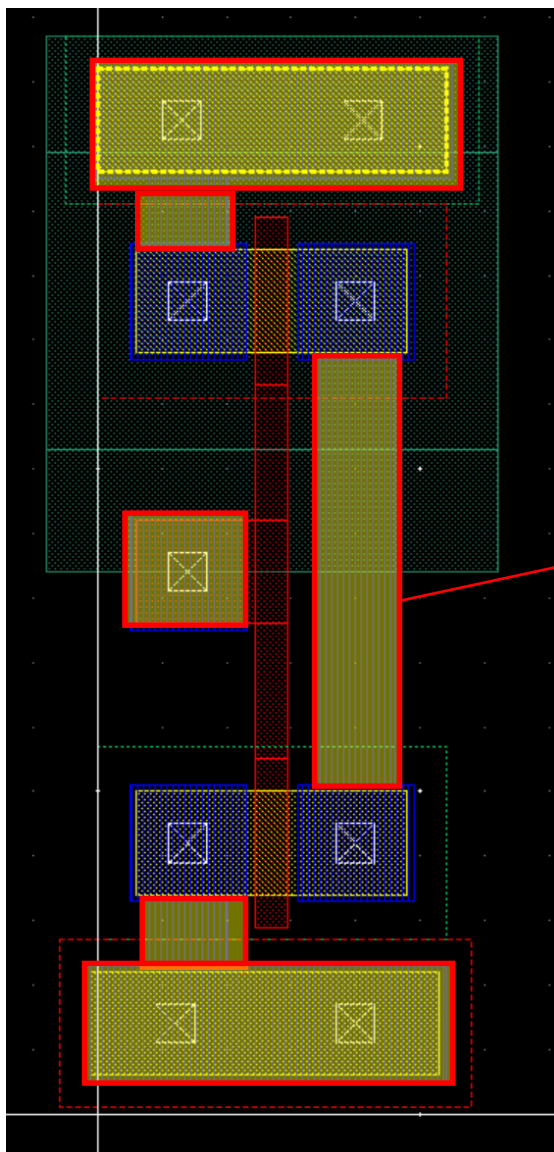
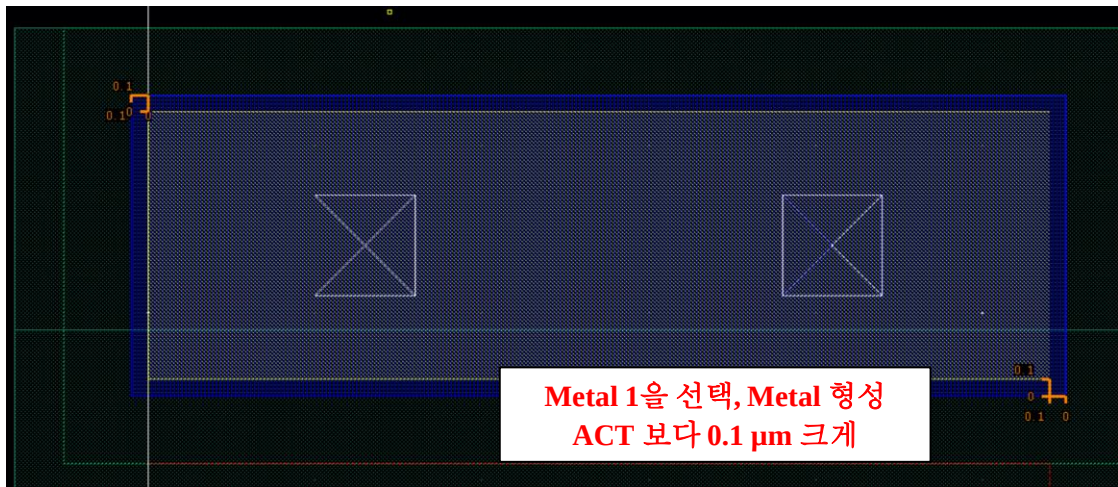
VDD : NSD → ACT → CONT 순서로 설계



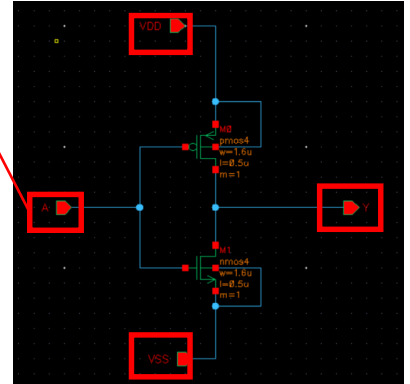
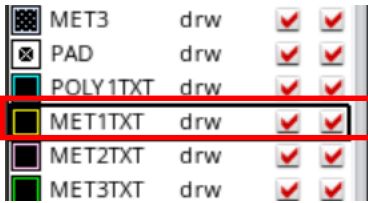
Poly 형성
(기존 형성된 소자의 Poly와도 연결)
→ Contact 형성



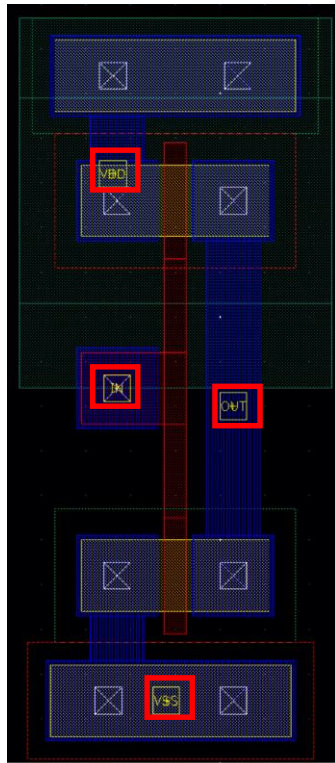
NWELL 추가 형성



⑥ Label 할당

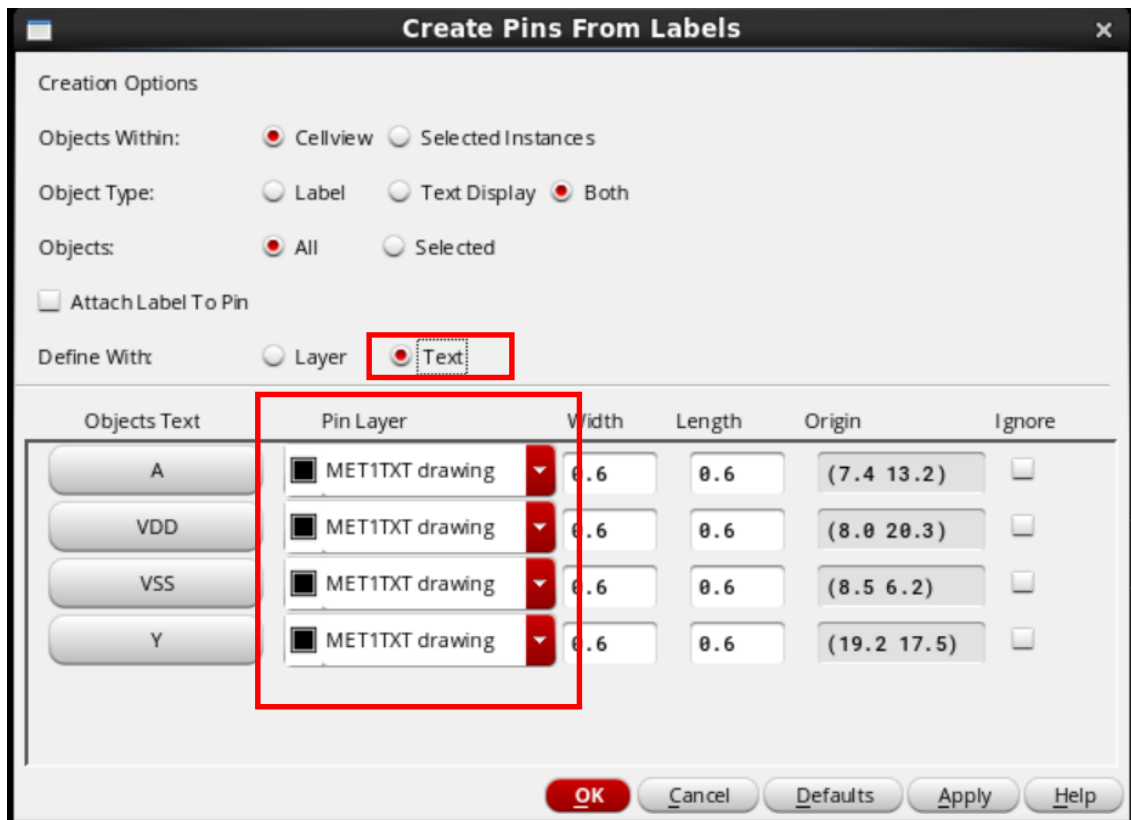


**Label 설정 : MET1TXT Layer 선택 → Create → Label (단축키 L)
→ Label명은 회로도와 동일하게 할당**



Metal1 안쪽으로 각각의 Label 배치

⑦ Label로 부터 Pin 할당



File → Save

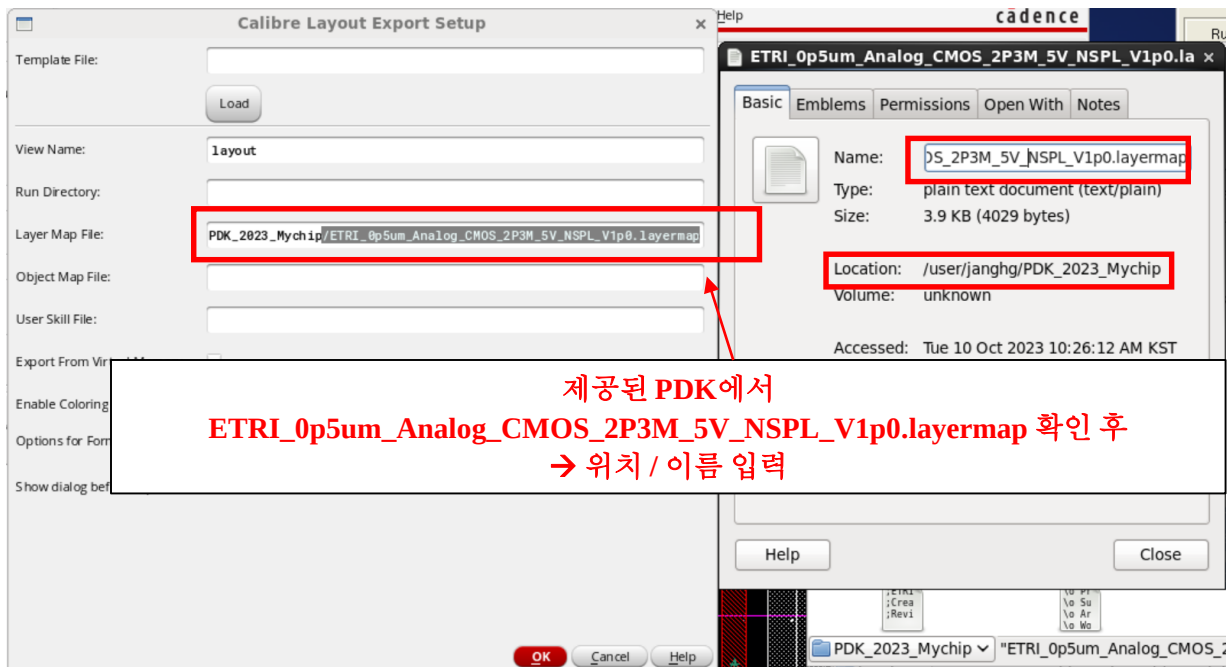
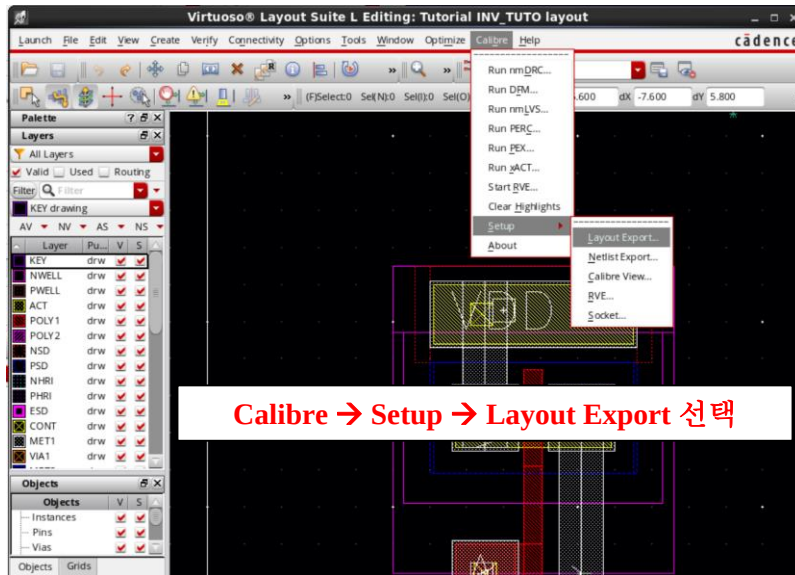
5. DRC (Design Rule Check)

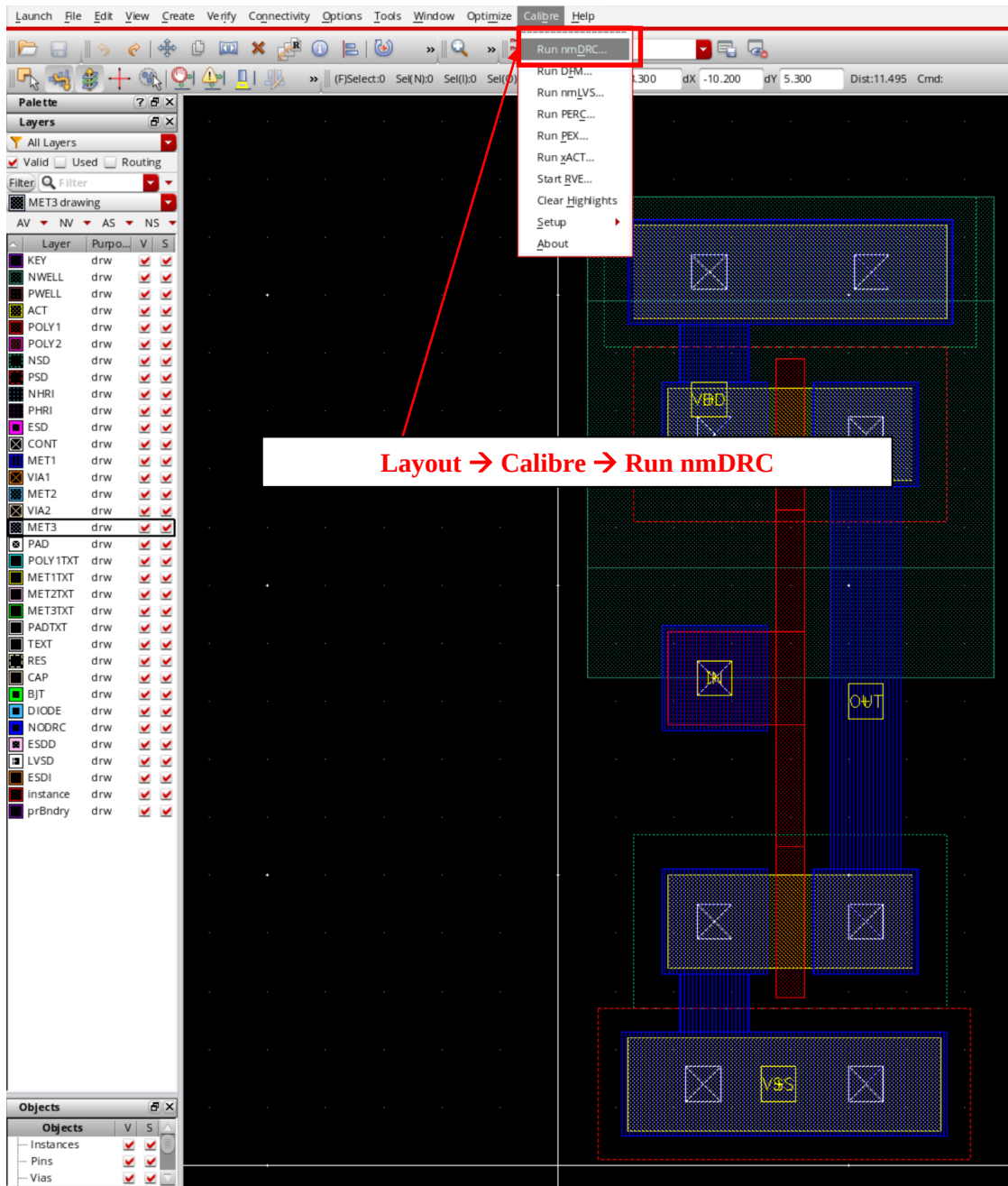
5. DRC

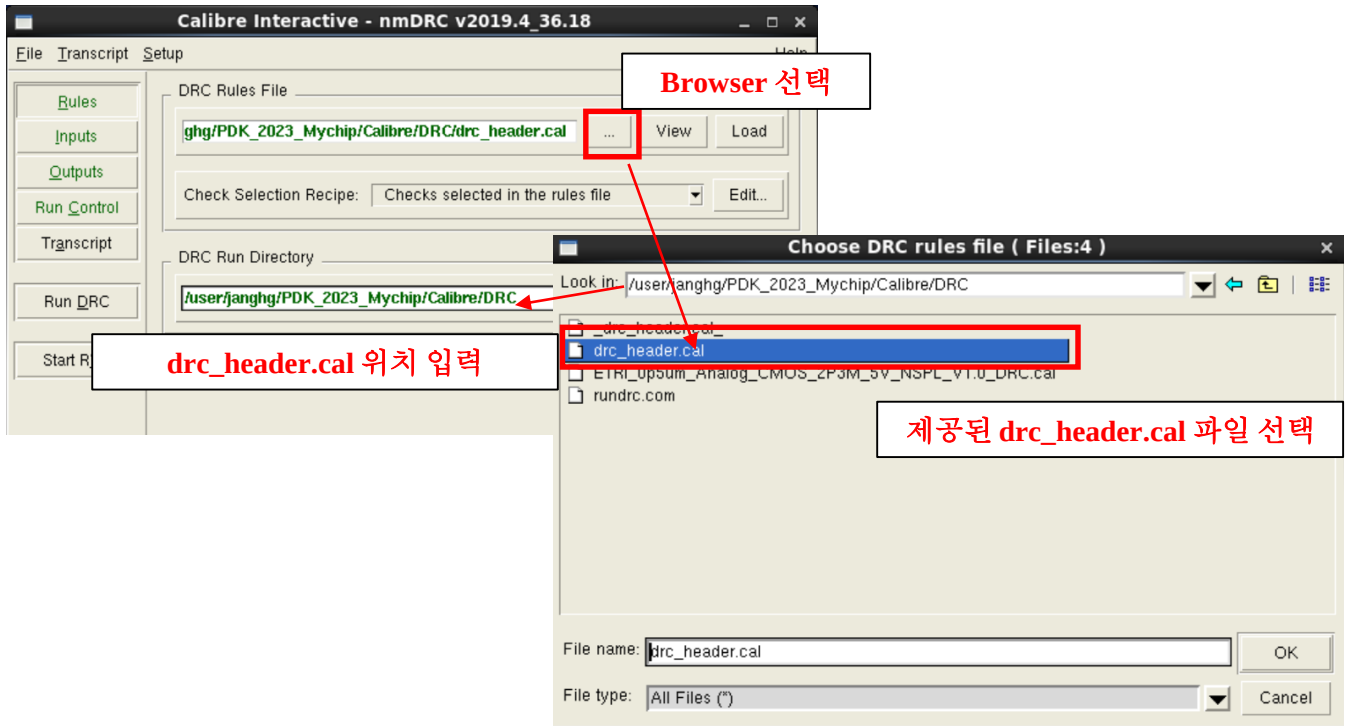
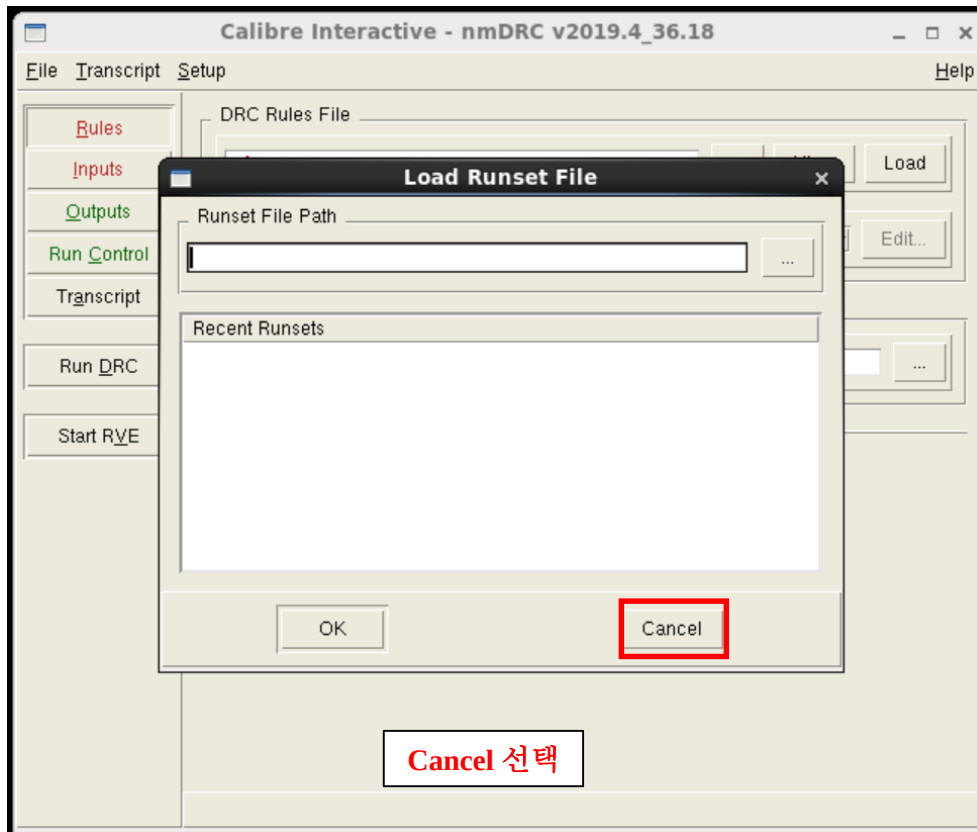
- 목적 : 설계된 레이아웃 디자인이 디자인룰에 적합한지 검토

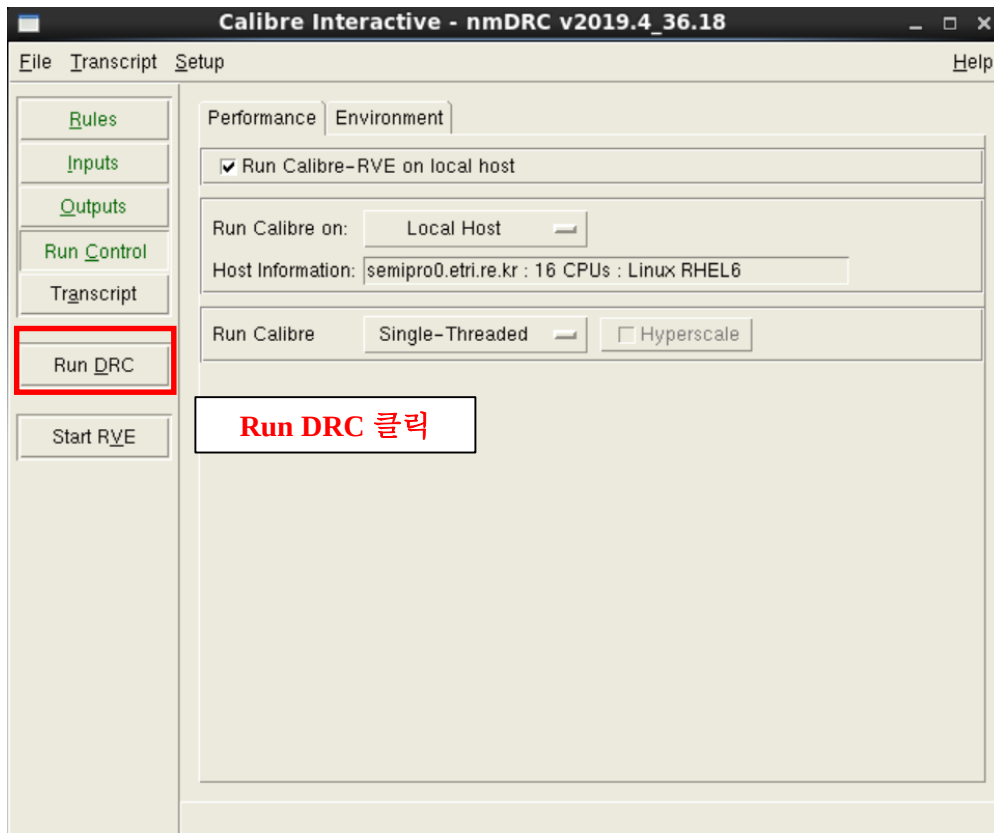
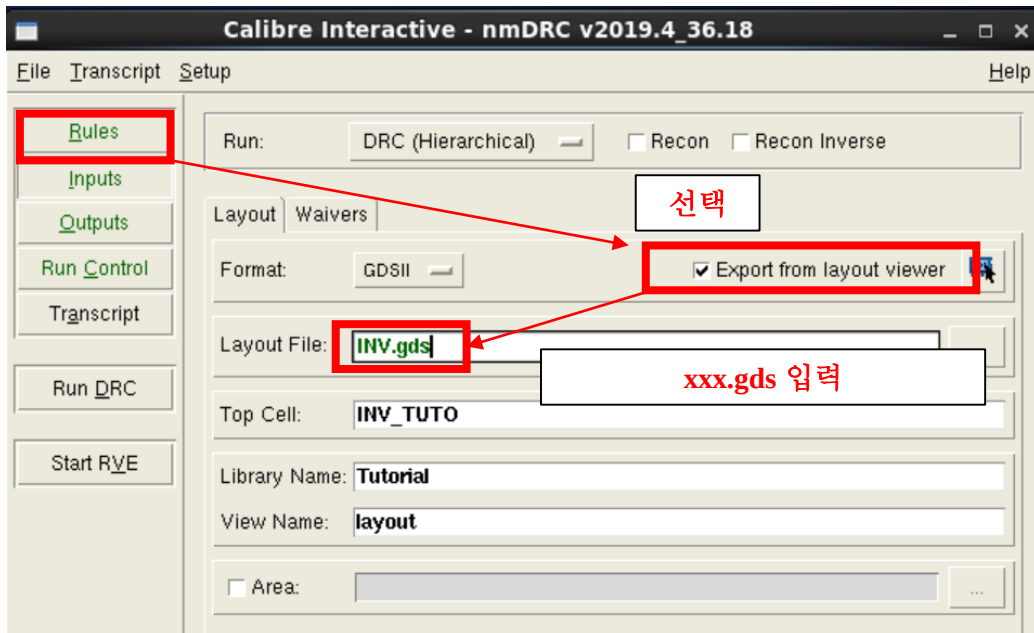
1) Calibre DRC 검토

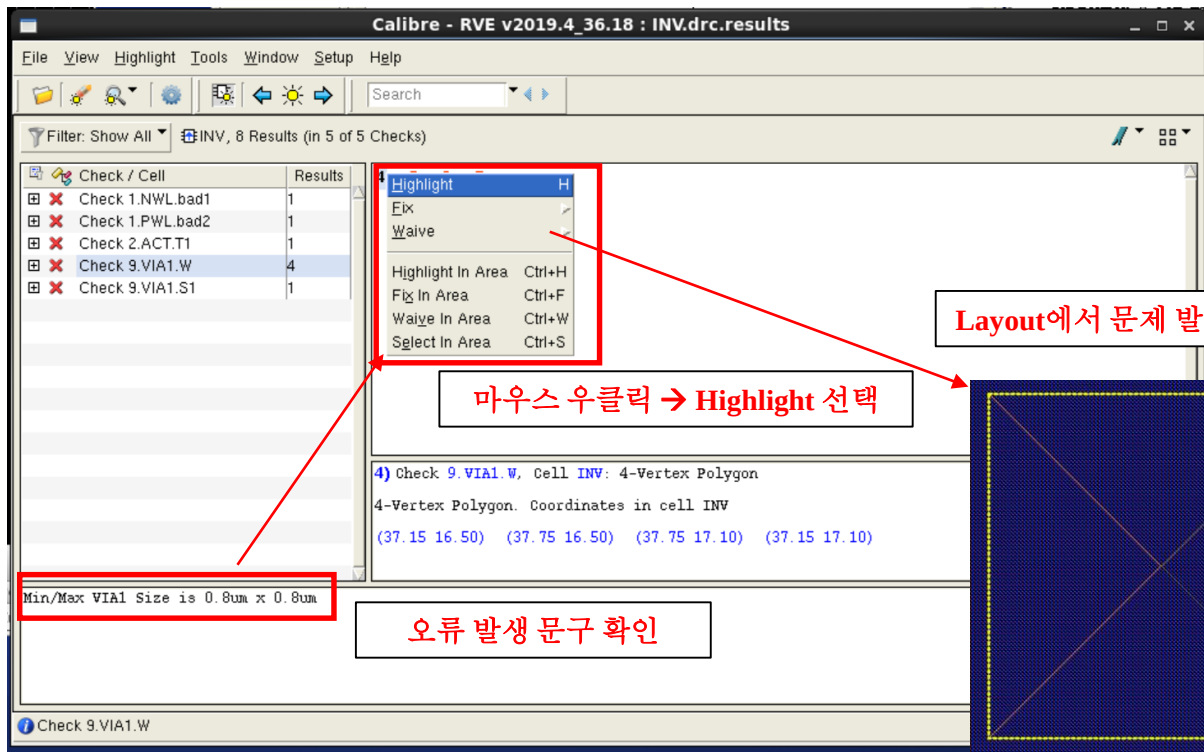
① DRC 검증 방법



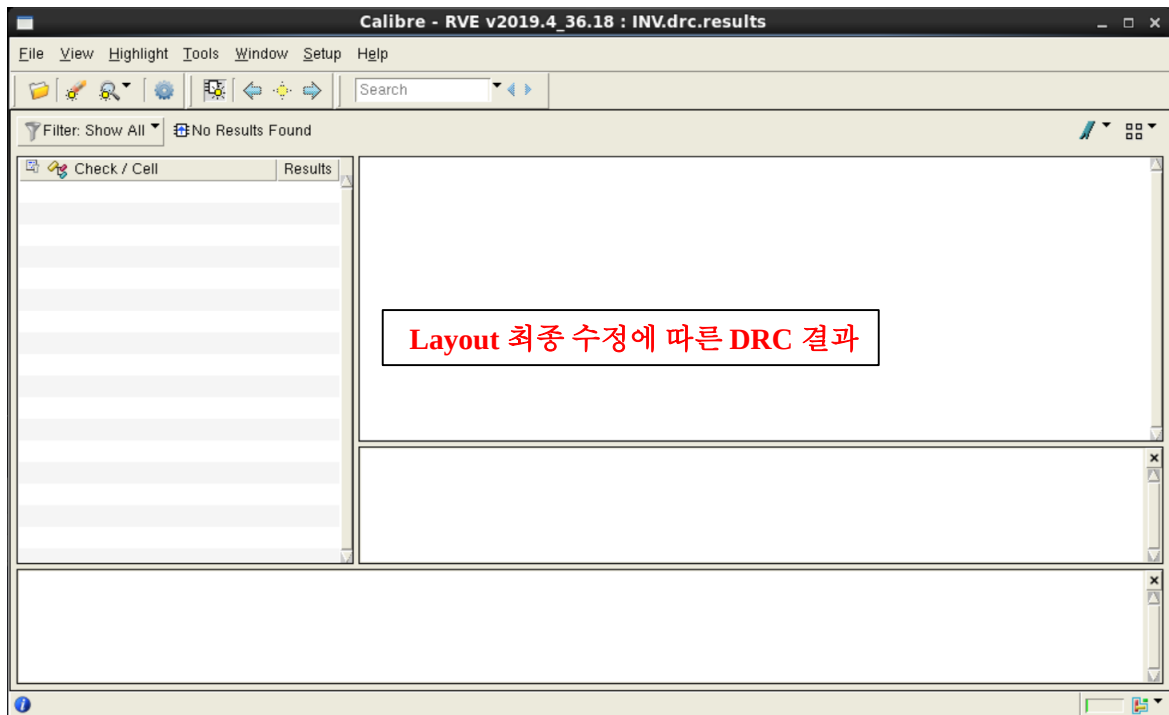








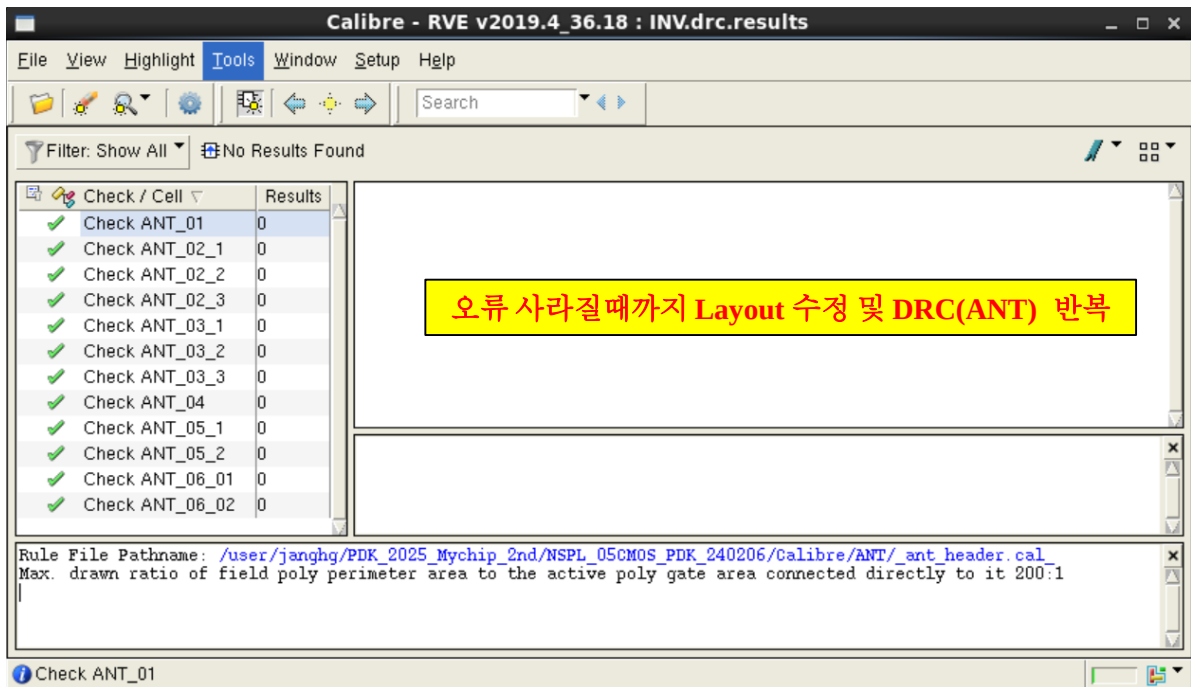
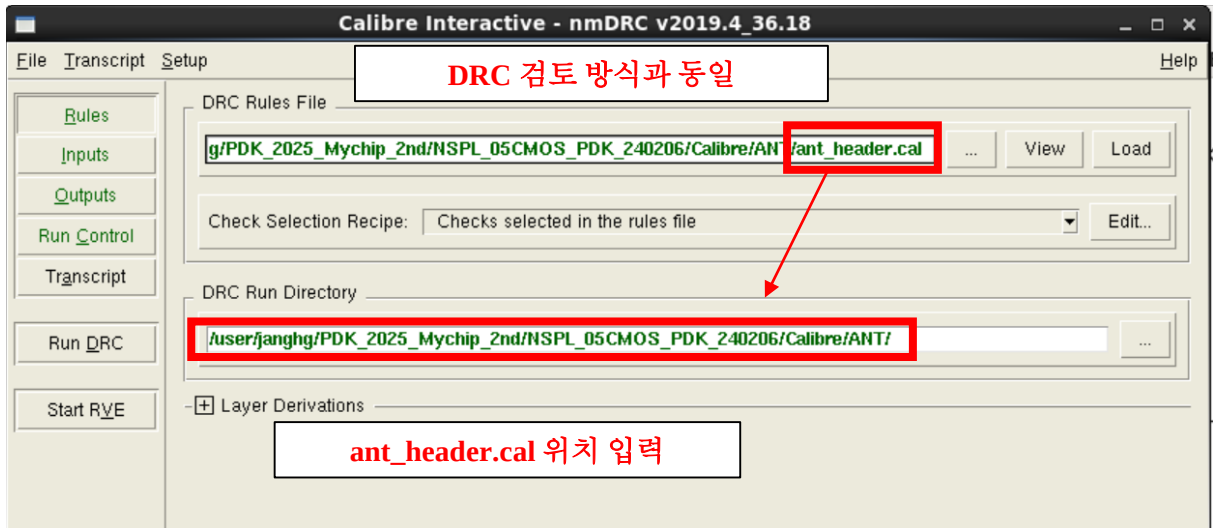
오류 사라질때까지 Layout 수정 및 DRC 반복



2) Calibre ANT 검토

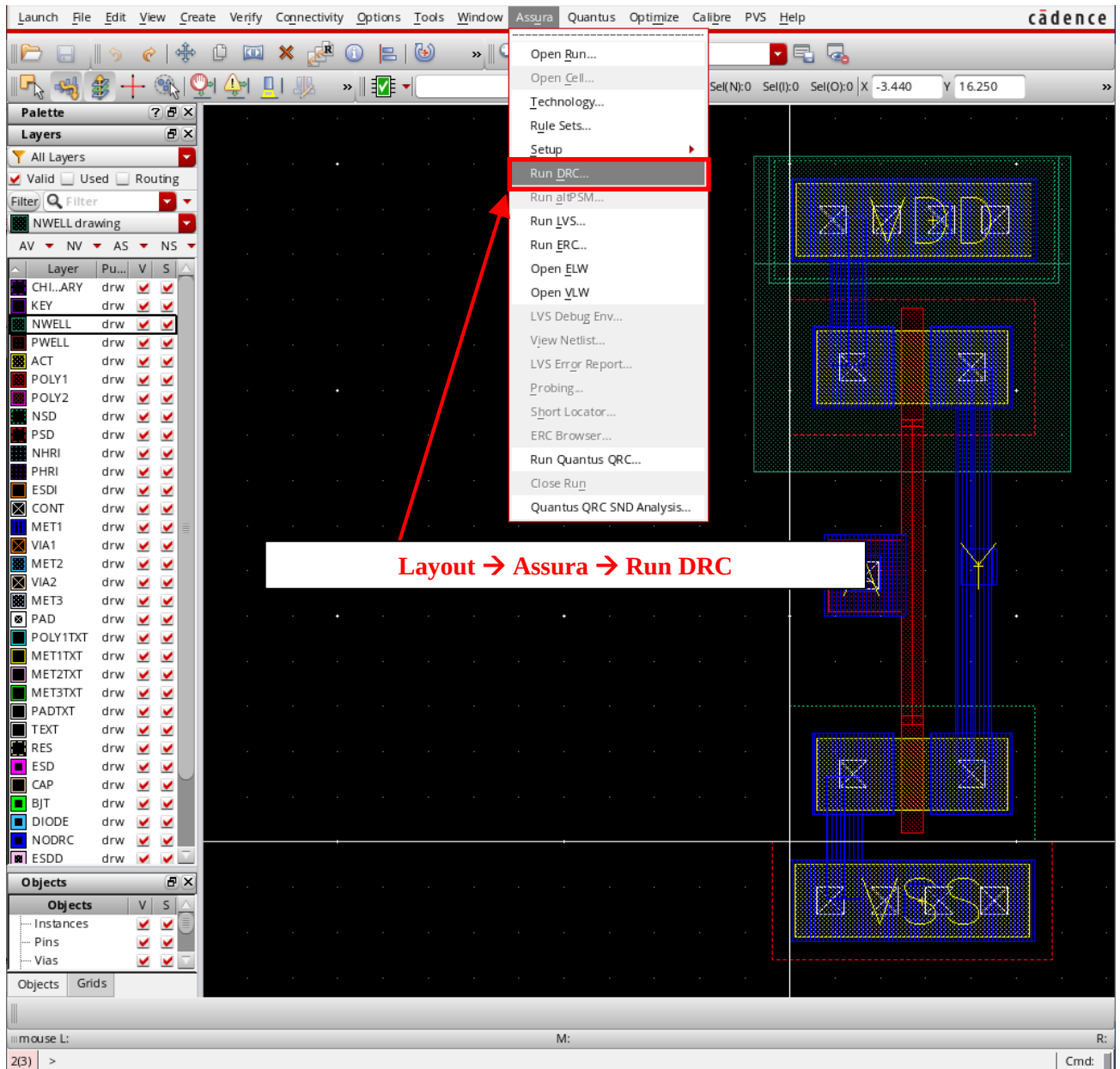
- 목적 : 설계된 레이아웃 디자인의 안테나룰에 적합 여부 검토

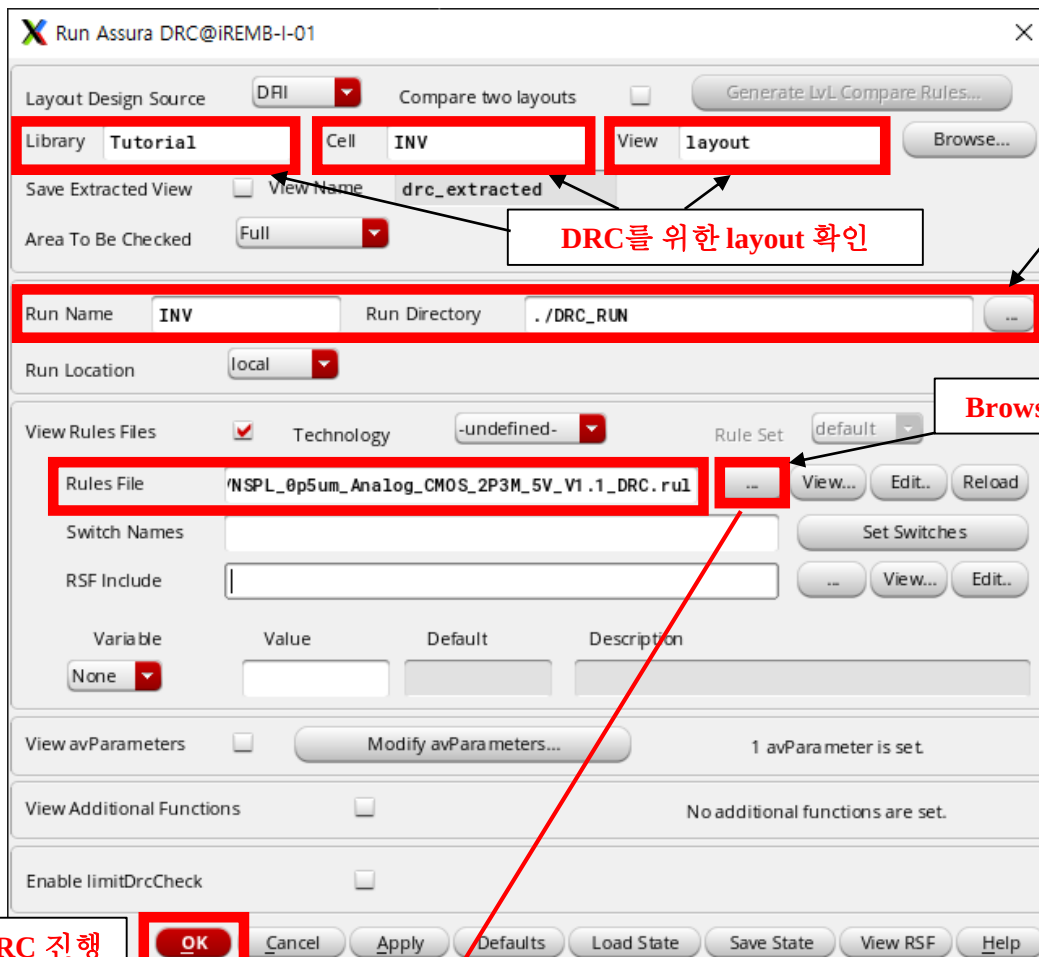
① ANT 검증 방법



2) Assura DRC 검토

① DRC 검증 방법





(Optional) Run 결과를 저장할 Directory 선택. 미선택시 Virtuoso 실행 디렉토리에 결과 저장됨.

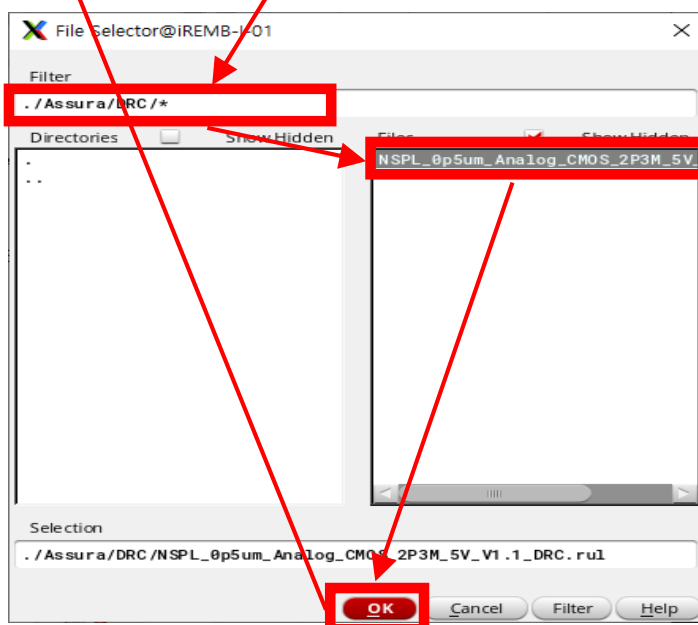
DRC를 위한 layout 확인

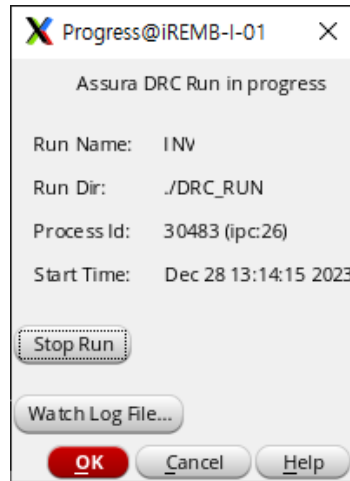
Browser 선택

Run DRC 진행

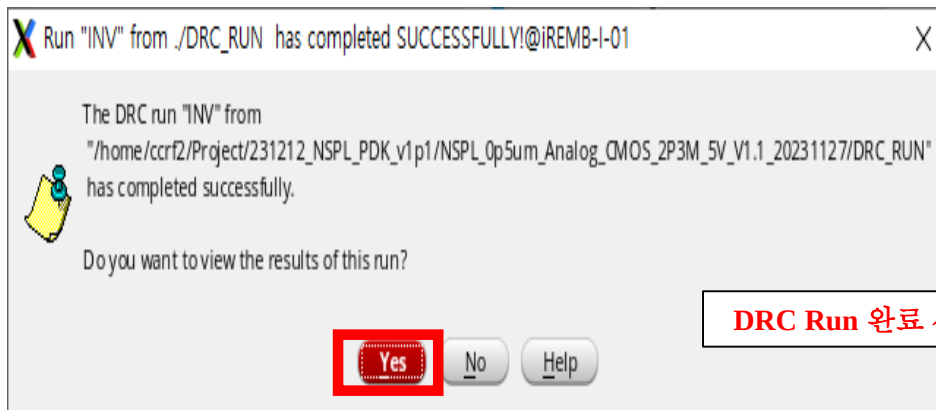
DRC Rule File 이 있는 경로로 이동.

DRC Rule File 선택.





DRC Run Progress 화면.
(Run 완료 후 자동으로 화면 사라짐)



DRC Run 완료 시, dialog box.

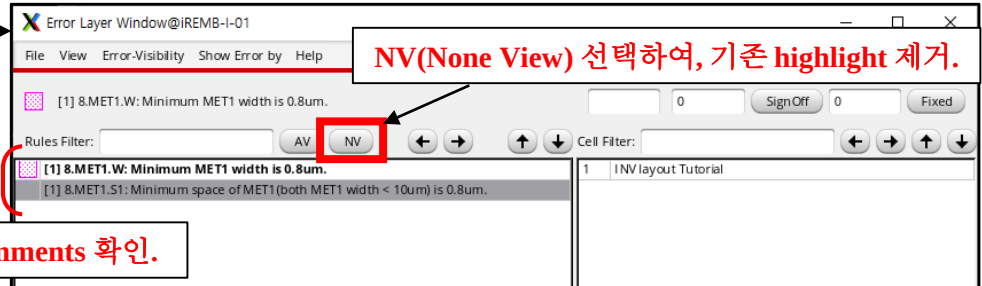


DRC pass

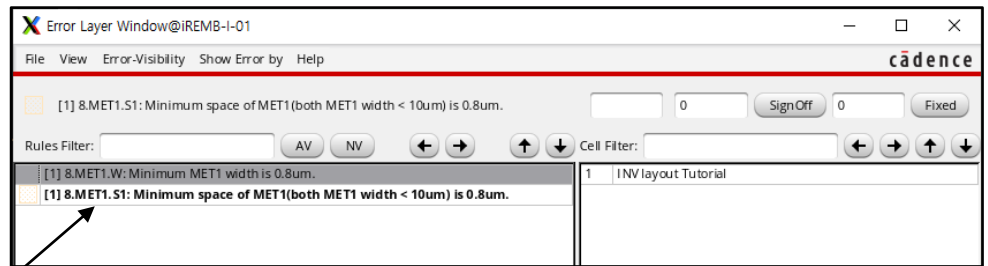
② DRC Debug

DRC error 발생. 예) Metal1 min. width / min. space error

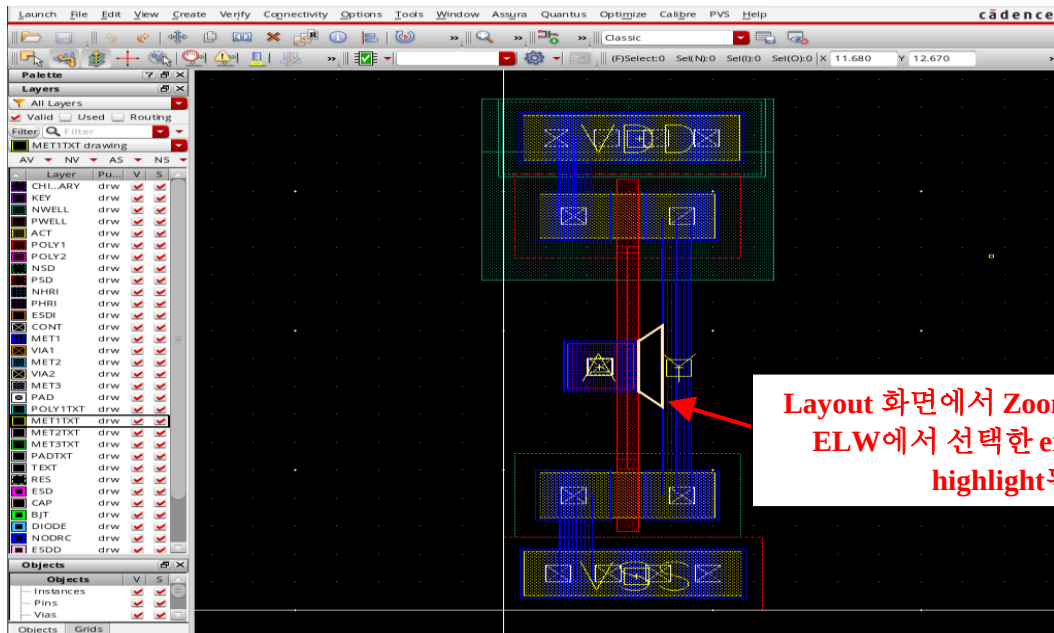
DRC error 발생 시,
ELW(Error Layer
Window)가 pop-up 됨.



Error Comments 확인.



수정할 DRC error item 선택.



Layout 화면에서 Zoom in/out 하면,
ELW에서 선택한 error 부분이
highlight됨.

오류 사라질때까지 Layout 수정 및 DRC 반복

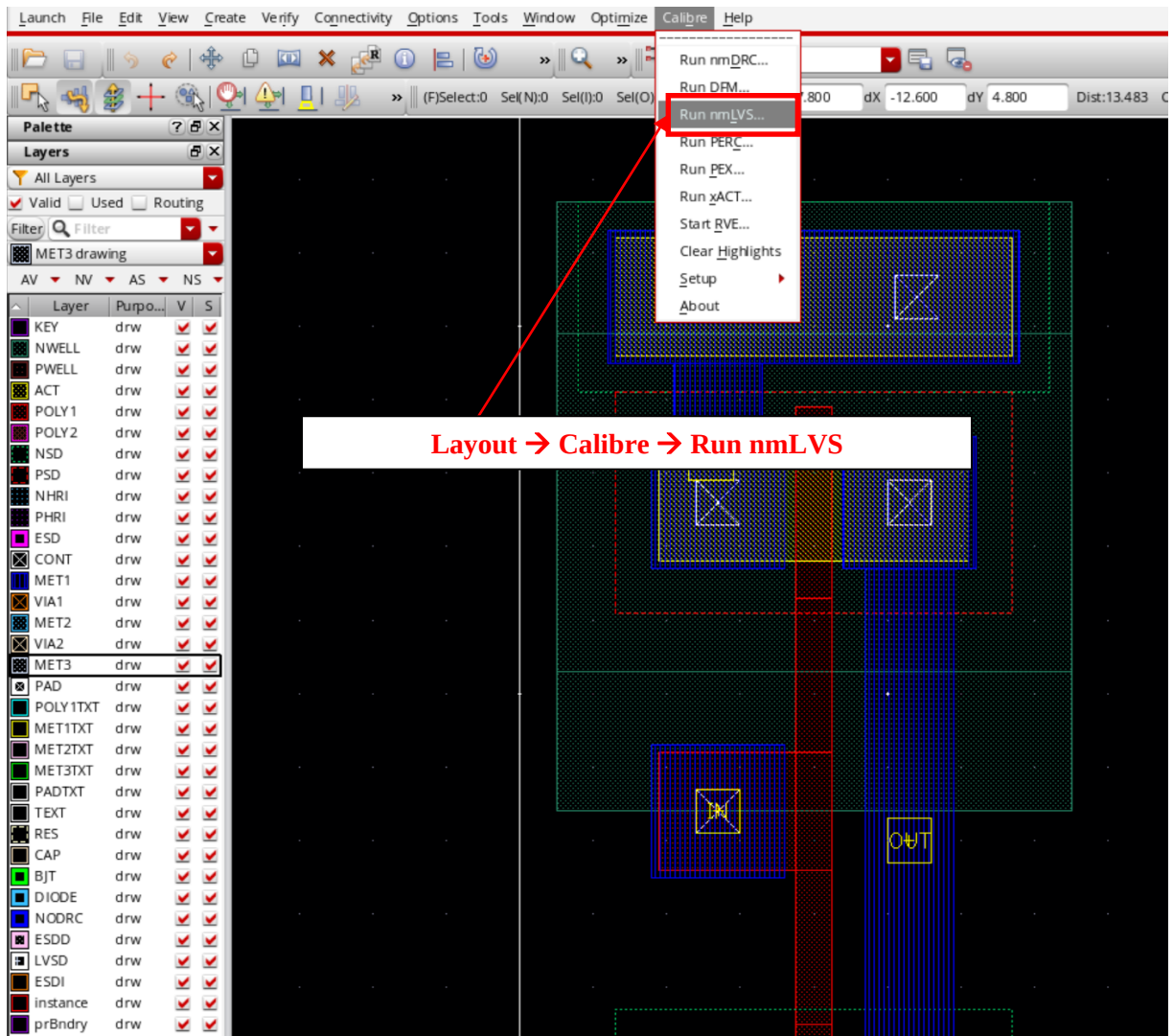
6. LVS (Layout Versus Schematic)

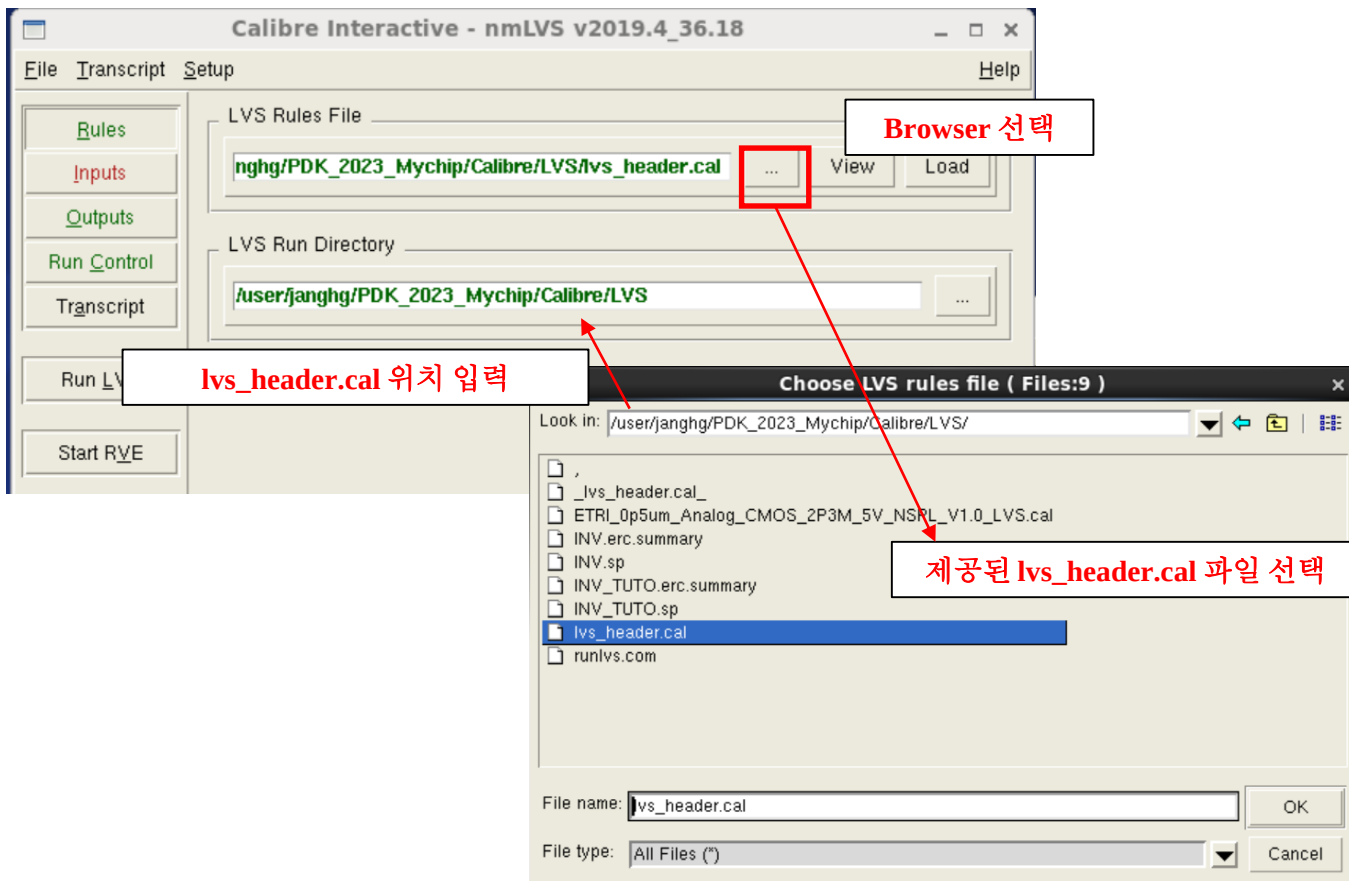
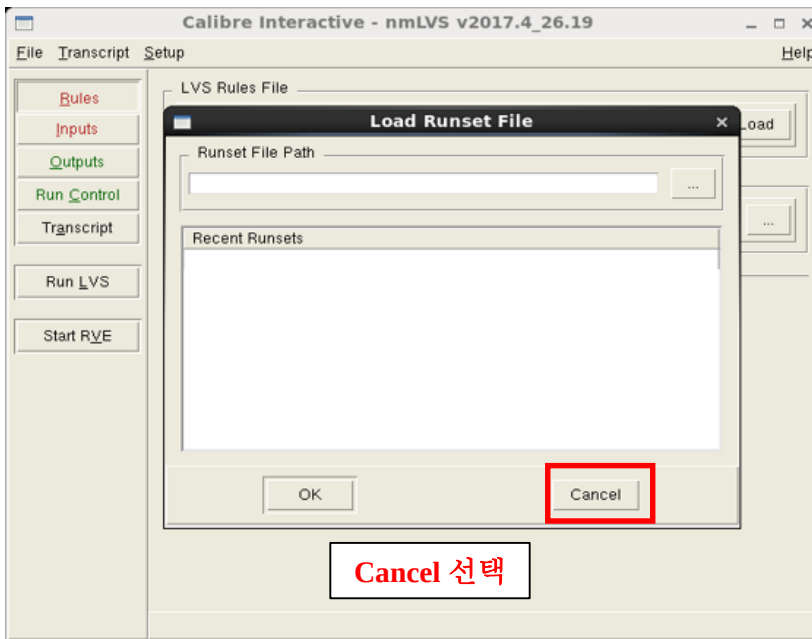
6. LVS

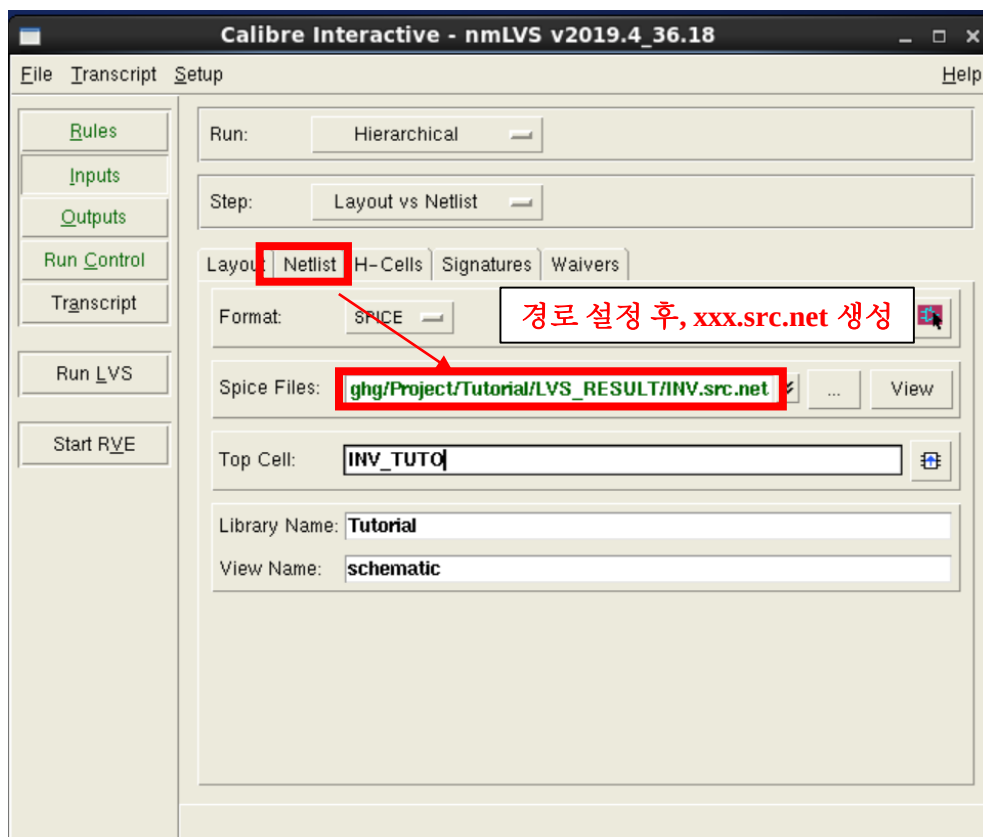
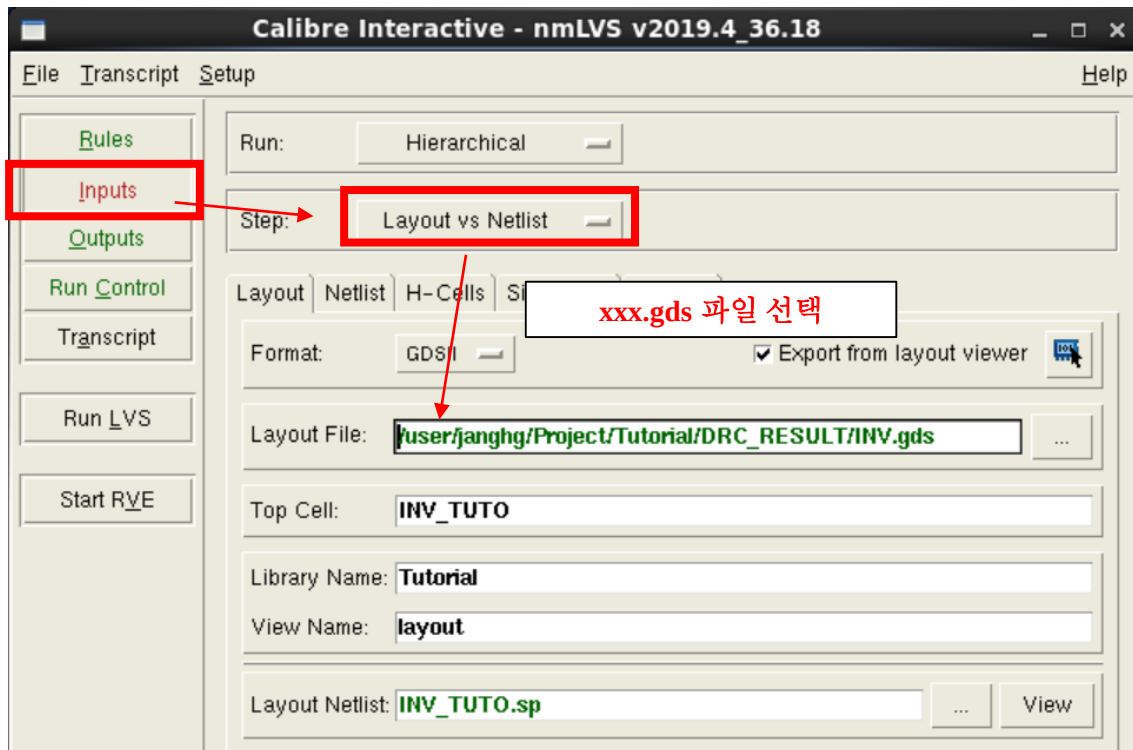
- 목적 : 설계한 Layout과 Schematic 비교

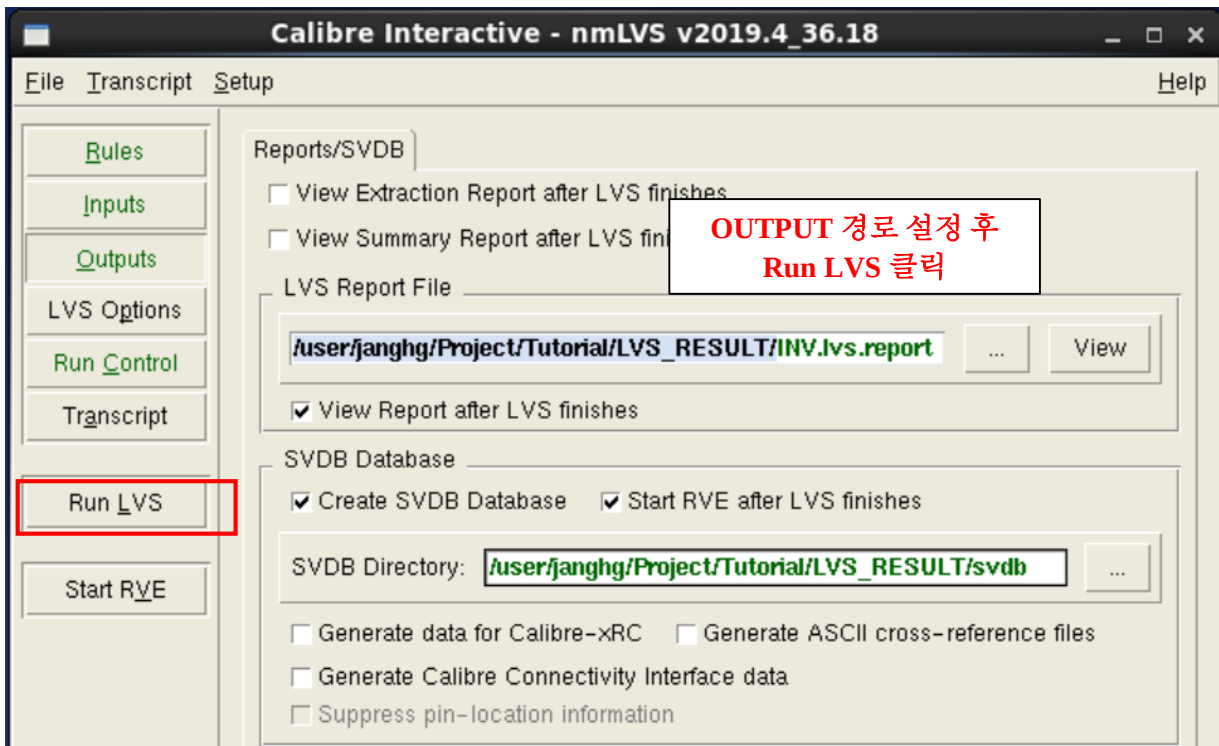
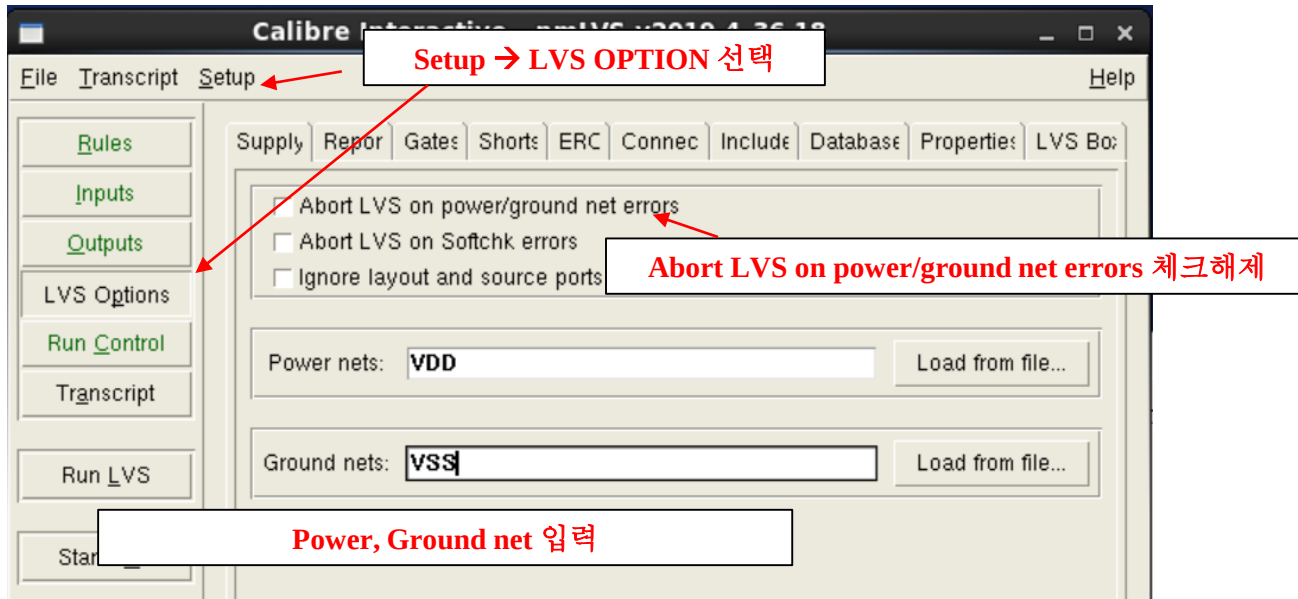
1) Calibre LVS 검토

① LVS 검증 방법



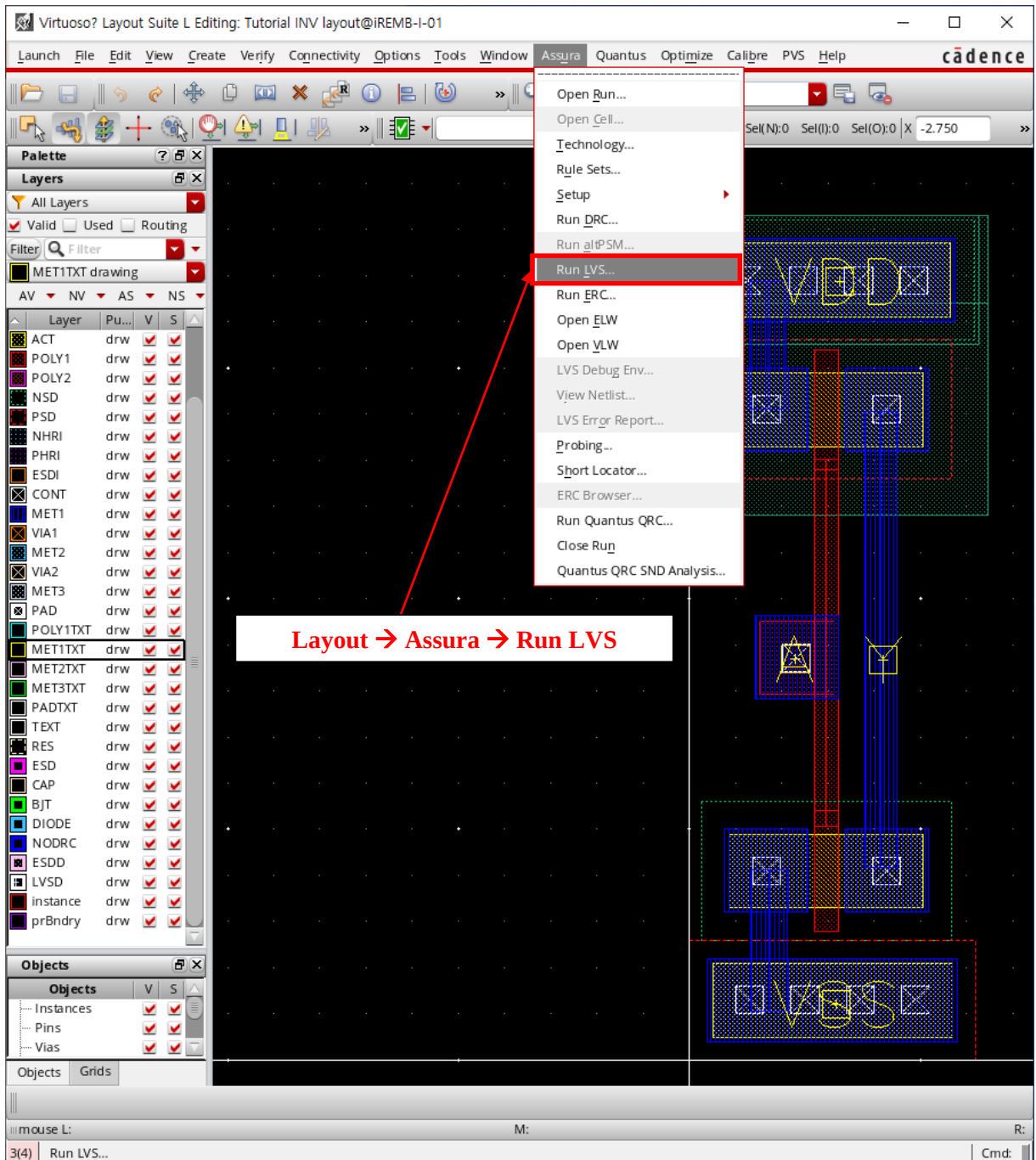


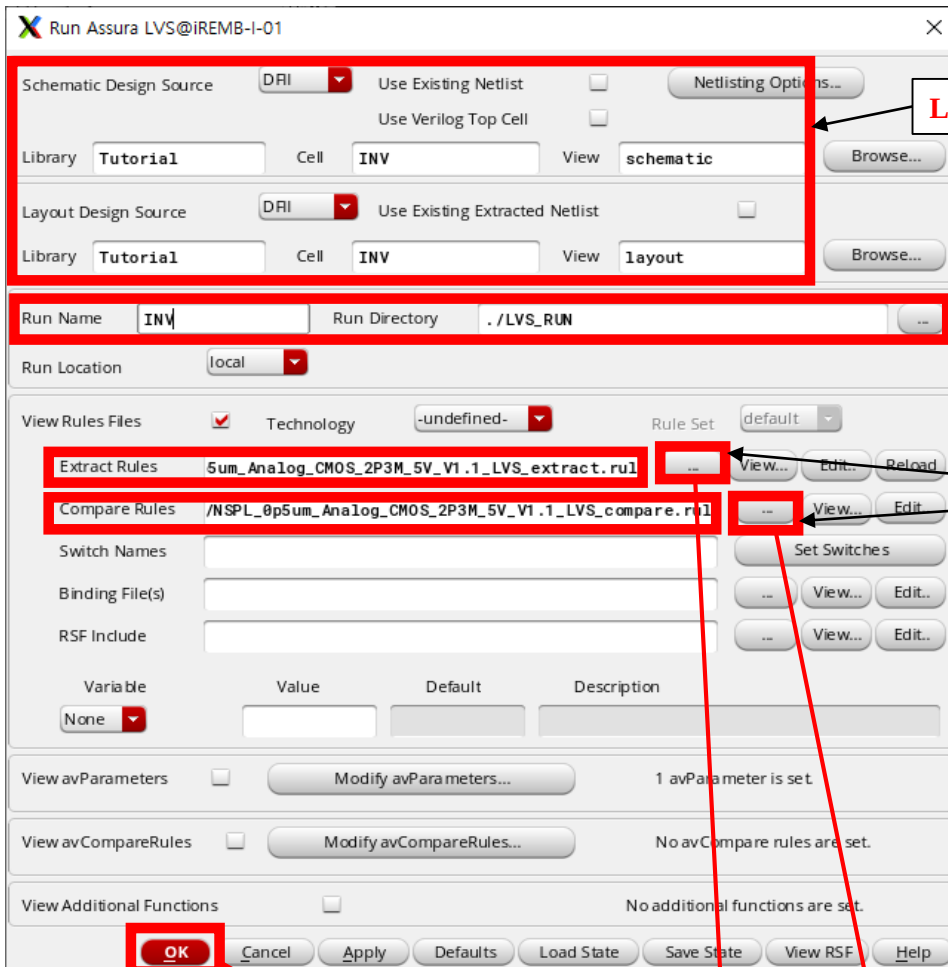




2) Assura LVS 검토

① LVS 검증 방법



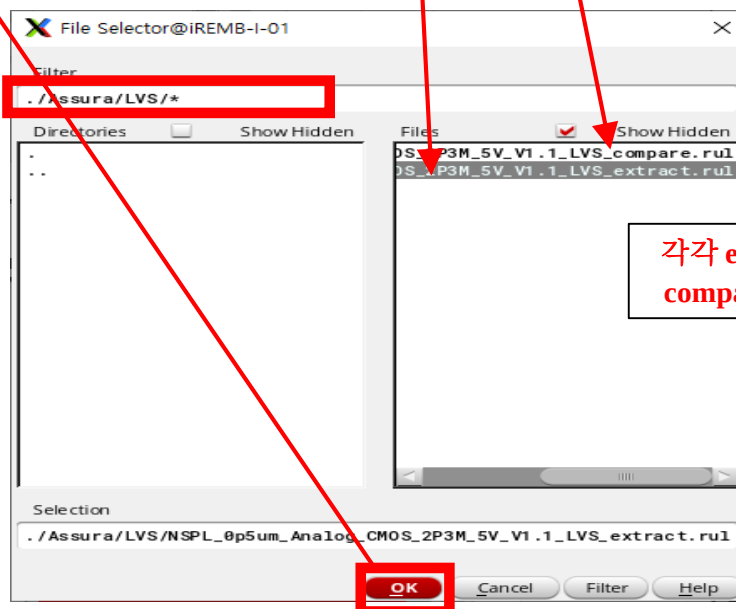


LVS를 위한 layout/schematic 확인.

(Optional) Run 결과를 저장할 Directory 선택.
미선택시 Virtuoso 실행 디렉토리에 결과 저장됨.

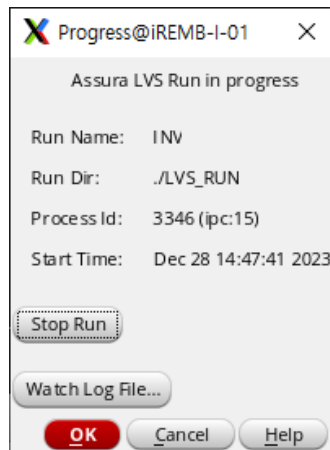
Browser 선택

Run LVS 진행

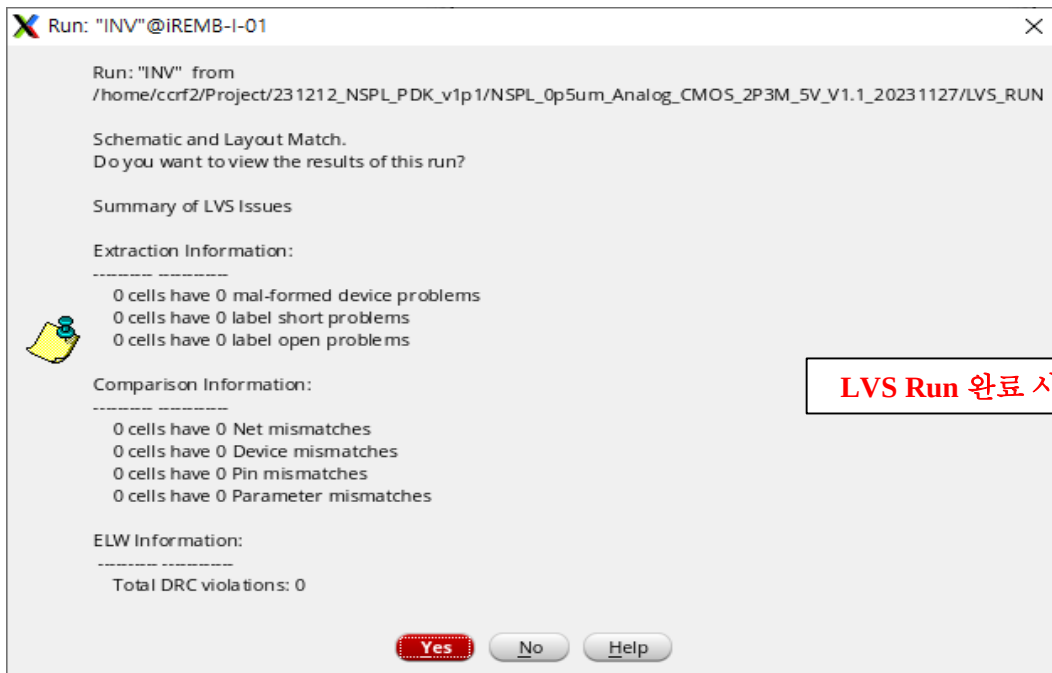


LVS Rule File이 있는 경로로 이동.

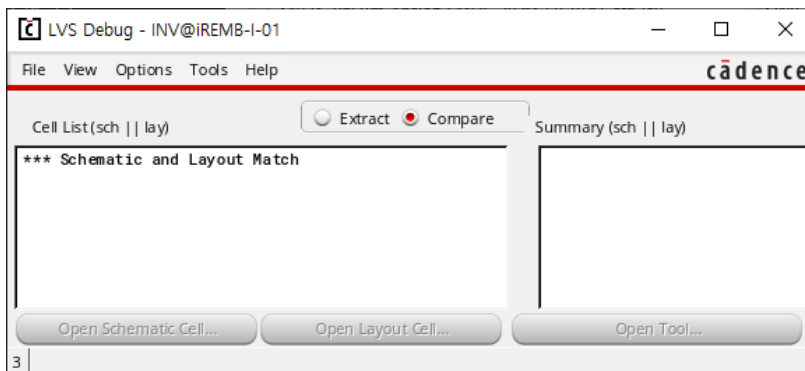
각각 extract rule file 및 compare rule file 선택.



LVS Run Progress 화면.
(Run 완료 후 자동으로 화면 사라짐)



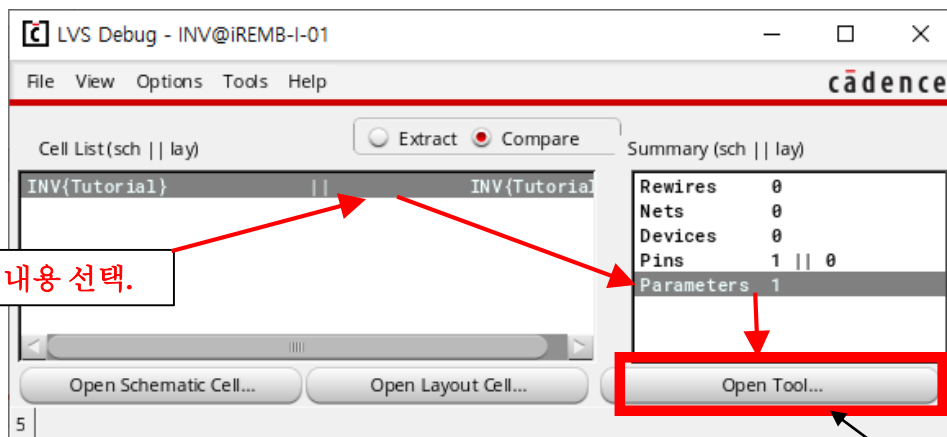
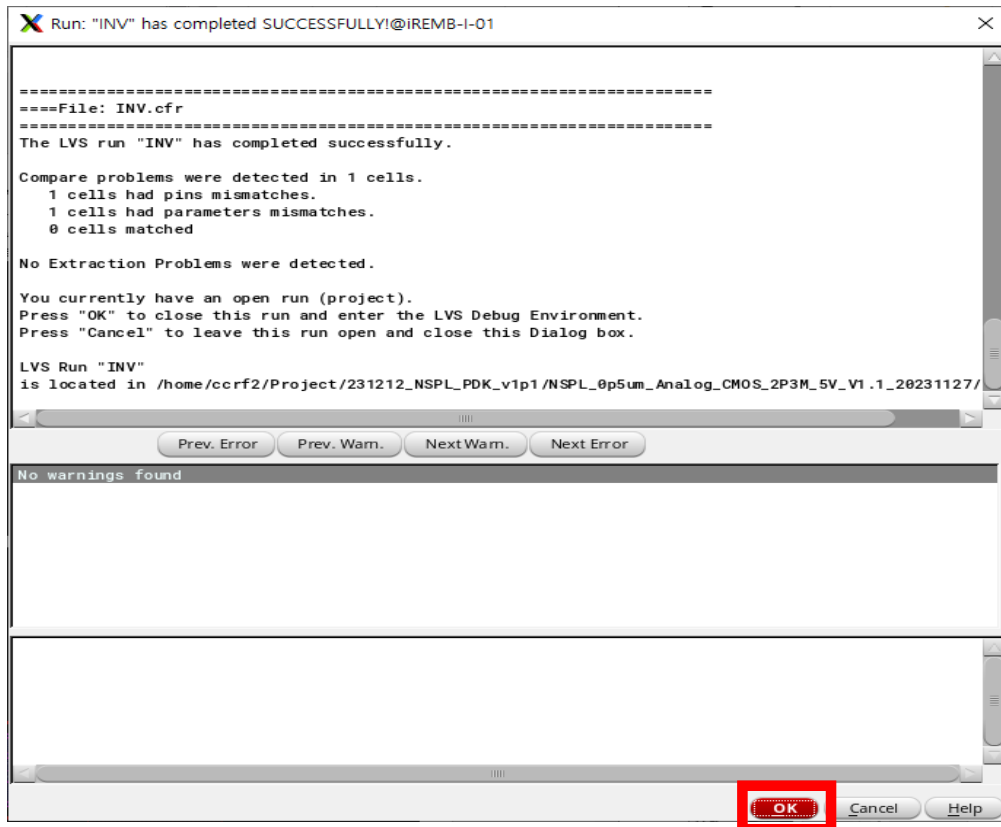
LVS Run 완료 시, dialog box.



LVS pass

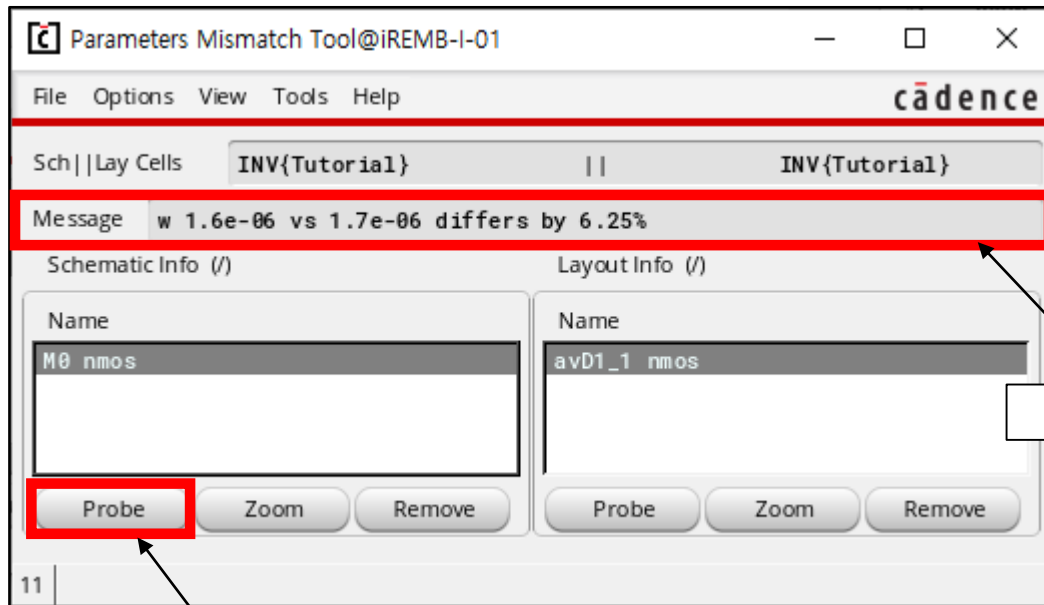
② LVS Debug

LVS mismatch 발생. 예) pin 및 parameter mismatch 발생.



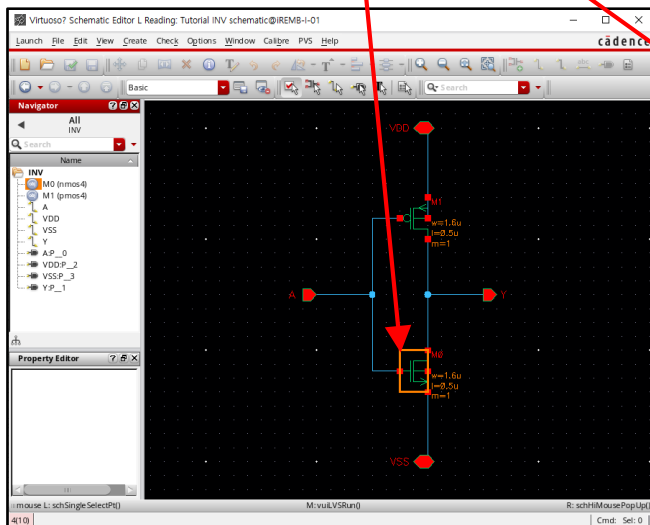
Mismatch 내용 선택.

Open Tool 클릭.

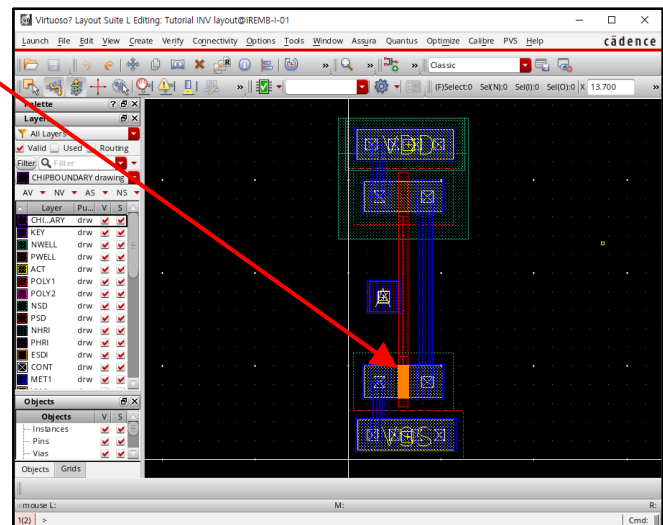


Mismatch 내용 확인.

Probe 클릭하여, schematic 및 layout에서 해당 mismatch 부분을 highlight.



< Schematic >



< Layout >

오류 사라질때까지 Layout 수정 및 LVS 반복

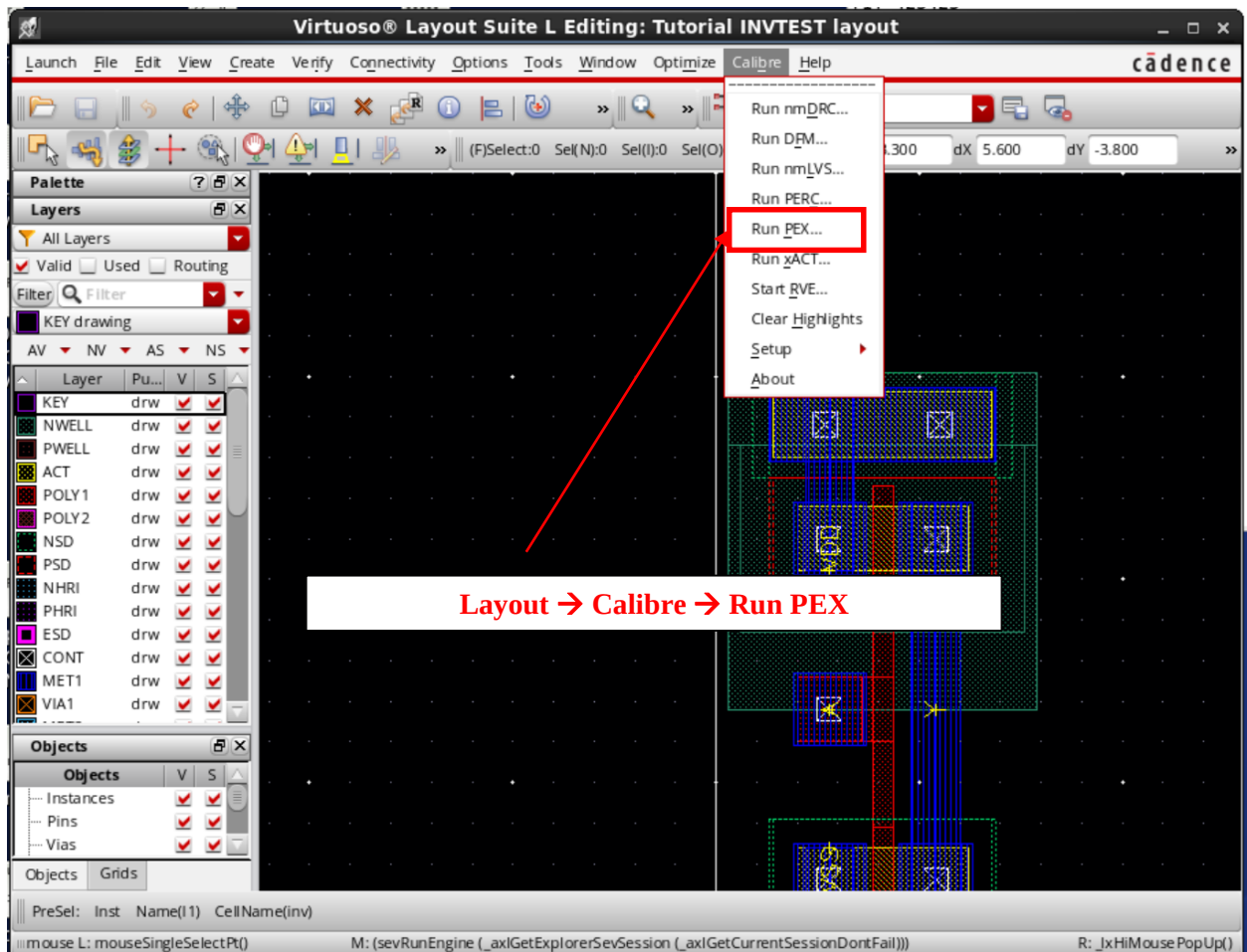
7. PEX (Parasitic EXtraction)

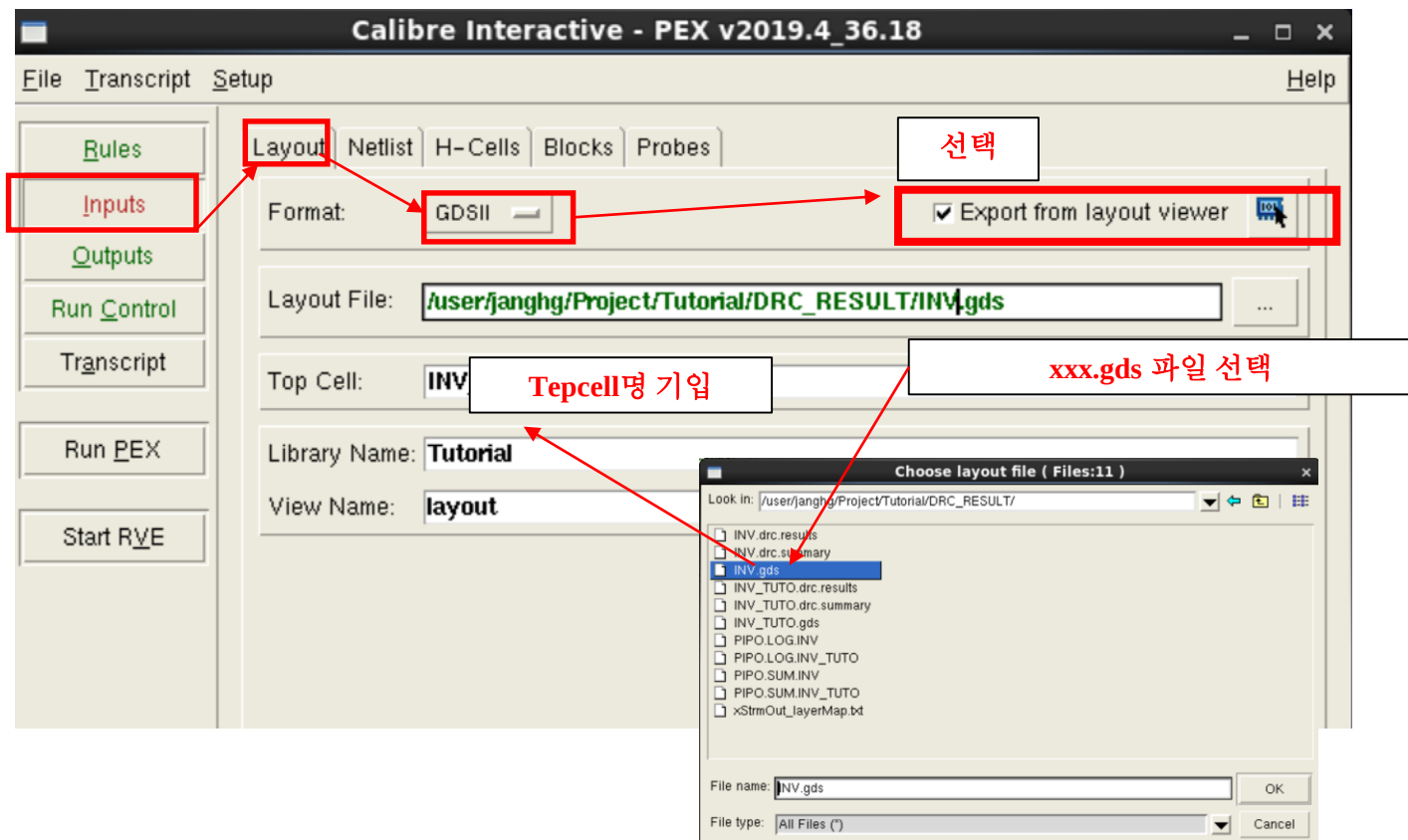
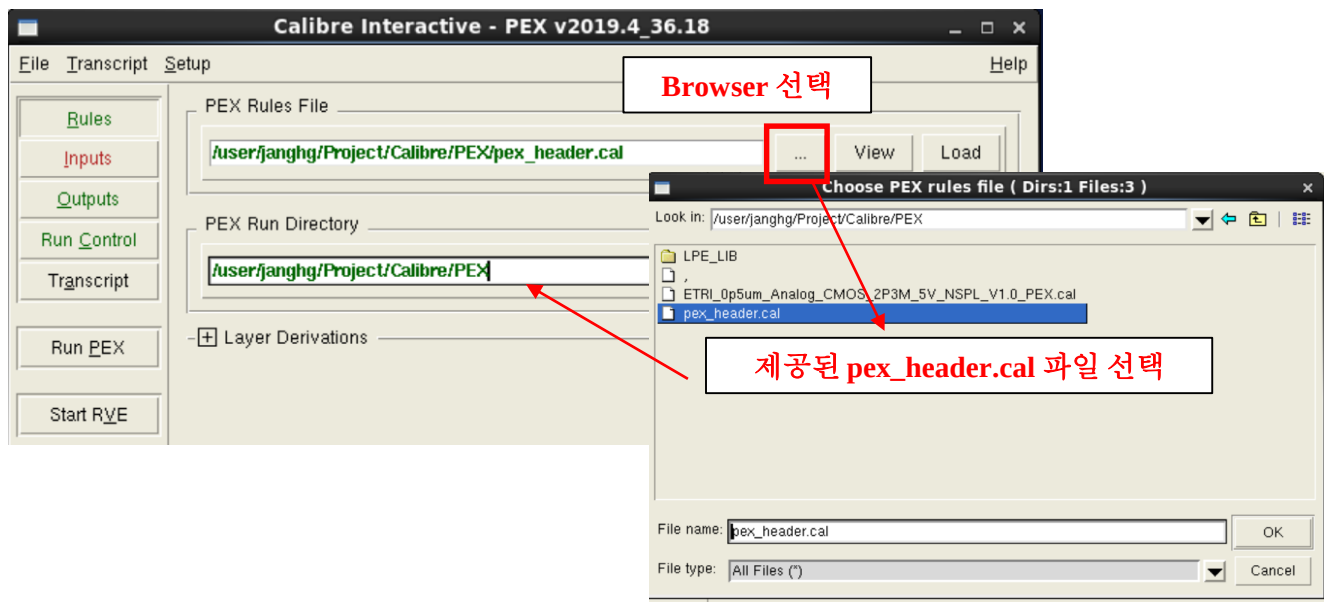
7. PEX

- 목적 : Parasitic 성분 추출

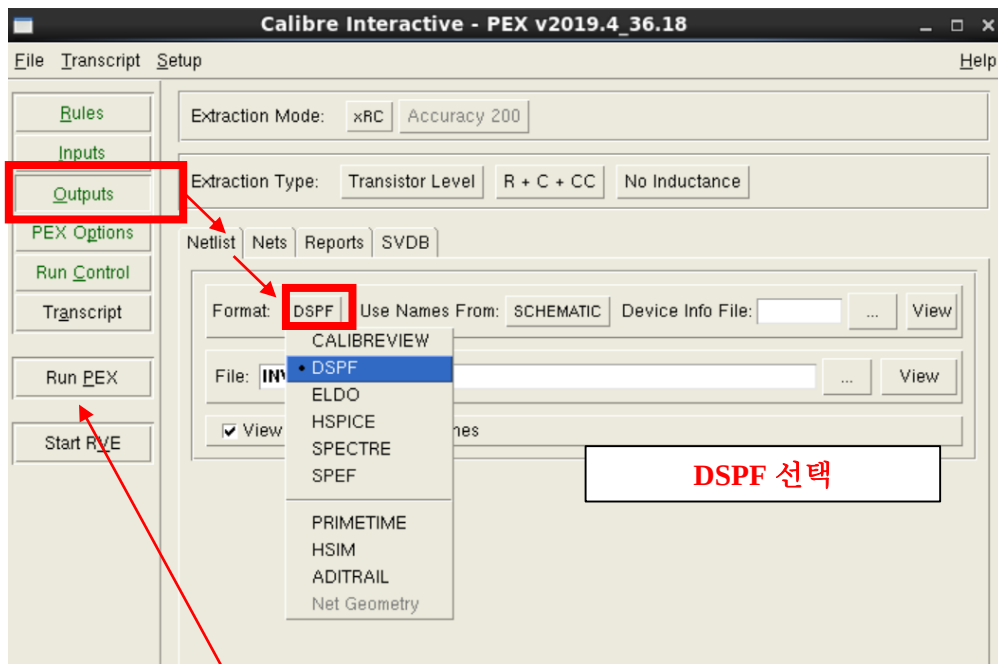
1) Calibre PEX 검토

① PEX 진행 방법

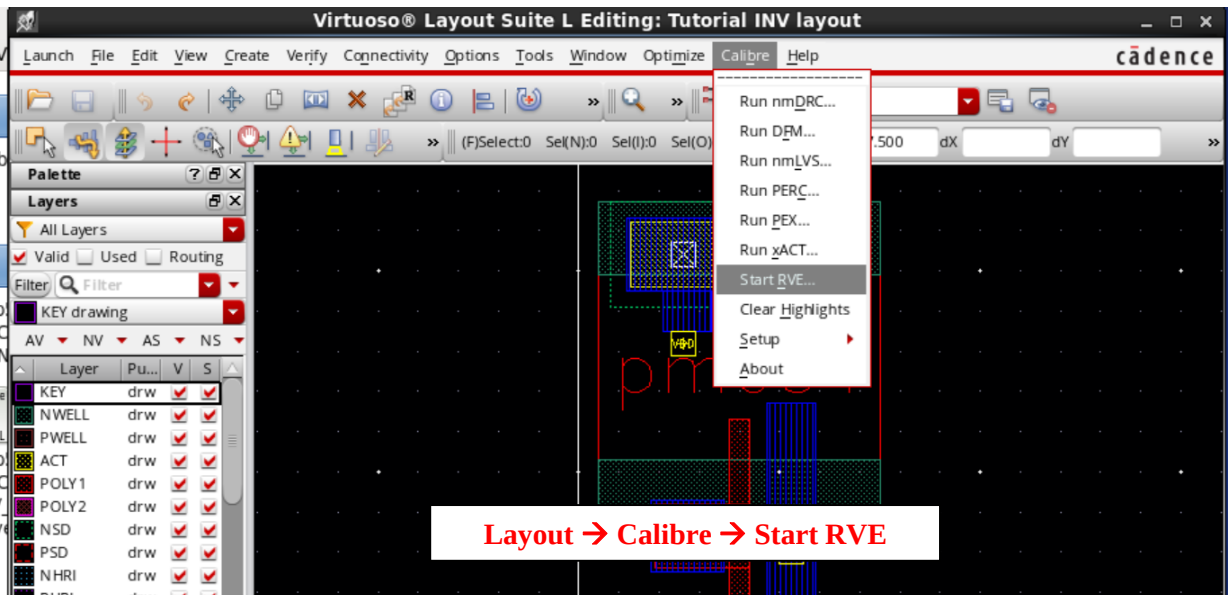
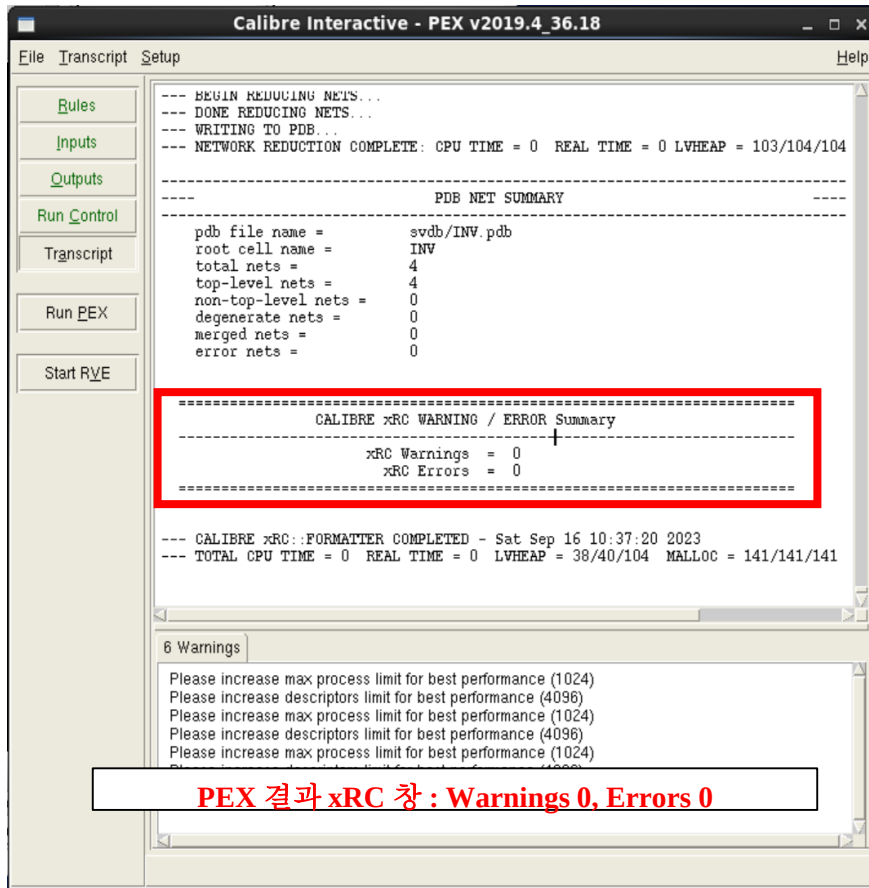


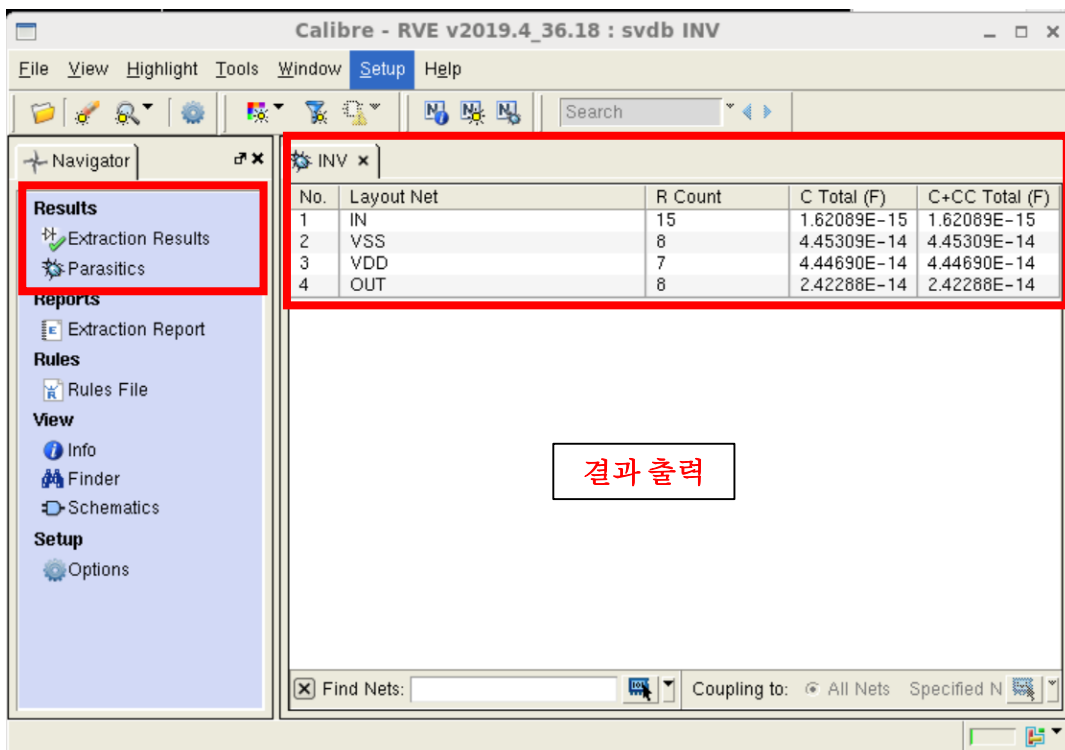
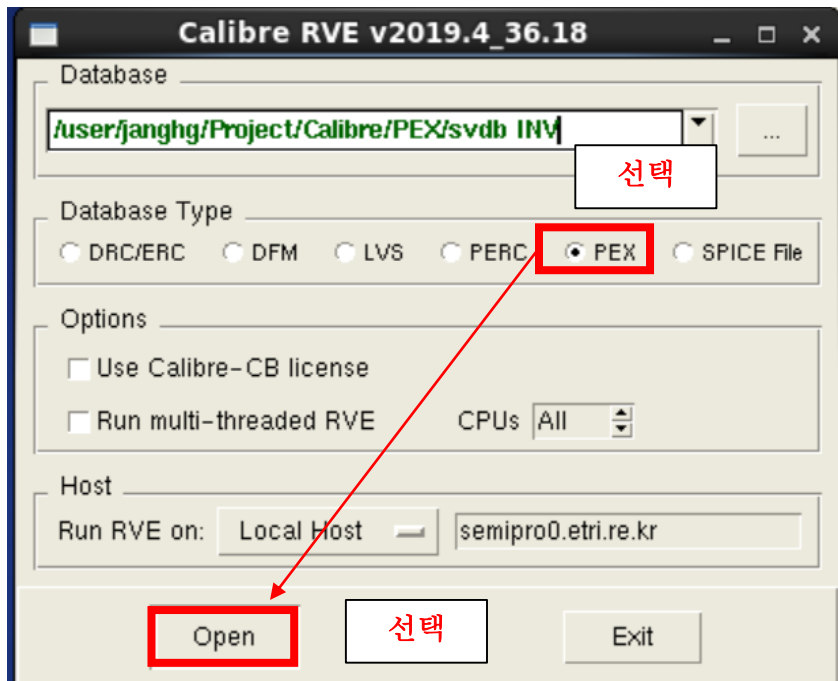


② PEX-DSPF 방식

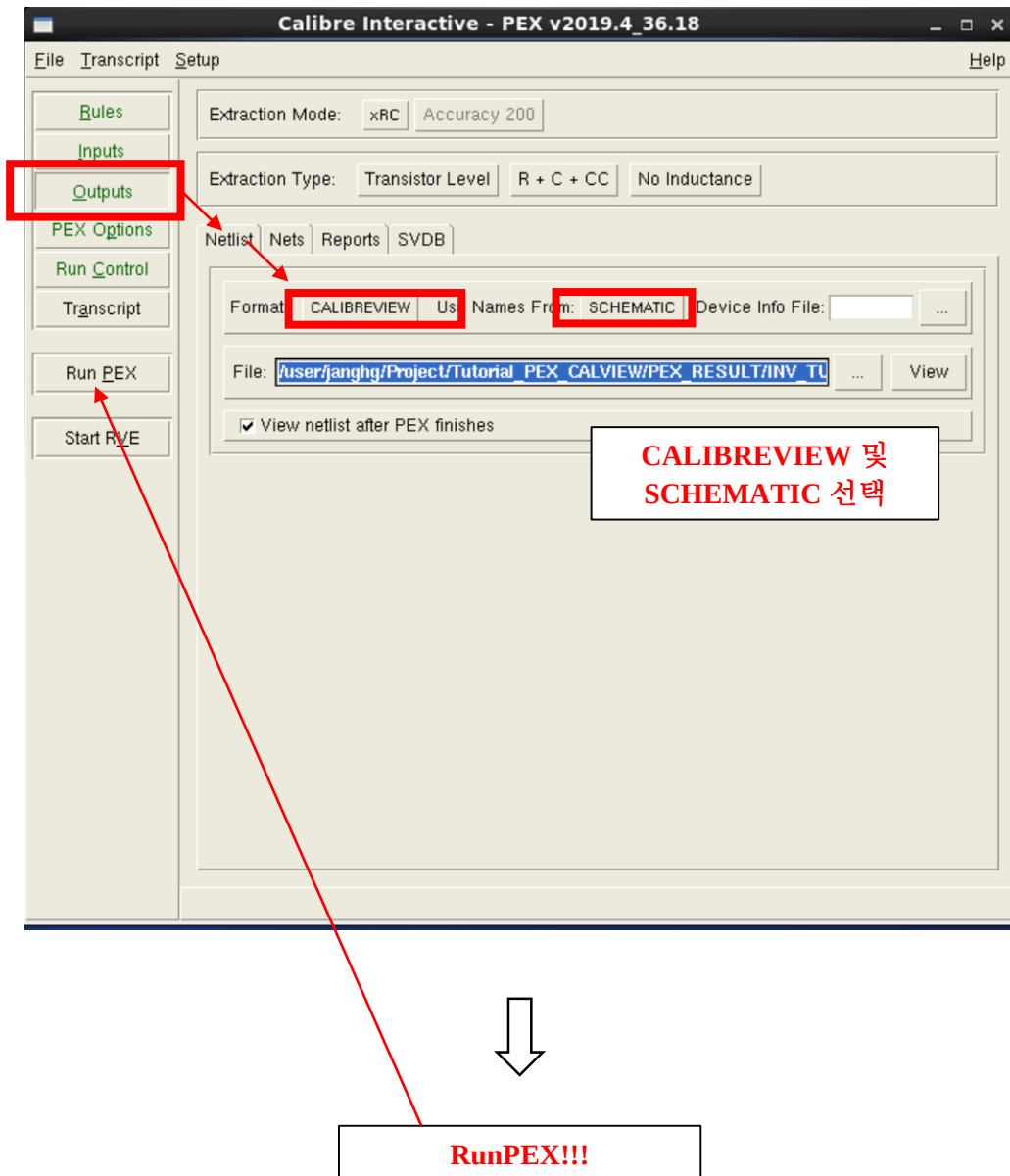


RunPEX!!!





③ PEX-CALIBREVIEW 방식



Calibre View Setup

CalibreView Setup File: Browse View Load

CalibreView Netlist File: Browse

Output Library:

Schematic Library:

Cellmap File: View Edit Browse

Log File:

Calibre View Name:

Calibre View Type: ☐ maskLayout ☒ schematic

Create Terminals: ☐ if matching terminal exists on symbol ☒ Create all terminals

Preserve Device Case ☒

Execute Callbacks ☒

Suppress Notes ☐

Reset Properties: 체크

Magnify Instances By:

Device Placement: ☐ Layout Location ☒ Arrayed

Parasitic Placement: ☐ Layout Location ☒ Arrayed

Show Parasitic Polygons ☐

Open Calibre CellView: ☐ Read-mode ☐ Edit-mode ☒ Don't Open 체크

Generate SPECTRE Netlist ☒

Always Show Dialog ☒

선택

OK Cancel Save Help

제공된 PDK에서 calview.cellmap 선택



```

/user/janghg/simulation/INV_TUTO/spectre/calibre/netli
File Edit View Help cadence

// Generated for: spectre
// Generated on: Oct 19 09:50:25 2023
// Design library name: Tutorial_CALPEX_RE
// Design cell name: INV_TUTO
// Design view name: calibre
simulator lang=spectre
global 0

// Library name: Tutorial_CALPEX_RE
// Cell name: INV_TUTO

MM1 (MM1_d MM1_g MM1_s MM1_b) nmos4 m=1 w=1.6e-06 l=5e-07
MM0 (MM0_d MM0_g MM0_s MM0_b) pmos4 m=1 w=1.6e-06 l=5e-07

cc_26 (c_36_p MM1_b) capacitor c=8.74478e-19
cc_27 (c_26_n c_46_n) capacitor c=8.74478e-19
cc_26 (c_36_p c_46_n) capacitor c=1.15378e-16
cc_25 (c_40_p VSS) capacitor c=3.75809e-15
cc_24 (c_26_n VSS) capacitor c=7.17844e-19
cc_23 (c_36_p VSS) capacitor c=8.81661e-17
cc_22 (c_37_p VSS) capacitor c=2.33005e-16
cc_21 (c_36_p MM1_b) capacitor c=7.25193e-19
cc_20 (c_15_n c_46_n) capacitor c=8.72308e-19
cc_19 (MM0_s c_27_n) capacitor c=1.65126e-17
cc_18 (c_15_n c_27_n) capacitor c=8.74478e-19
cc_17 (VDD c_27_n) capacitor c=7.17844e-19
cc_16 (MM0_s c_29_n) capacitor c=8.74478e-19
cc_15 (c_15_n c_29_n) capacitor c=1.15378e-16
cc_14 (VDD c_29_n) capacitor c=8.79236e-17
cc_13 (MM0_b c_29_n) capacitor c=7.25193e-19
cc_12 (MM1_g MM1_s) capacitor c=3.51981e-17
cc_11 (A MM1_s) capacitor c=2.65446e-19
cc_10 (A c_46_n) capacitor c=4.1187e-17
cc_9 (A c_45_n) capacitor c=2.96264e-19
cc_8 (A c_25_n) capacitor c=7.99899e-17
cc_7 (MM0_g c_27_n) capacitor c=3.51981e-17
cc_6 (MM1_g c_26_n) capacitor c=3.51981e-17
cc_5 (MM0_g MM0_s) capacitor c=3.51981e-17
cc_4 (A MM0_s) capacitor c=2.65446e-19
cc_3 (A c_15_n) capacitor c=4.12808e-17
cc_2 (A VDD) capacitor c=2.95497e-19
cc_1 (A c_13_n) capacitor c=2.96264e-19
ciVSS_19 (MM1_b VSS) capacitor c=3.20851e-14
ciVSS_18 (c_45_n VSS) capacitor c=2.347e-16
ciVSS_17 (VSS_3 VSS) capacitor c=1.38282e-16
ciVSS_16 (VSS VSS) capacitor c=1.21307e-14
ciVSS_15 (c_46_n VSS) capacitor c=1.87226e-16
ciVSS_14 (MM1_s VSS) capacitor c=1.12232e-14
ciY_28 (Y_2 VSS) capacitor c=1.11674e-16
ciY_27 (Y_4 VSS) capacitor c=5.23929e-16
ciY_26 (c_37_p VSS) capacitor c=1.15015e-15
ciY_25 (Y VSS) capacitor c=1.13959e-15
ciY_24 (c_36_p VSS) capacitor c=1.61822e-16
ciY_23 (c_26_n VSS) capacitor c=1.12218e-14
ciY_22 (c_29_n VSS) capacitor c=1.70118e-16
ciY_21 (c_27_n VSS) capacitor c=1.11429e-14
ciY_20 (c_28_n VSS) capacitor c=1.72678e-16
ciY_19 (c_40_p VSS) capacitor c=9.69942e-16
ciVDD_17 (MM0_b VSS) capacitor c=3.20822e-14
ciVDD_16 (c_13_n VSS) capacitor c=3.72982e-16
ciVDD_15 (VDD VSS) capacitor c=3.73709e-16
ciVDD_14 (c_15_n VSS) capacitor c=1.78127e-16
ciVDD_13 (MM0_s VSS) capacitor c=1.11444e-14
ciA_18 (A_8 VSS) capacitor c=1.47512e-16
ciA_17 (A_13 VSS) capacitor c=1.47512e-16
ciA_16 (A_20 VSS) capacitor c=3.49264e-16
ciA_15 (A_23 VSS) capacitor c=9.44456e-17
ciA_14 (A VSS) capacitor c=2.08242e-16
ciA_13 (MM1_g VSS) capacitor c=1.88888e-16
ciA_12 (MM0_g VSS) capacitor c=1.88888e-16

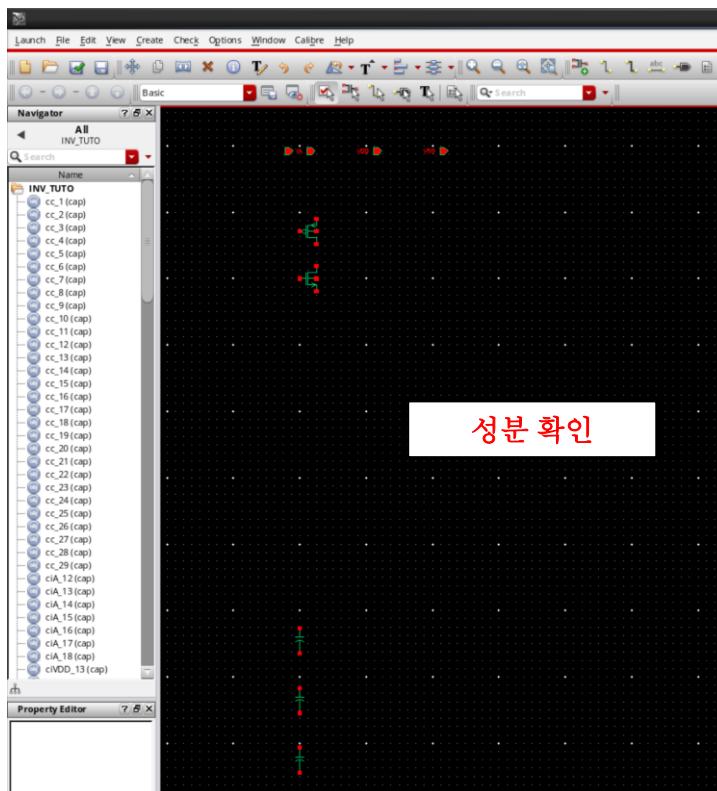
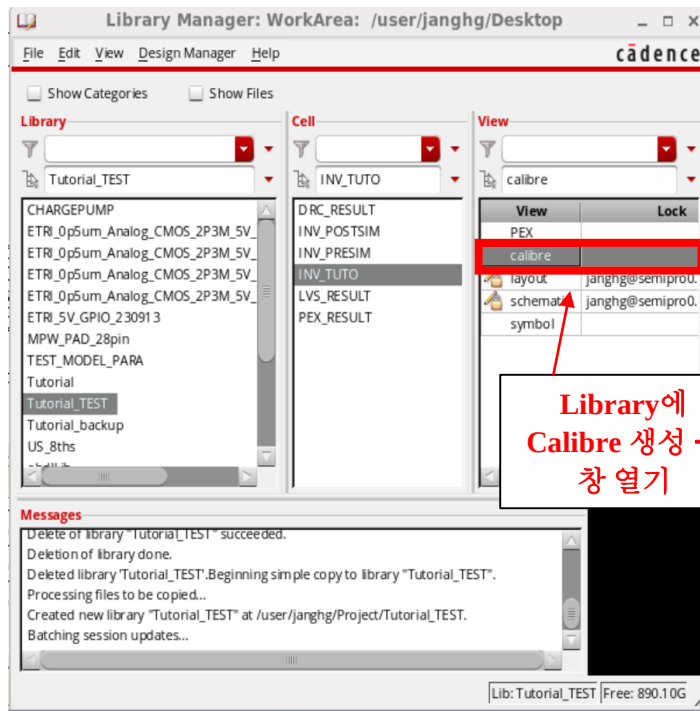
rVSS_26 (c_45_n VSS_3) resistor r=0.0096667
rVSS_25 (c_45_n VSS_14) resistor r=0.0264734
rVSS_24 (VSS_3 c_46_n) resistor r=0.0489097
rVSS_23 (VSS_5 VSS) resistor r=0.0177778
rVSS_22 (VSS_5 VSS_14) resistor r=0.0205904
rVSS_21 (c_46_n VSS_18) resistor r=20.8333
rVSS_20 (VSS_18 MM1_s) resistor r=42.6562
rY_43 (Y_1 Y_2) resistor r=0.0533333
  
```

소자 및 기생
성분 포함
정보 팝업

MOS

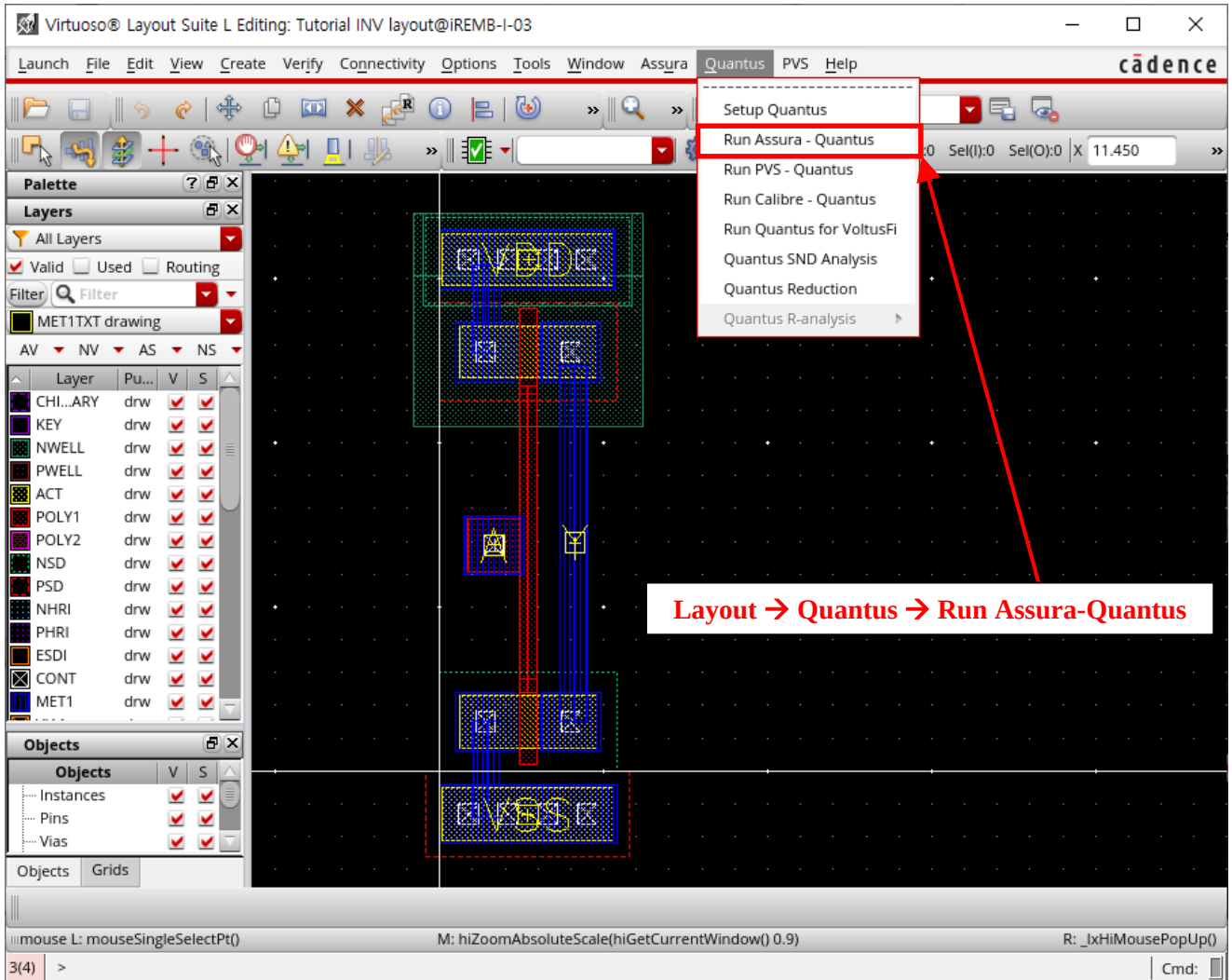
기생 C

기생 R

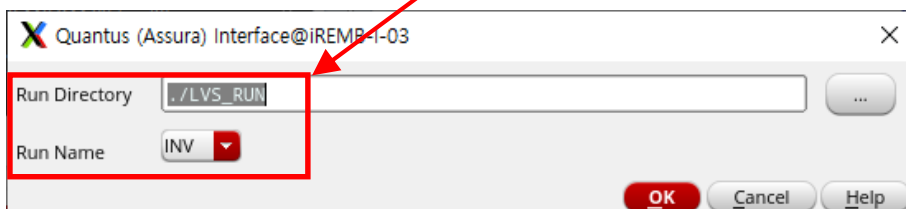


2) Assura QRC 검토

① QRC 진행 방법



Assura LVS 실행에서 Run 결과를 저장한 디렉토리 및 'Run Name' 선택('Assura LVS 검증 방법' 참고).



RuleSet 선택(rcx_typical / worst / best).

Quantus (Assura) Parasitic Extraction Run Form - Version 21.2.2-p045@iREMB-I-03

Setup Extraction Filtering Netlisting Run Details Substrate

Technology NSPL_0p5um_PDK_av RuleSet rcx_typical

☐ p2lvsSet NONE Use Multi-rule Sets ☐

☐ Setup Dir I0S_PDK_240925/NSPL_05CM0S_PDK_240206/Assura/QRC

Template .rcx_setup.tp1 Load

☐ Include Command File View Edit

Rule Command File Include View Edit

☐ Tech Cmd File User View Edit

☐ LPE Config File View Edit

☐ Library Cell Mapping File View Edit

Library Definitions File I206/cds.lib View Edit

☐ Probe Text File View Edit

extracted cell views during hierarchical extraction of an extracted view.

Library Directory: Specify a directory for writing local libraries created during the hierarchical extraction of an extracted view.

OK Cancel Apply Defaults Load State Save State View Command File Help

'Technology' 및 'RuleSet' 선택 시, Template이 자동으로 load 됨.

Output을 Extracted View 선택.

필요시 'View' 이름 변경 가능. 예) rc_extracted

Quantus (Assura) Parasitic Extraction Run Form - Version 21.2.2-p045@iREMB-I-03

Setup Extraction Filtering Netlisting Run Details Substrate

Output **Extracted View**

Lib Tutorial Cell INV View **av_extracted**

Enable CellView Check ☐

Parasitic Res Component presistor Prop Id r

Parasitic Cap Component pcapacitor Prop Id c

Parasitic Ind Component pinductor Prop Id l

Parasitic M Component pmind Prop Id k

Inductance L1 Prop Id ind1 Inductance L2 Prop Id ind2

Parasitic CCVS Component ccvs Hgain Prop Id hgain Vref Prop Id vref

Parasitic VS Component vsource Prop Id vr

☒ Call Procedure ...

Substrate Extract ☐ Substrate Profile NONE

Add LVS MOS Diffusion Res ☐ Extract MOS Diffusion Res ☐

Auto Accuracy Downgrade ☒ Extract MOS Diffusion High NONE

extracted cell views during hierarchical extraction of an extracted view.

Library Directory: Specify a directory for writing local libraries created during the hierarchical extraction of an extracted view.

OK Cancel Apply Defaults Load State Save State View Command File Help

Extraction Type 선택(C Only 또는 RC).

'Decoupled' 선택.

Quantus (Assura) Parasitic Extraction Run Form - Version 21.2.2-p045@iREMP-I-03

Setup **Extraction** Filtering Netlisting Run Details Substrate

Extraction Type **RC** Name Space Layout Names

Max fracture length infinite microns Temperature 25.0 C Edit

Max fracture length by layer Edit

☐ From File

Cap Coupling Mode Decoupled

Advanced Coupling ☐ Mult Factor 1.0

Ref Node VSS Ground Substrate Floating Nets ☐

Extend Checking Distance default Diffusion Equation R ☒

Ladder Network ☐

Select... User Region View Edit

Net Selection Mode Full Chip All Nets Layer Setup Customization Edit...

Field Solver

splitting pins.

HRCX Cells: Specify a list of cells which appear in the output hierarchy, requires cell name, with optional view and lib names (cell, view, lib).

OK Cancel Apply Defaults Load State Save State View Command File Help

해당 회로의 ground reference node 기재.

해당 회로의 power net/ground net 기재(schematic/layout에서 define한 pin 이름 기재).

Quantus (Assura) Parasitic Extraction Run Form - Version 21.2.2-p045@iREMB-I-03

Setup Extraction **Filtering** Netlisting Run Details Substrate

Exclude Self Capacitance: No
Ext. Self Cap Recursive: ☐
Exclude Floating Nets: ☐
Exclude Float Limit: 2000

Merge Via for all layer:
☒ Size: auto um
☐ Count: 1

Merge Via for selected layers:
☐ Size Edit
☐ Count Edit

Array Vias Spacing: auto Microns

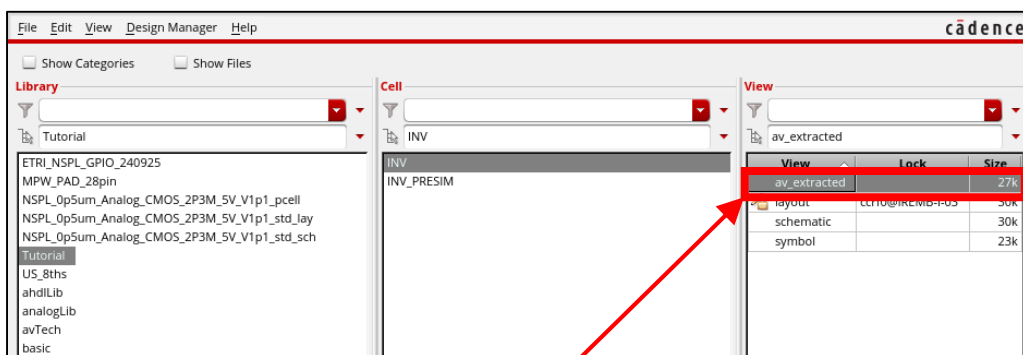
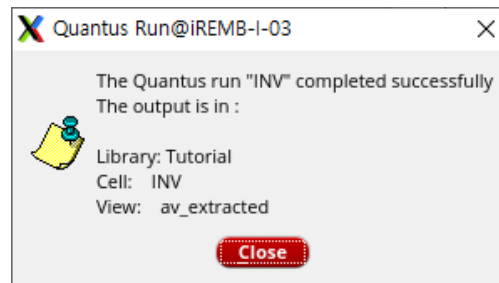
Enter Power Nets: VDD
From File: ☐
SelfFromSch

Enter Ground Nets: VSS
From File: ☐
SelfFromSch

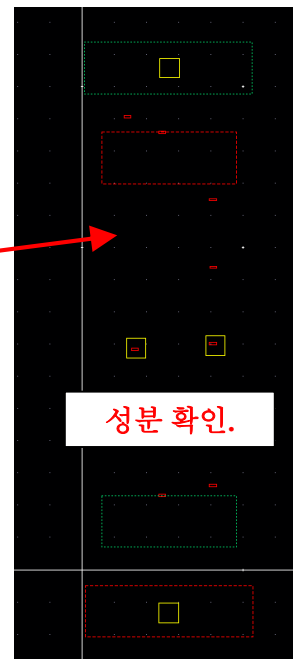
for exclusion from parasitic resistance extraction.
Ground Nets: Specify Ground Nets that are not defined as global by LVS extraction for exclusion from parasitic resistance extraction.

OK Cancel **Apply** Defaults Load State Save State View Command File Help

Apply(또는 OK) 눌러서 QRC 실행.



av_extracted 생성 확인.



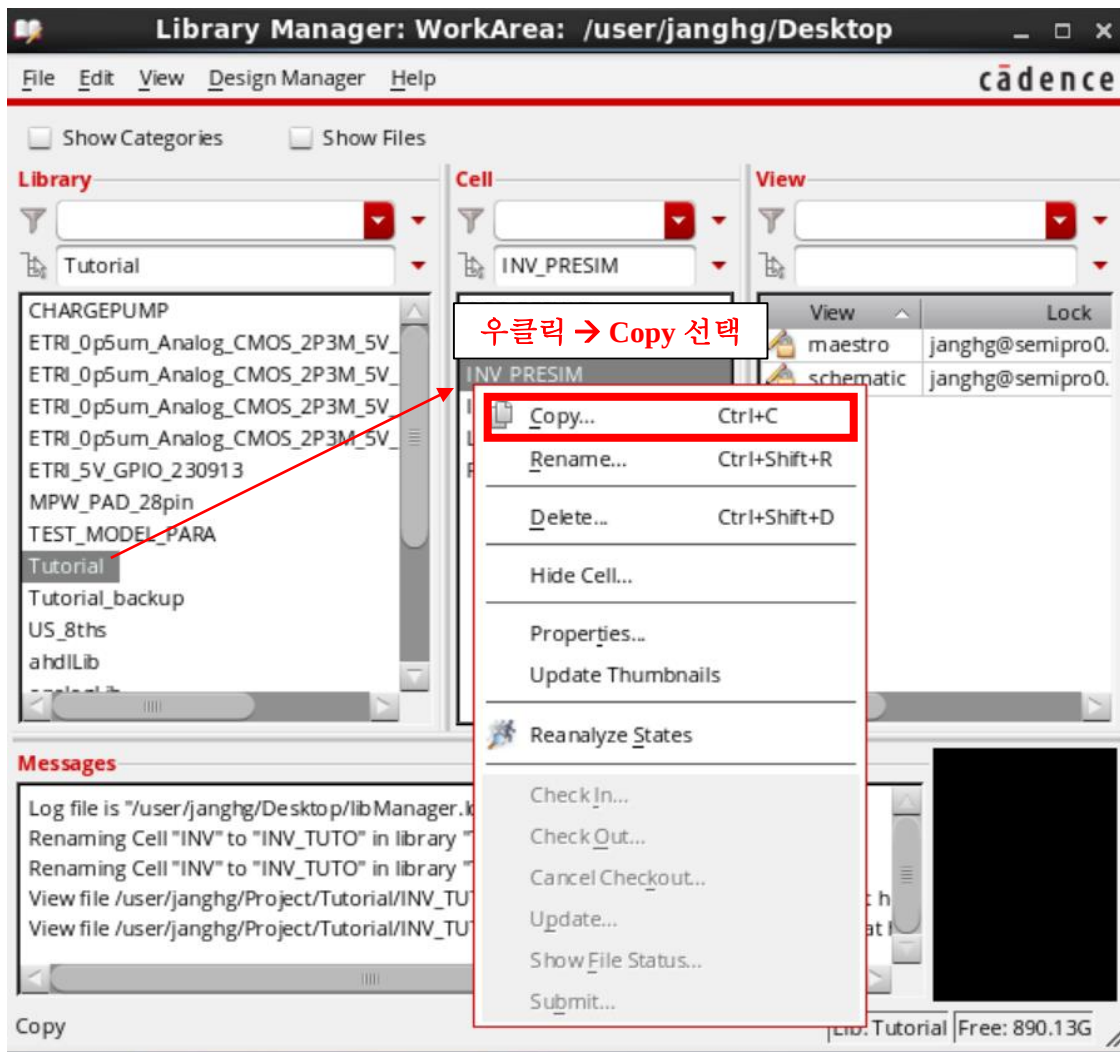
8. 회로 설계 및 검증 (Post-simulation)

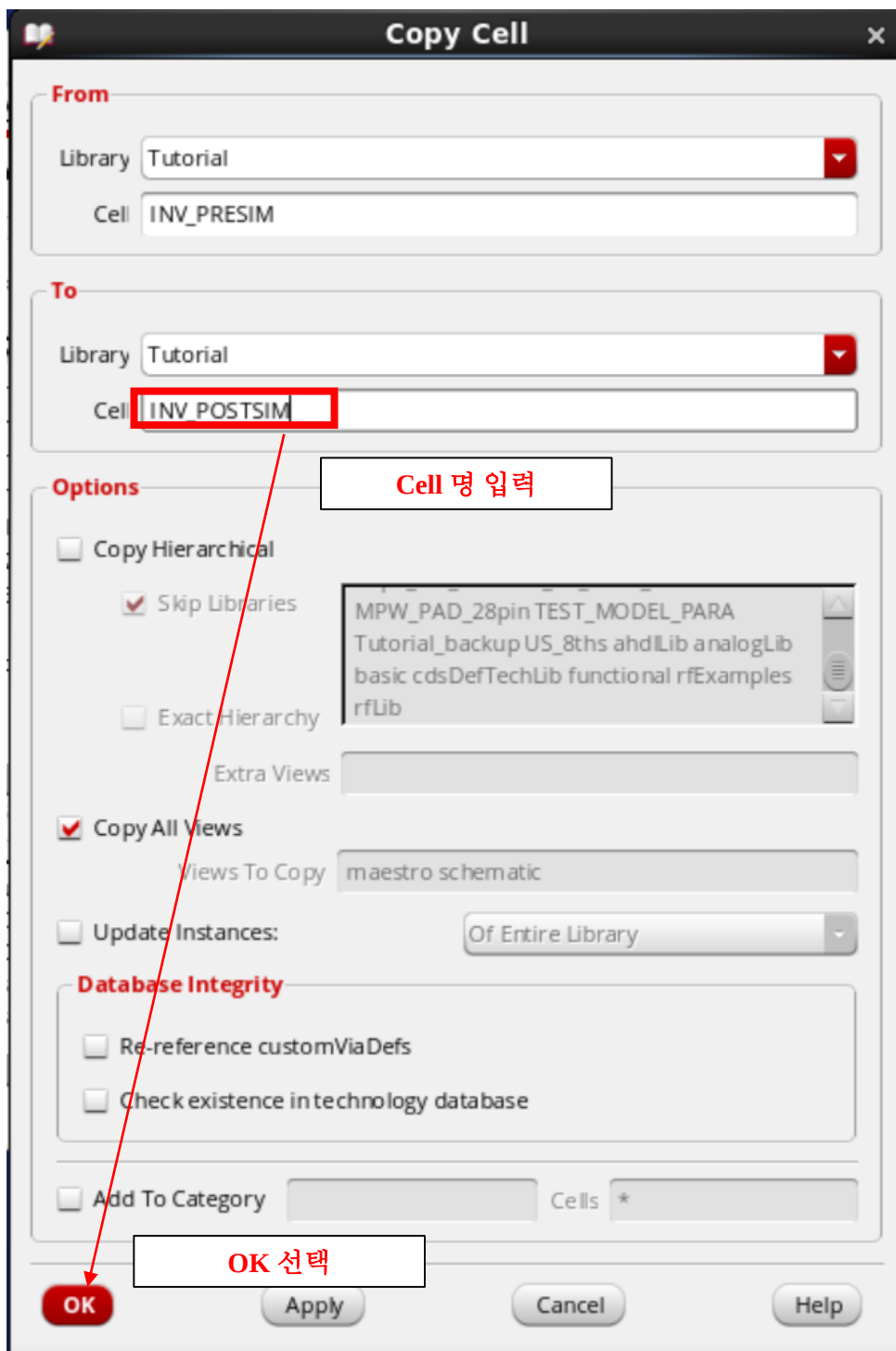
8. 회로 설계 및 검증 (Post-simulation)

- 목적 : Parasitic 성분 반영한 시뮬레이션을 통해 Pre-sim.과 결과 비교

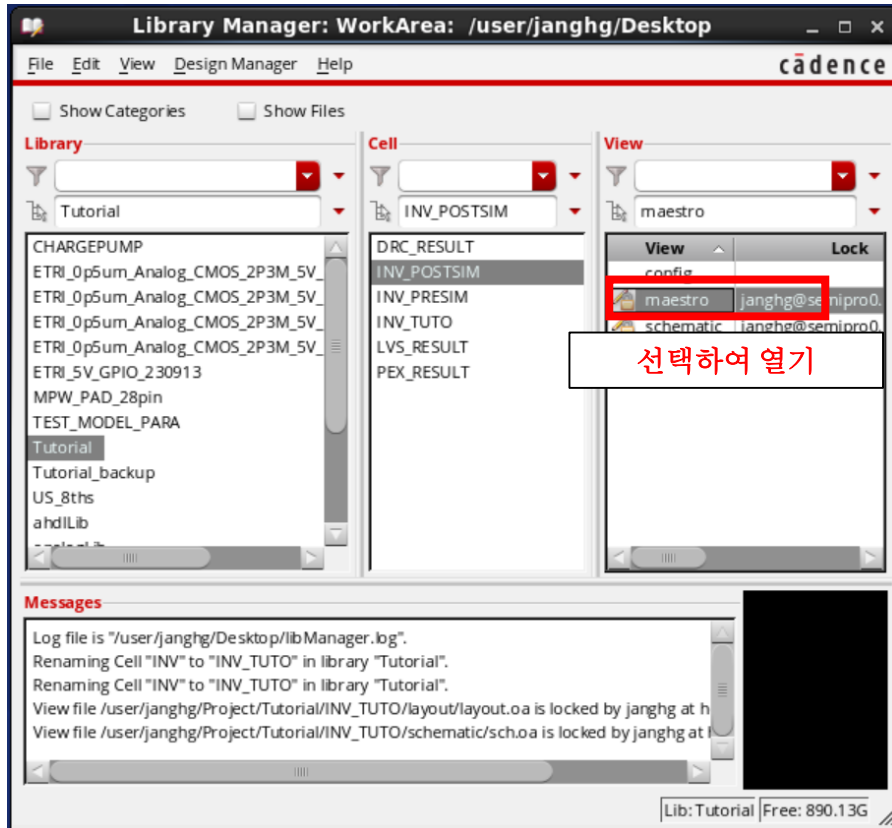
1) Post-simulation

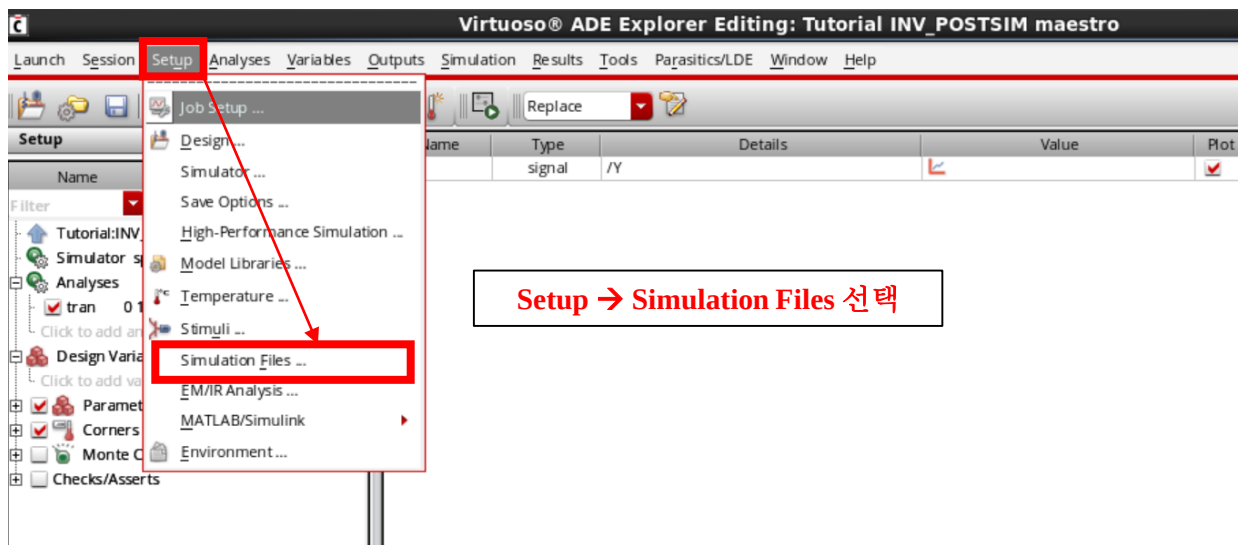
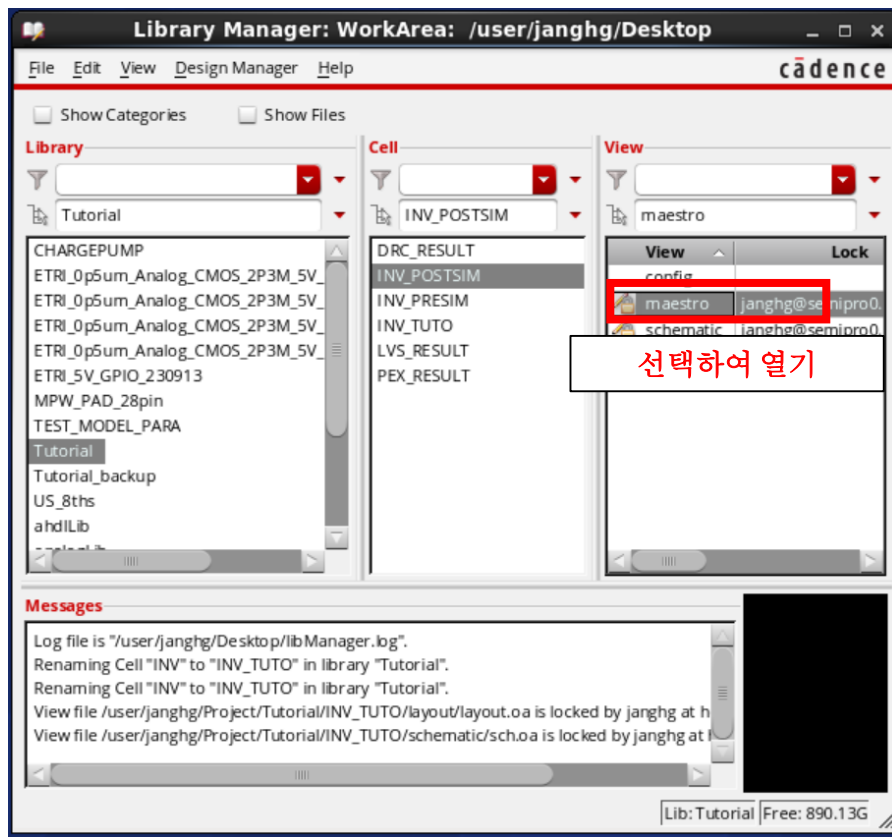
① Schematic 설계 프로그램 실행

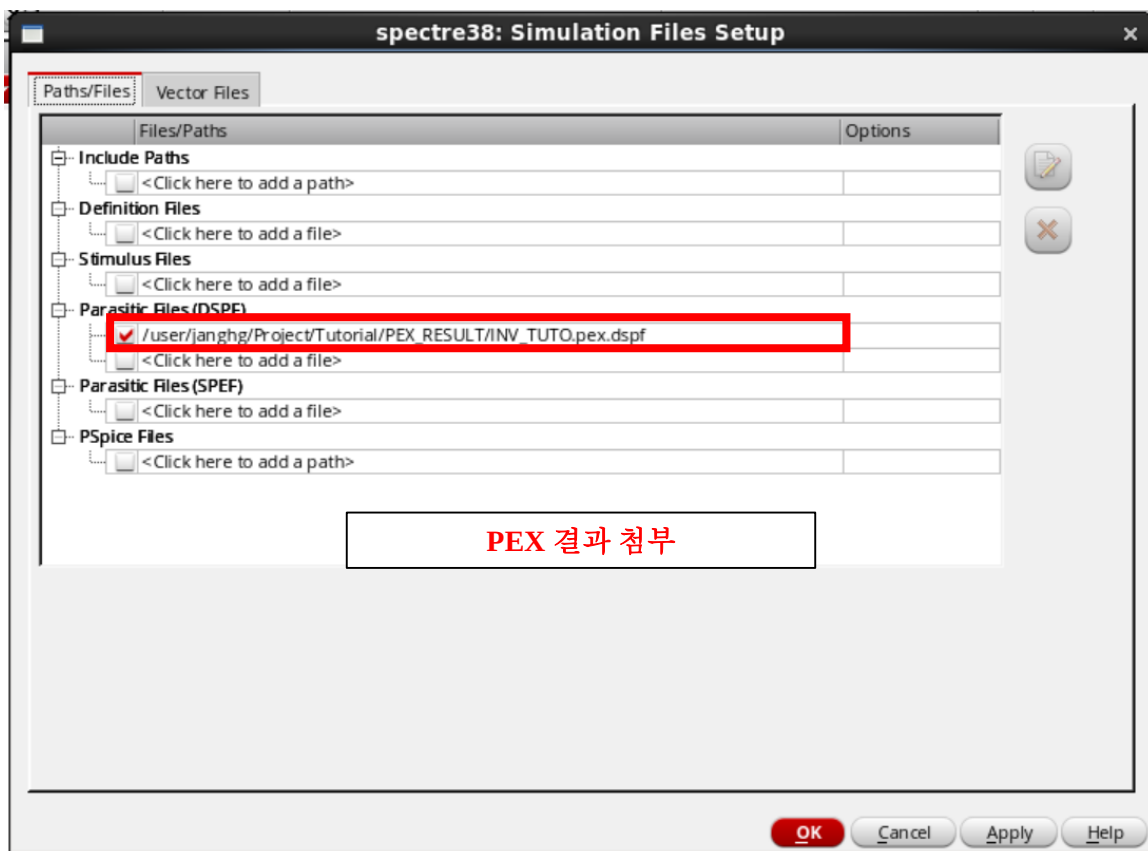




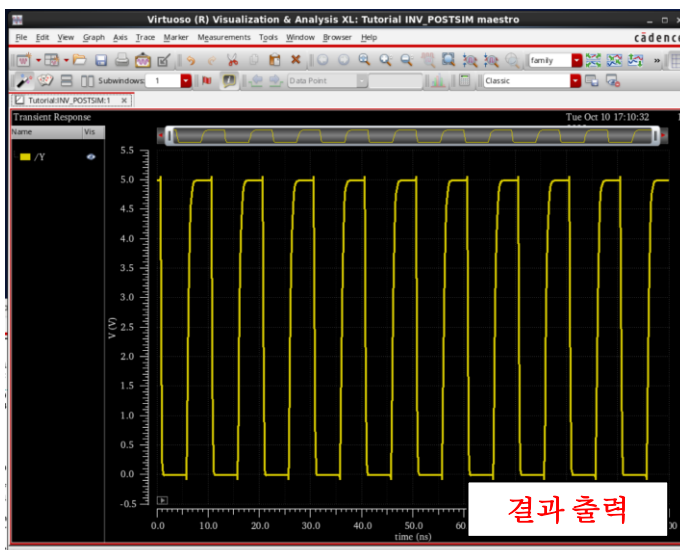
2) Post-simulation; DSPF 방식



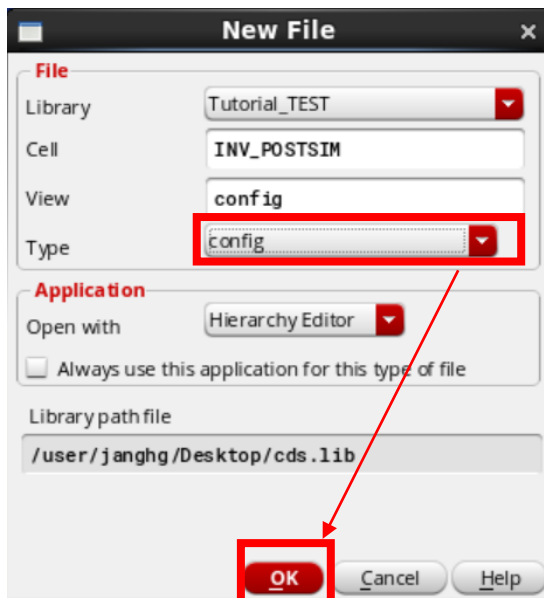
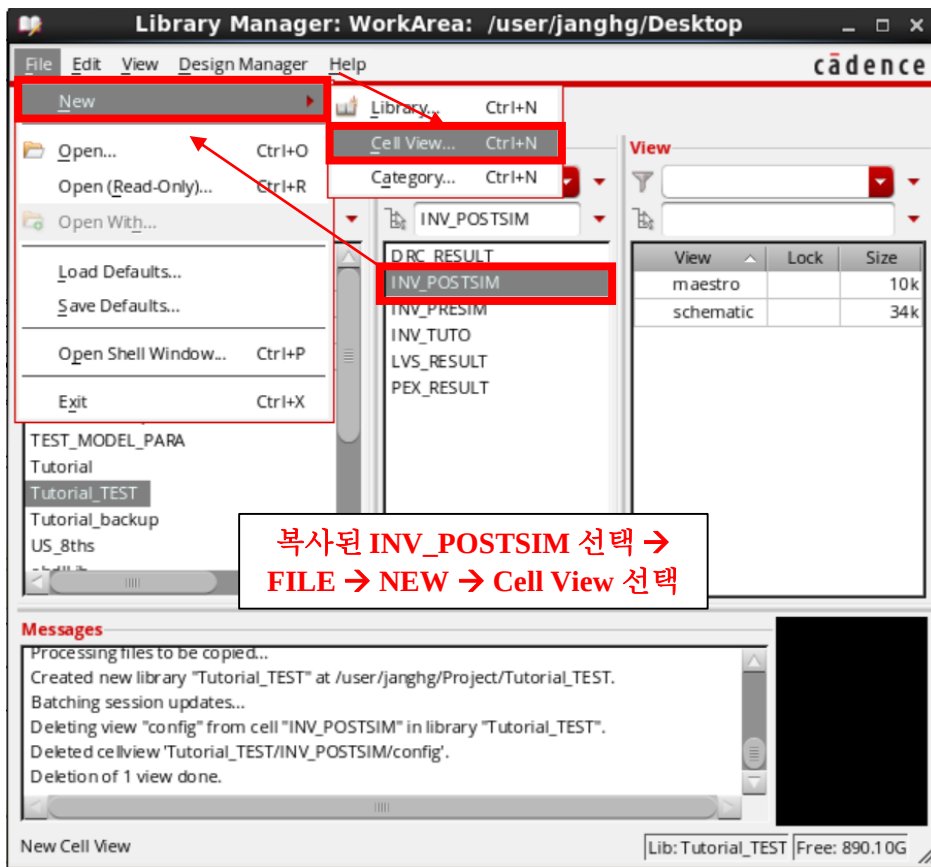


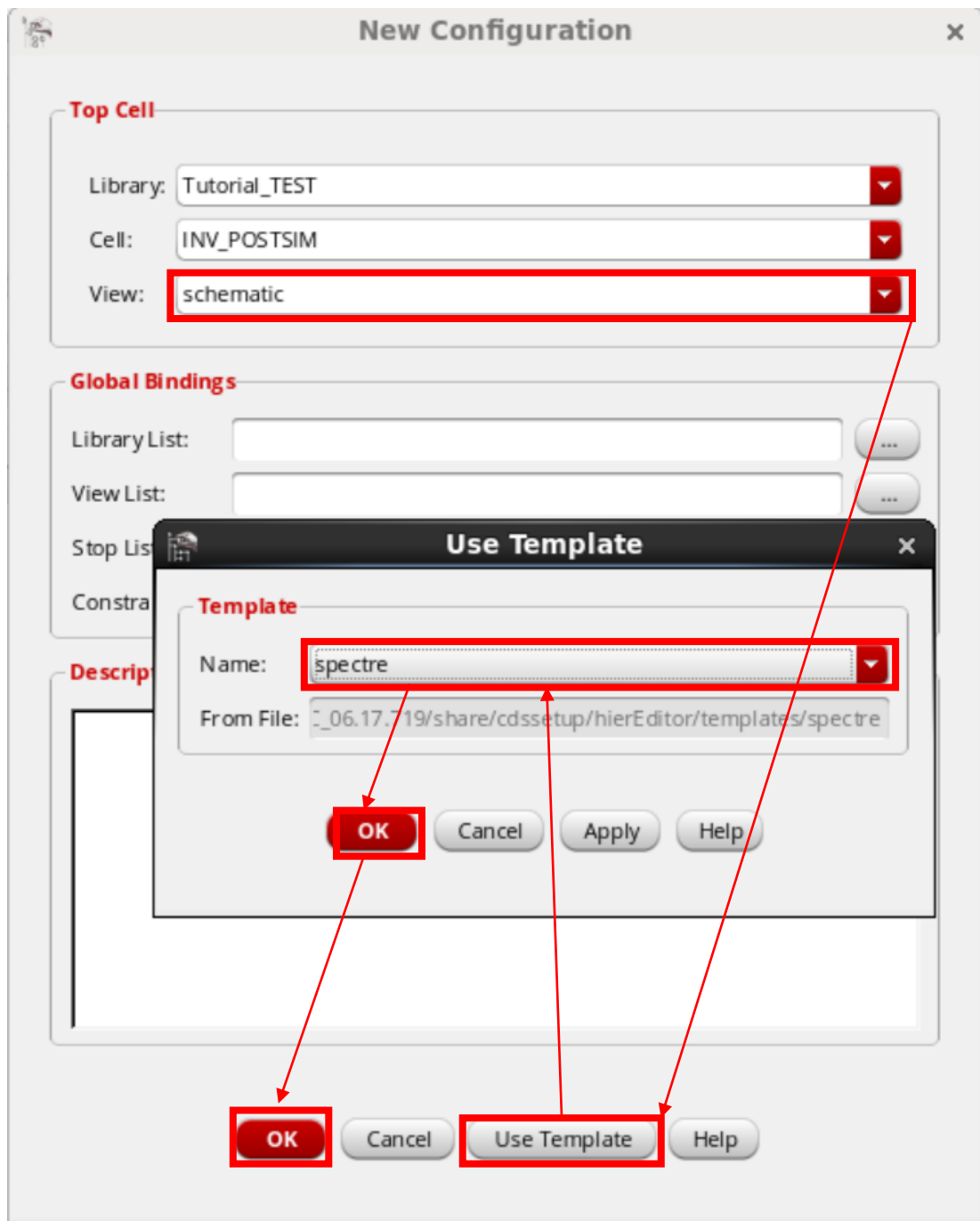


Run Simulation!!!

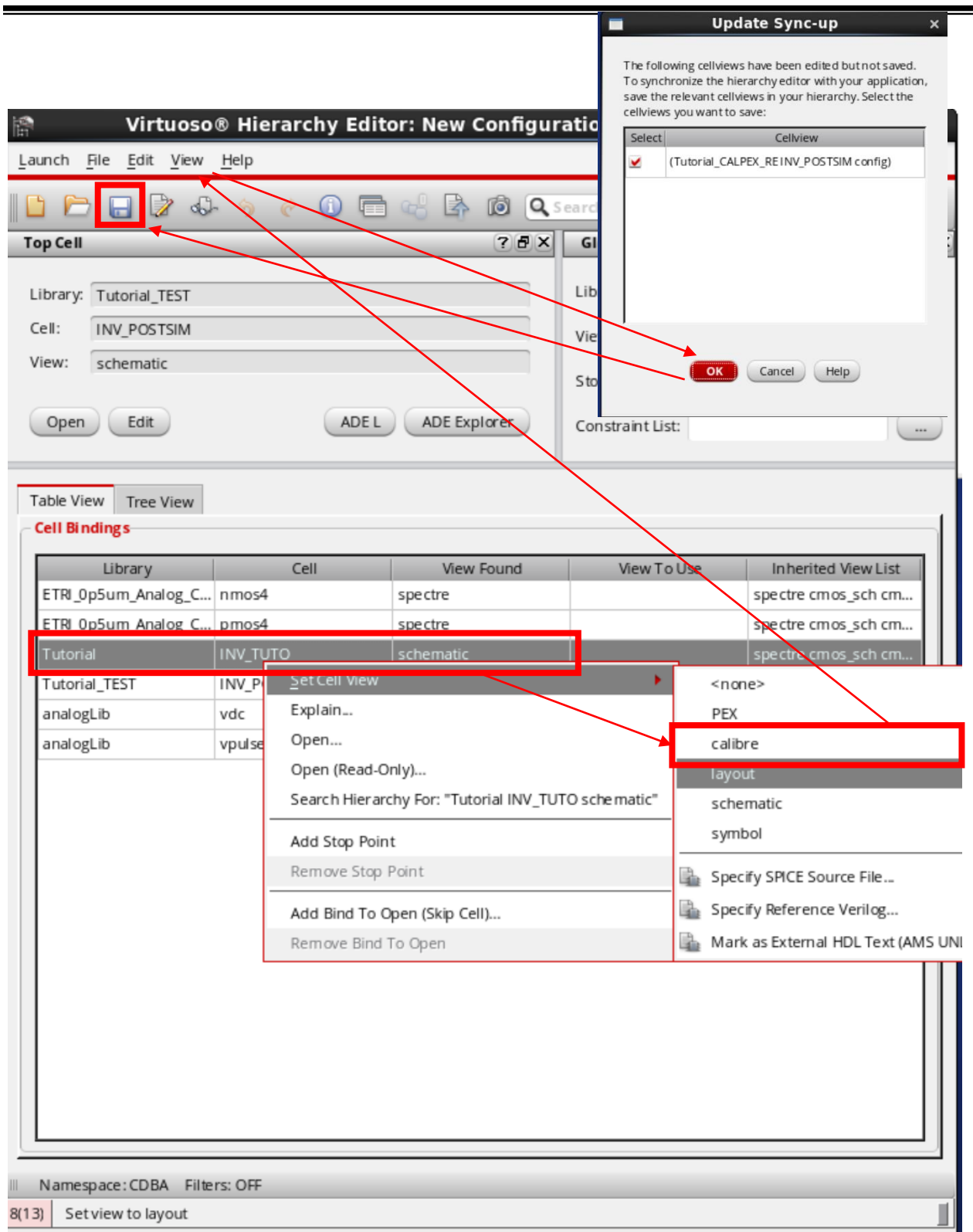


3) Post-simulation; Calibre 방식

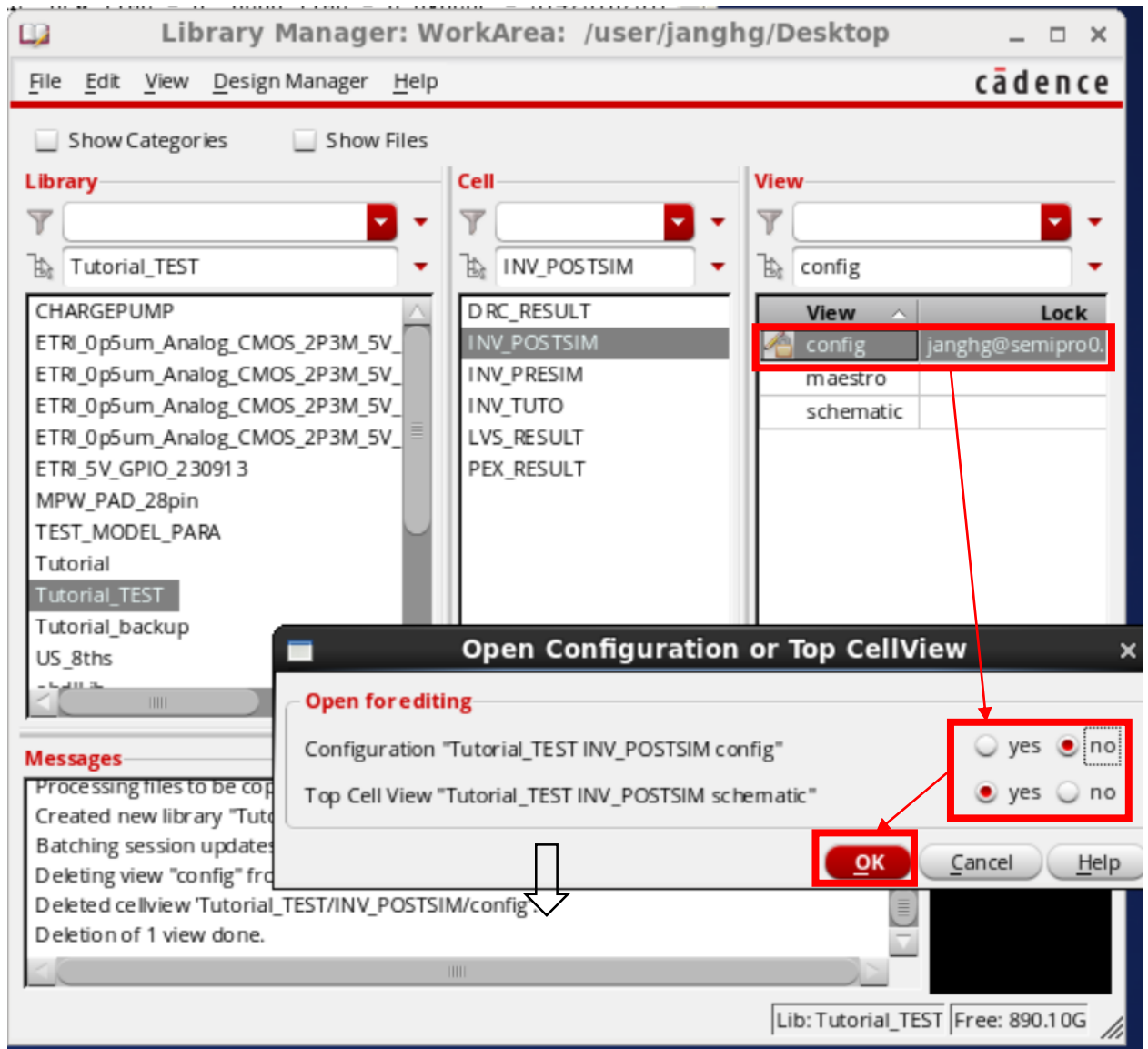




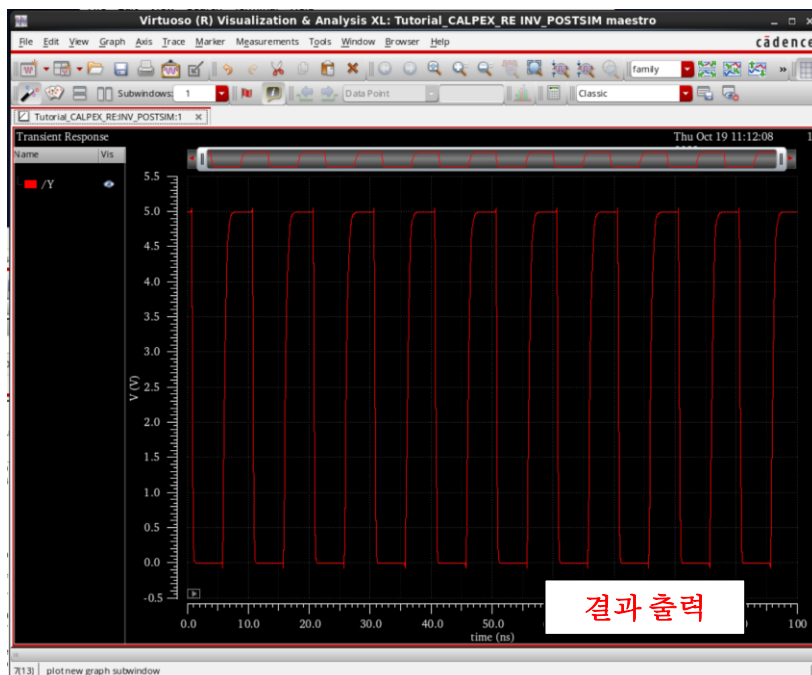
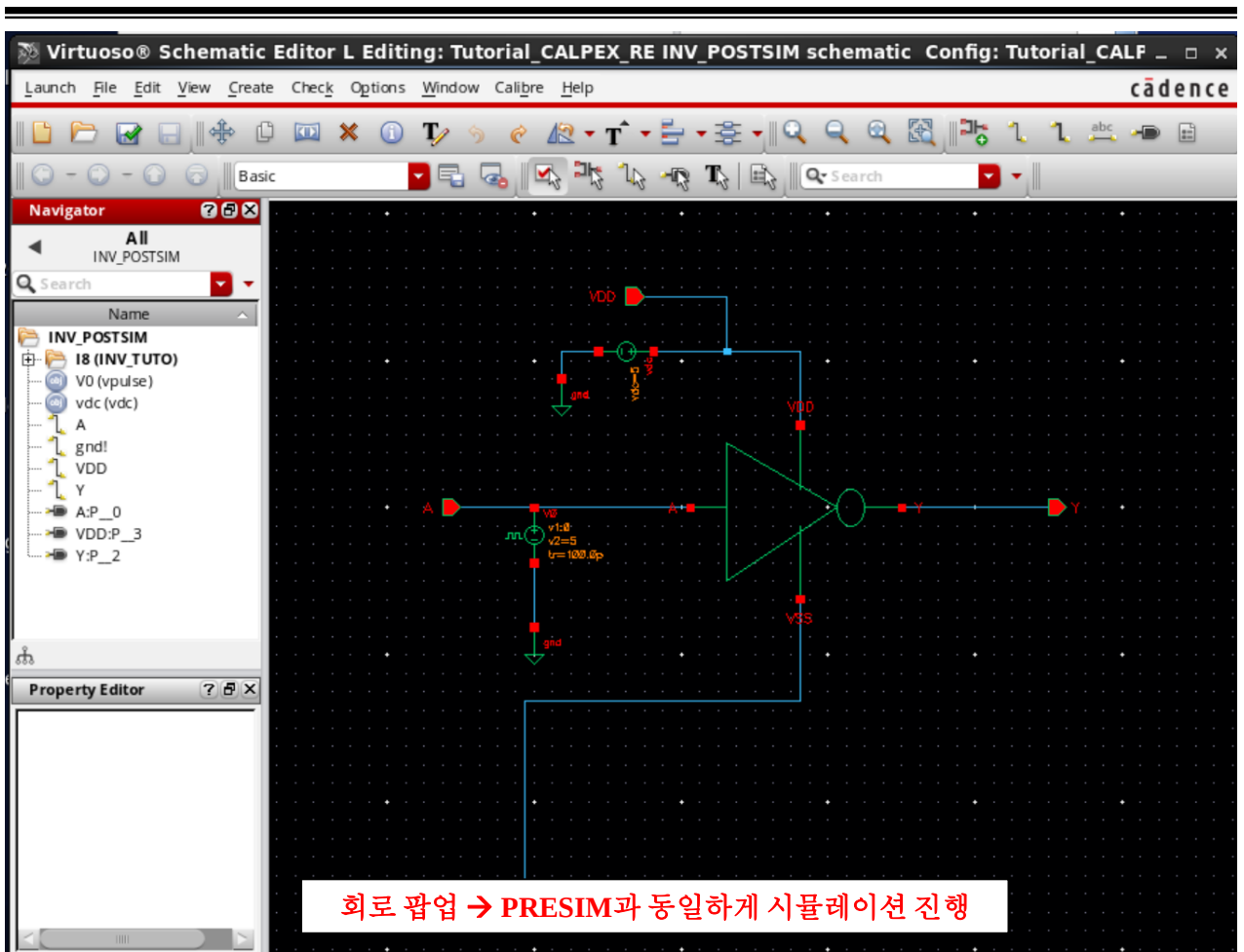
Top Cell / VIEW에서 Schematic 선택
→ Use Template 선택
→ Tempalate / name에서 Specture 선택
→ OK → OK 클릭



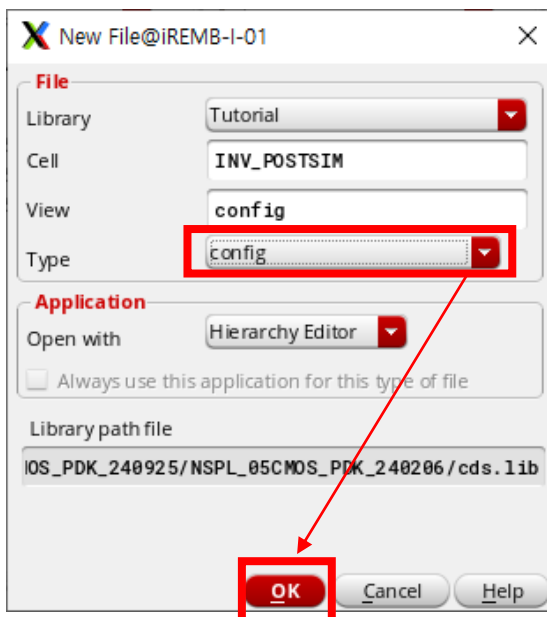
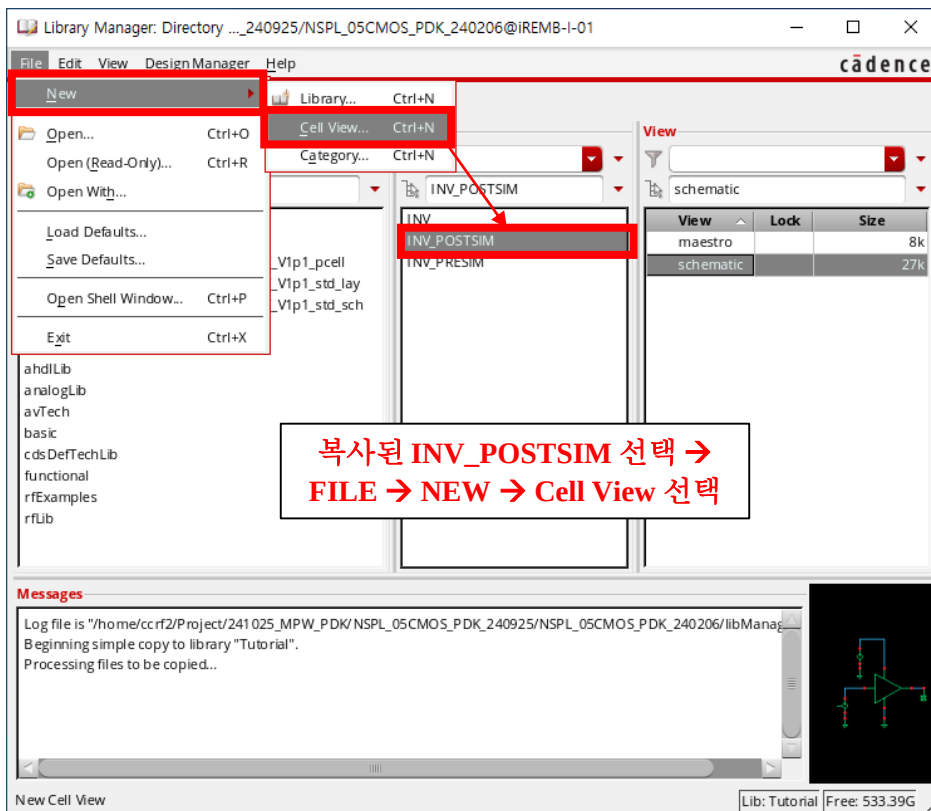
Cell 우클릭 선택 → Setcell view 선택
→ Calibre 선택 → View / Update(needed)
선택 → OK 선택 → Save



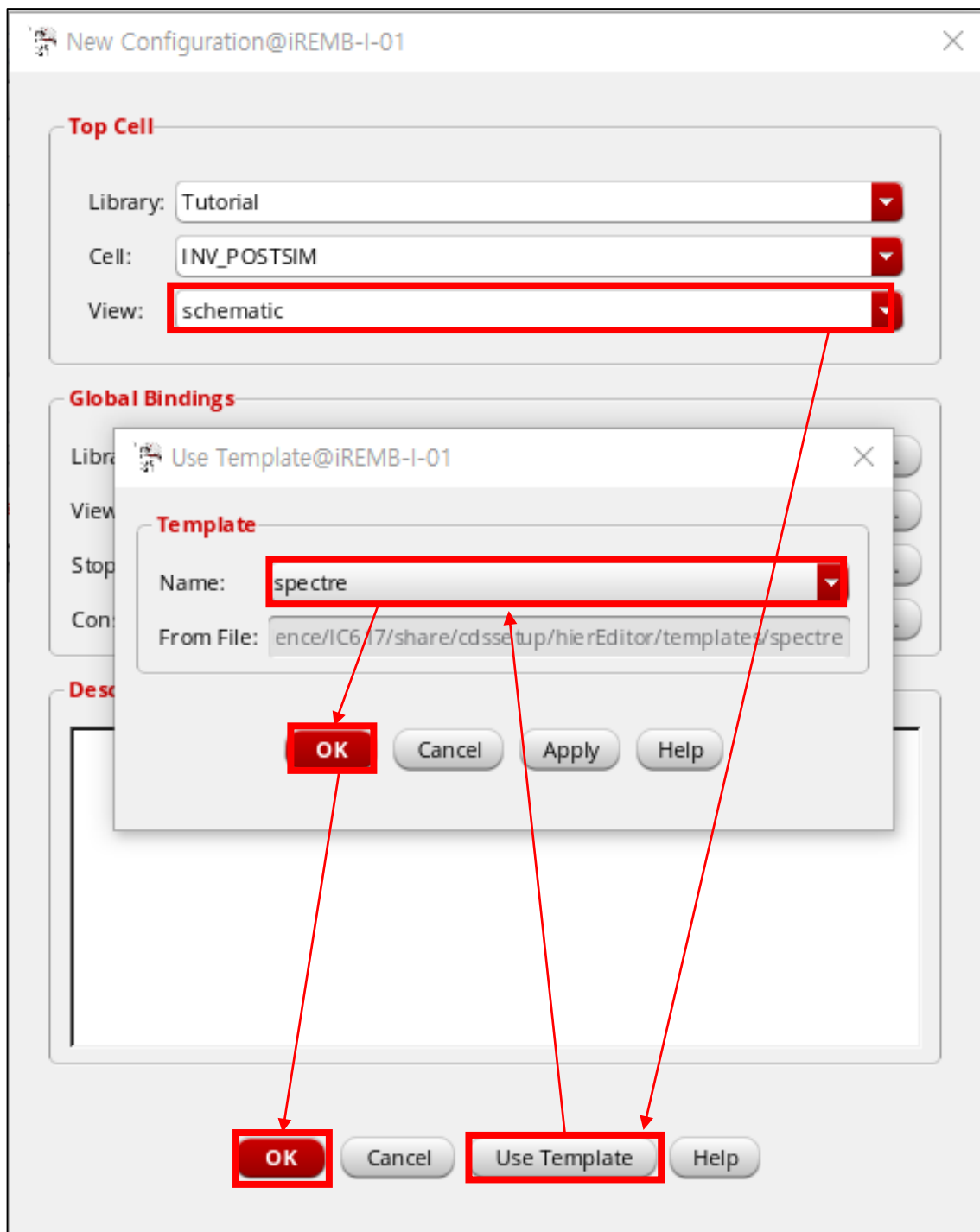
Library manager → Config 더블클릭
→ 설정 후 OK 클릭



3) Post-simulation; Extracted View 방식



Type에서 Config 선택
→ OK 클릭



Top Cell / VIEW에서 Schematic 선택
→ Use Template 선택
→ Template / name에서 Specture 선택
→ OK→ OK 클릭

Virtuoso? Hierarchy Editor: New Configuration (Save Needed)@IREMB-I-01

Launch File Edit View Help

Top Cell

Library: Tutorial
Cell: INV_POSTSIM
View: schematic

Open Edit ADE L ADE Explorer

Update Sync-up@IREMB-I-01

The following cellviews have been edited but not saved. To synchronize the hierarchy editor with your application, save the relevant cellviews in your hierarchy. Select the cellviews you want to save:

Select	Cellview
☒	(Tutorial INV_POSTSIM config)

OK Cancel Help

Table View Tree View

Cell Bindings

Library	Cell	View Found	View To Use	Inherited View List
NSPL_0p5um_Analog_...	nmos4	spectre		spectre cmos_sch cm...
NSPL_0p5um_Analog_...	pmos4	spectre		spectre cmos_sch cm...
Tutorial	INV	schematic	schematic	spectre cmos_sch cm...
Tutorial	INV_POSTSIM	schematic		spectre cmos_sch cm...
analogLib	vdc	spectre		spectre cmos_sch cm...

Cell: INV 칸에서 마우스 오른쪽 버튼 클릭.

Set Cell View

- Explain...
- Open...
- Open (Read-Only)...
- Search Hierarchy For: "Tutorial INV schematic"
- Add Stop Point
- Remove Stop Point
- Add Bind To Open (Skip Cell)...
- Remove Bind To Open

<none>

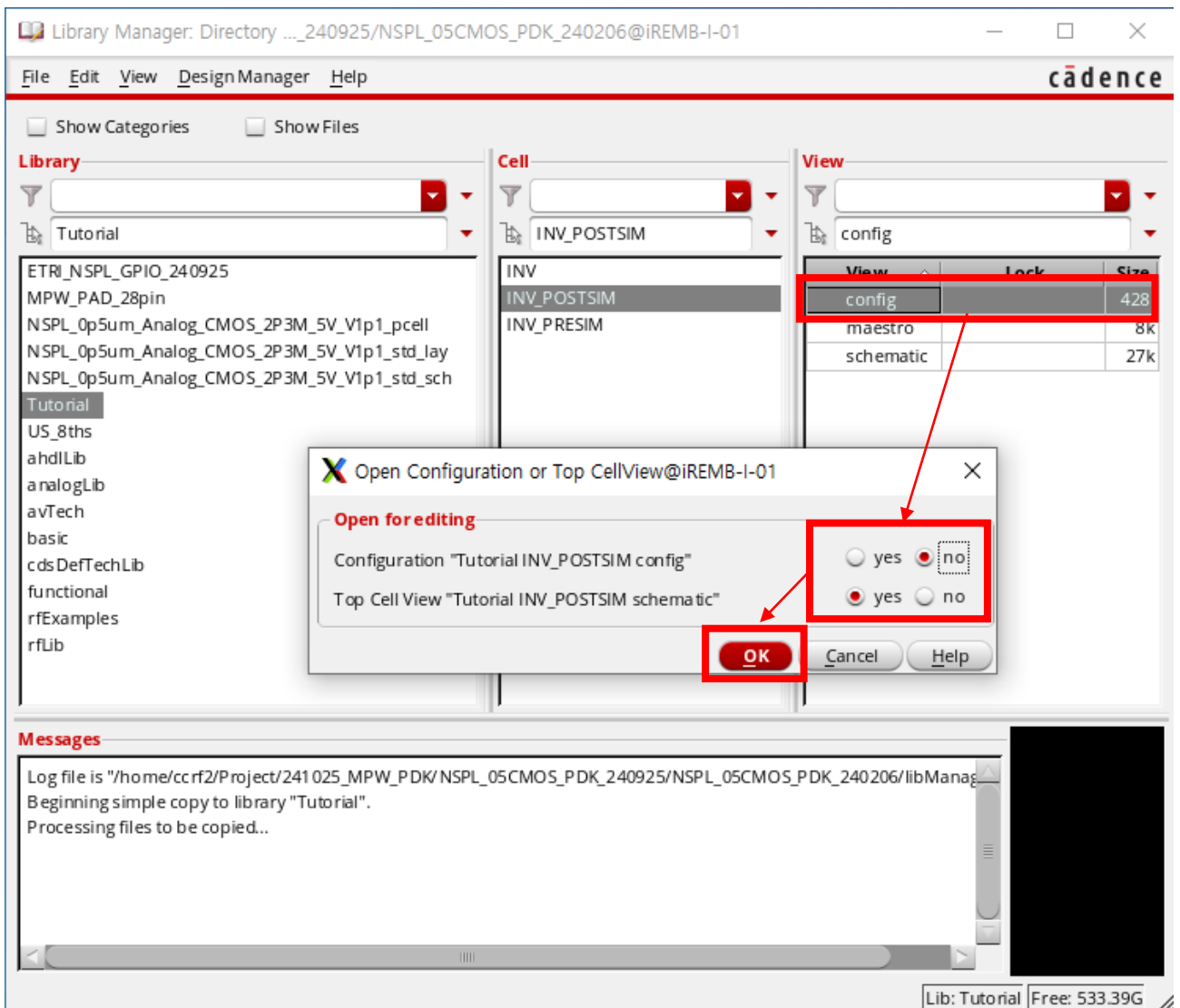
- av_extracted
- layout
- schematic
- symbol
- Specify SPICE Source File...
- Specify Reference Verilog...
- Mark as External HDL Text (AMS UNL only)

RC 기생성분을 추출한 'av_extracted' 선택.

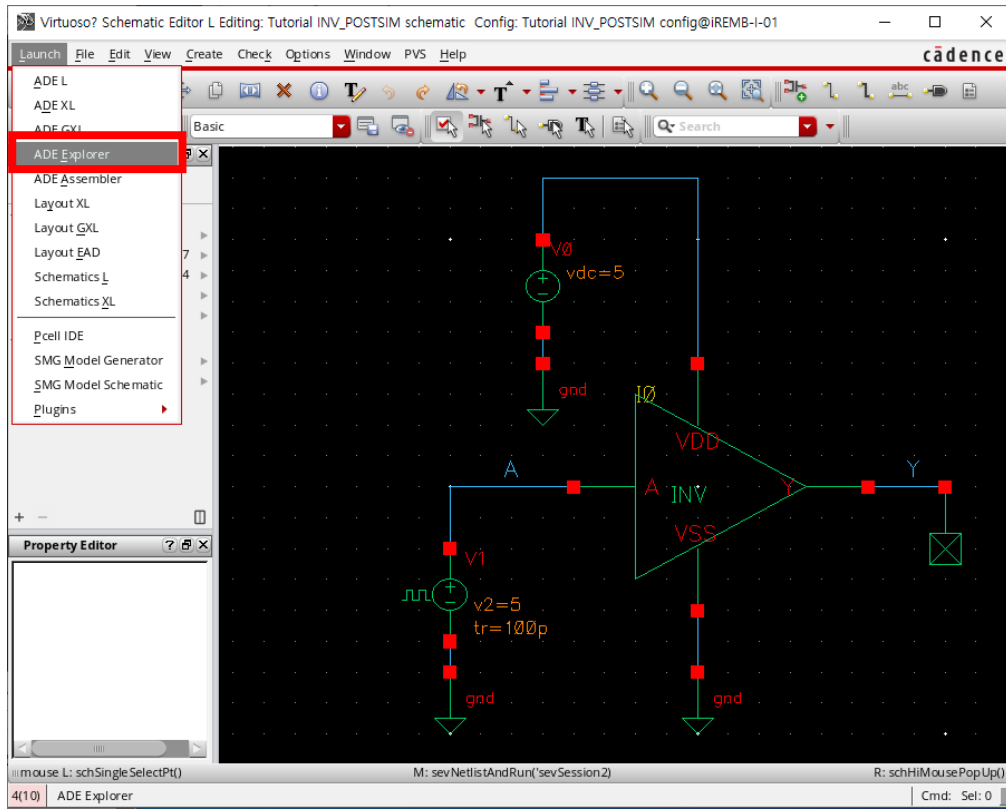
Namespace: CDBA Filters: OFF Update: Needed

3(8) Set view to av_extracted

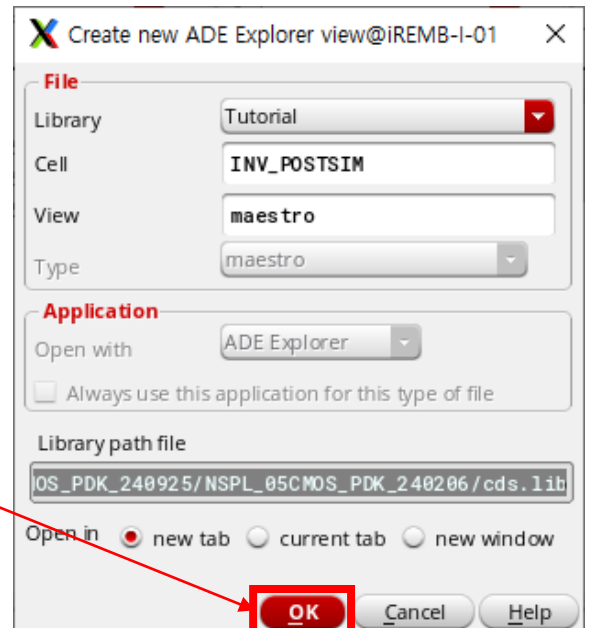
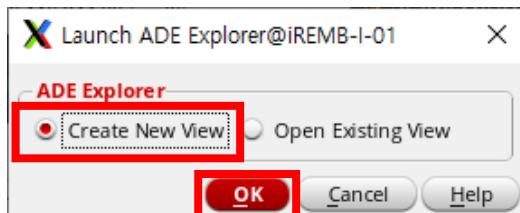
Cell 우클릭 선택 → Setcell view 선택
→ av_extracted 선택 → View / Update(needed) 선택 → OK 선택 → Save

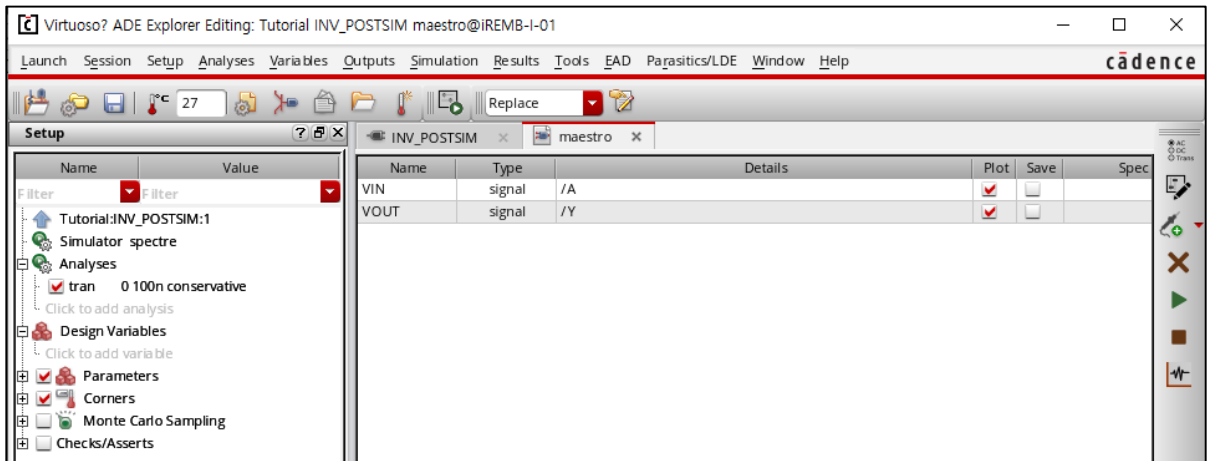


Library manager → Config 더블클릭
→ 설정 후 OK 클릭

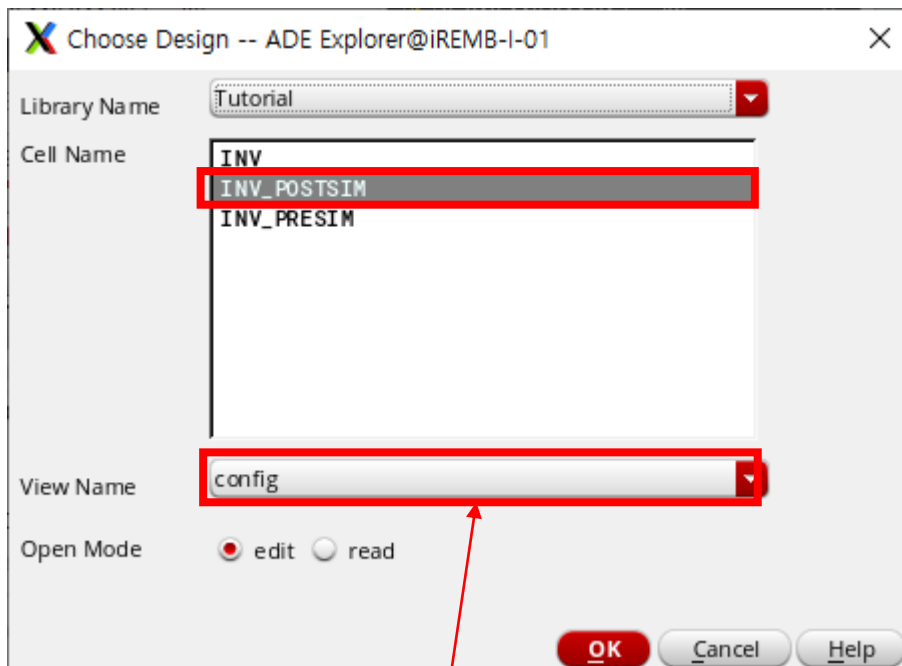


회로 팝업 → ADE Explorer(또는 Simulator) 실행.

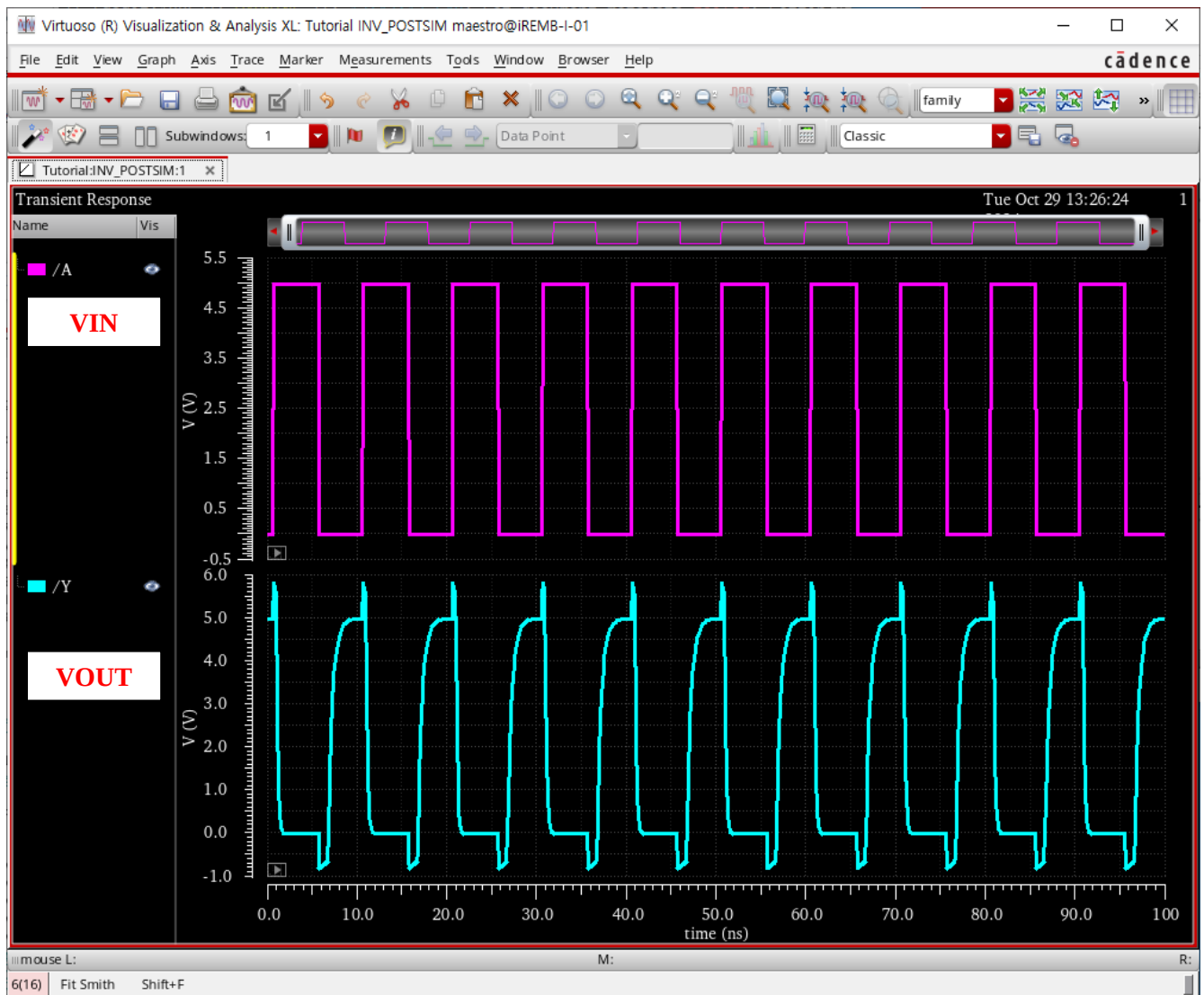




Setup - Model Library 설정 및 Analyses - Choose 설정 등 INV_PRESIM 설정과 동일하게 설정.

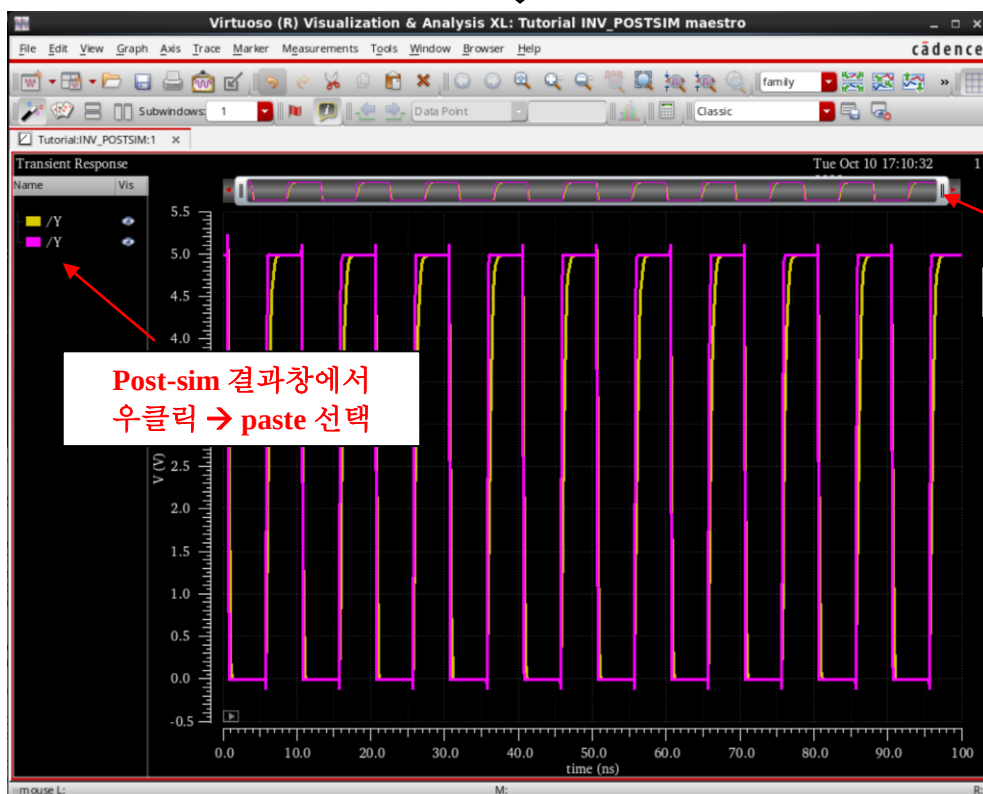
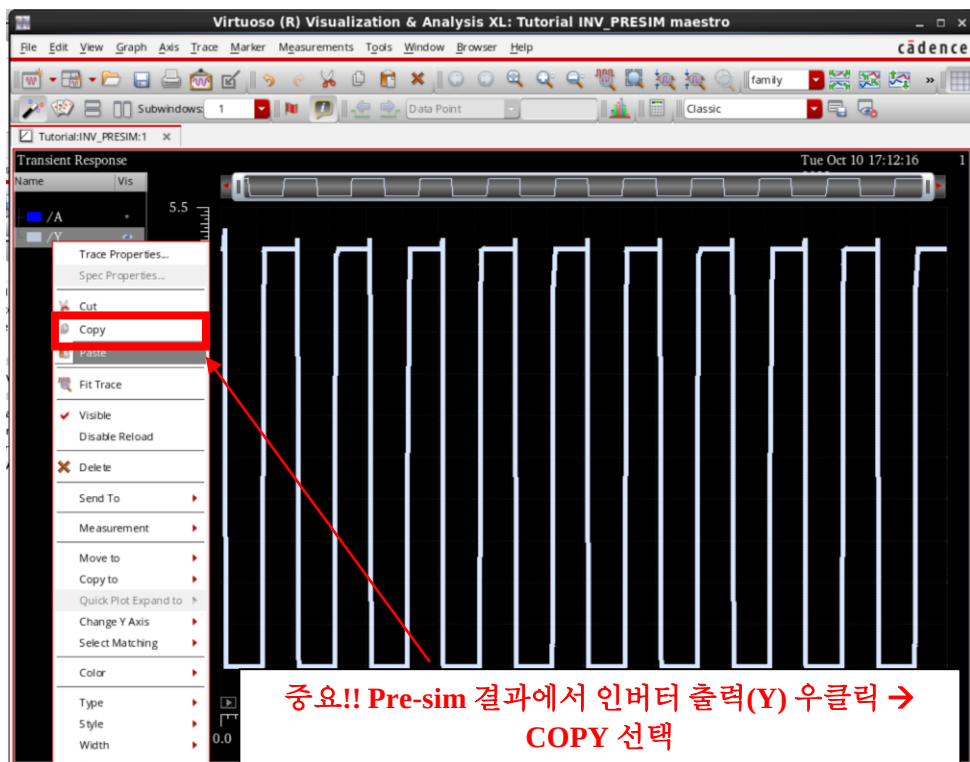


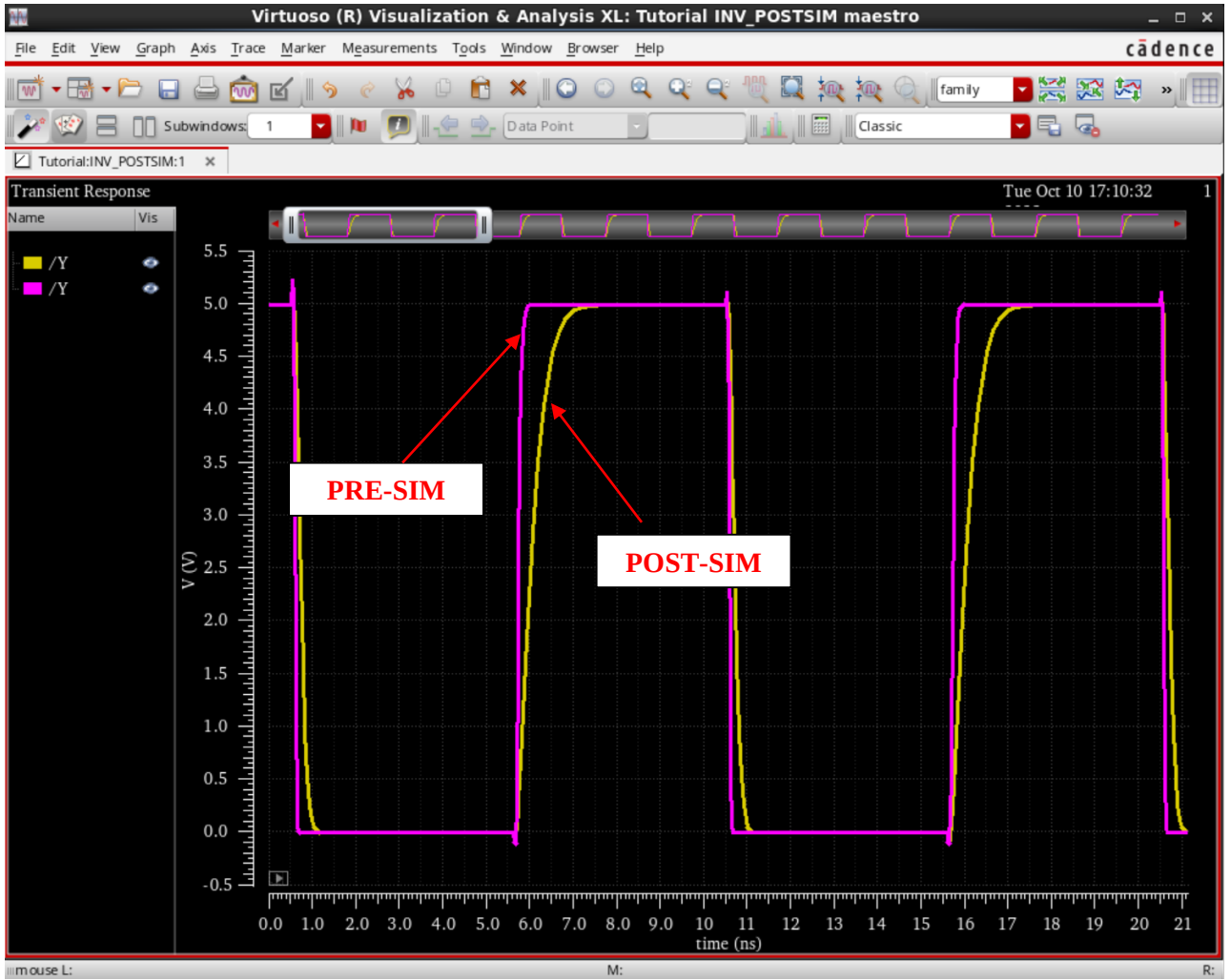
단, Setup - Design에서 'View Name'이 'config'로 설정되어 있는지 반드시 확인 필요.



POSTSIM 결과 파형 확인.

4) 결과비교





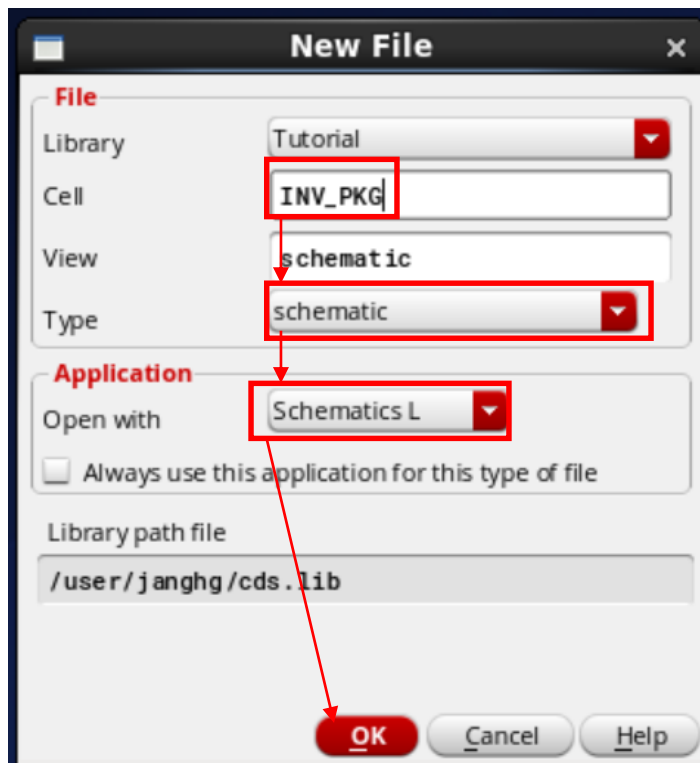
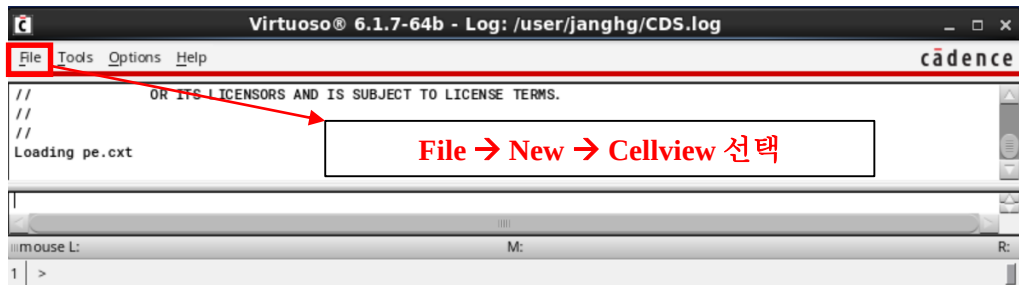
9. 28-pin Package Pads 및 GPIO

9. 28-pin Package Pads 및 GPIO

- 목적 : 패키지를 위한 패드 및 GPIO 연결
(Package 또는 GPIO 사용 안할 시 스킵 가능)
- 예: 설계된 인버터 회로 적용

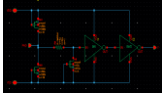
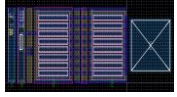
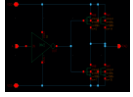
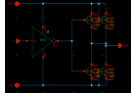
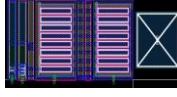
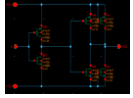
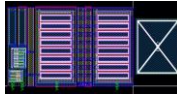
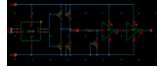
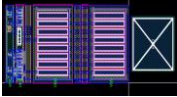
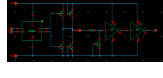
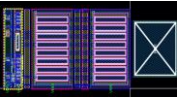
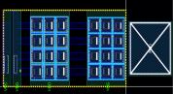
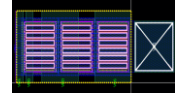
1) 설계 및 검증

① INV_PKG cell 생성 후 → schematic open

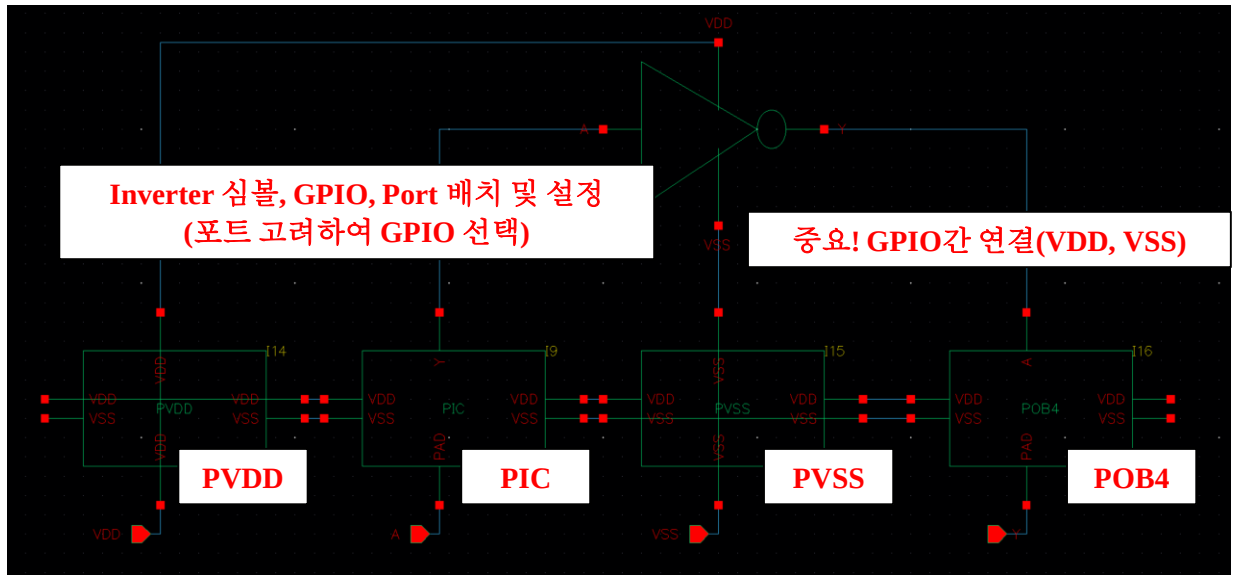


Cell 명 → Type 설정 → 실행 가능한 시뮬레이터 선정 → OK 클릭

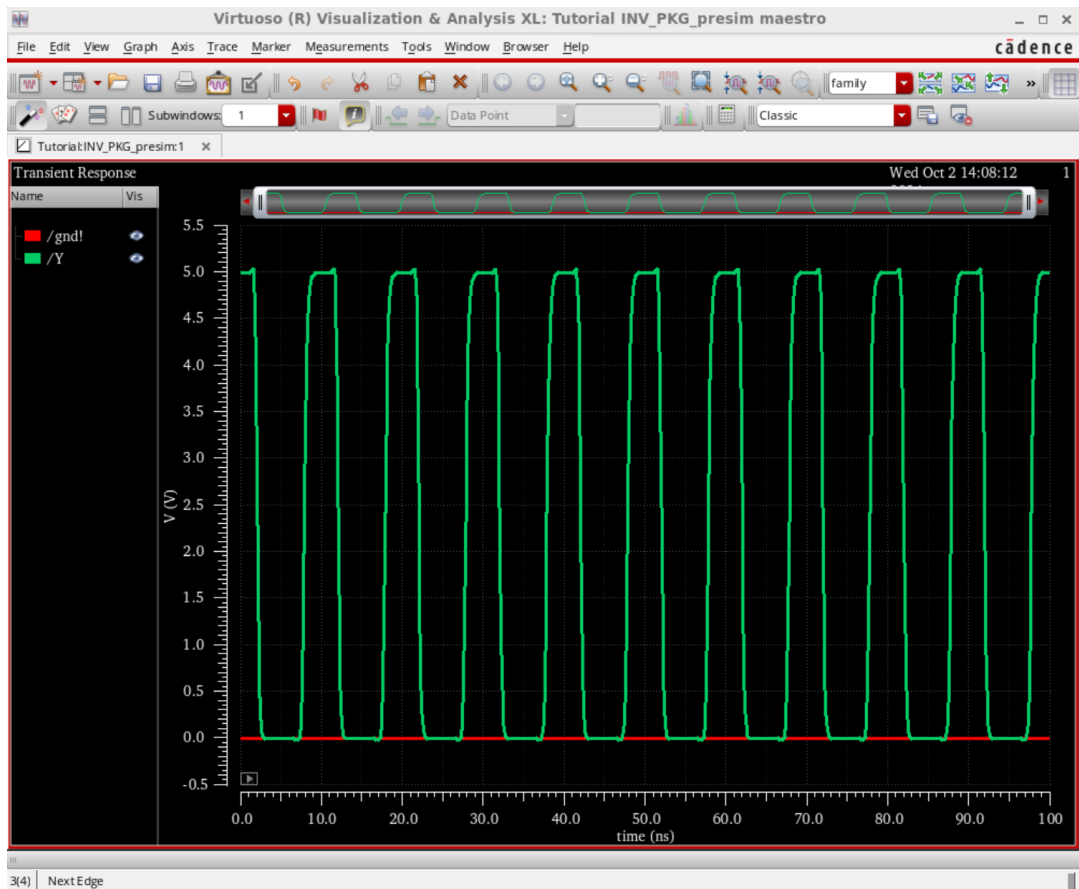
② 제공되는 GPIO 종류

No.	Name	Function	Schematic	Laout
1	PIC	Input Cell		
2	POB4	Output Cell with 4mA Driving Ability		
3	POB8	Output Cell with 8mA Driving Ability		
4	POB24	Output Cell with 24mA Driving Ability		
5	PBCT4	Bi directional Cell with 4mA Driving Ability, CMOS Input		
6	PBCT8	Bi directional Cell with 8mA Driving Ability, CMOS Input		
7	PANA	Analog Cell(N/P Diode Size : 30umx30umx4)		
8	PVDD	Power Clamp Cell		
9	PVSS	Ground Clamp Cell		
10	PCORNER	Corner Cell	—	
11	IOFILLER50	Filler Cell	—	
12	IOFILLER10	Filler Cell	—	
13	IOFILLER1	Filler Cell	—	

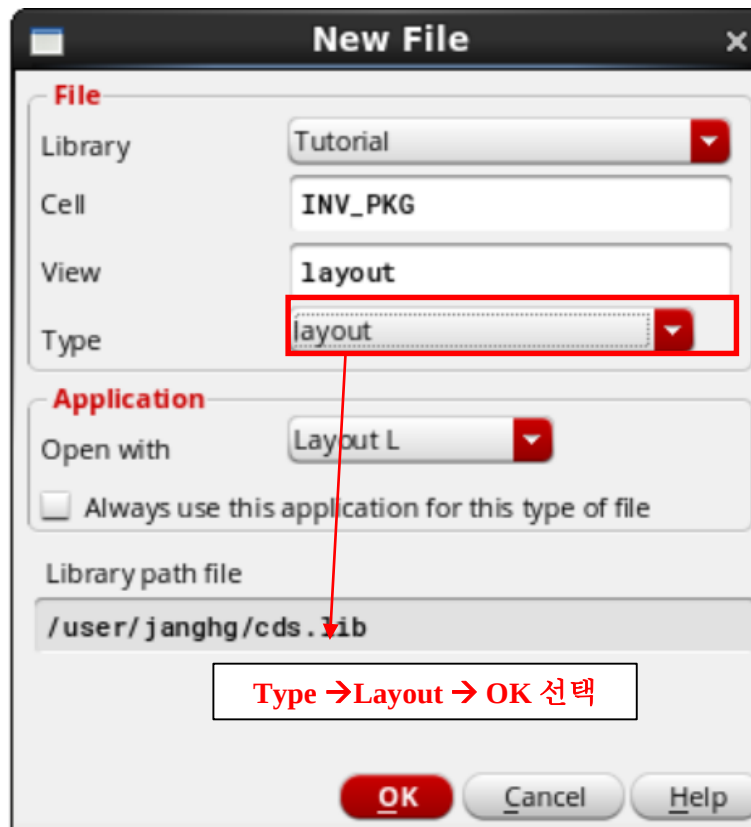
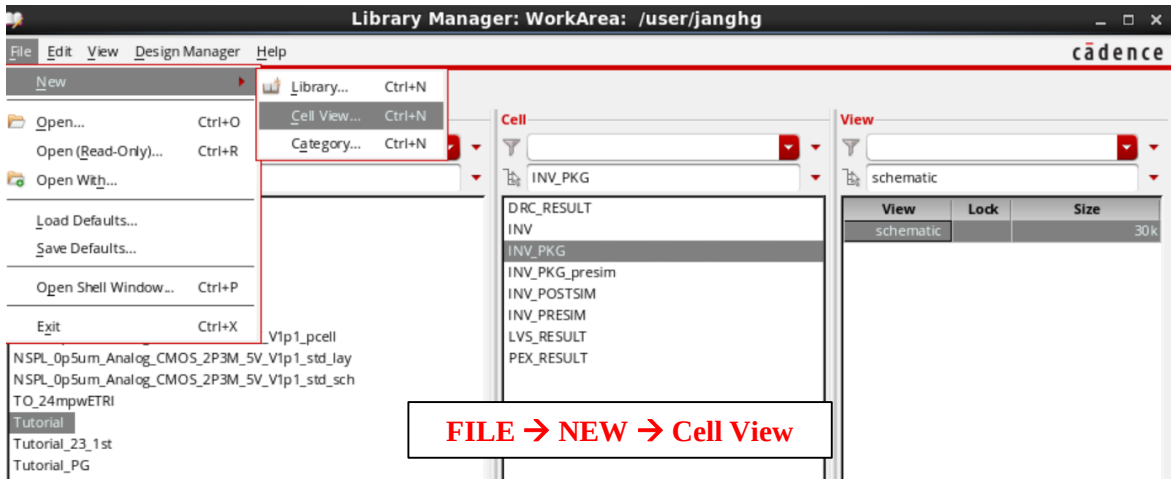
③ INV_PKG cell → schematic 설계

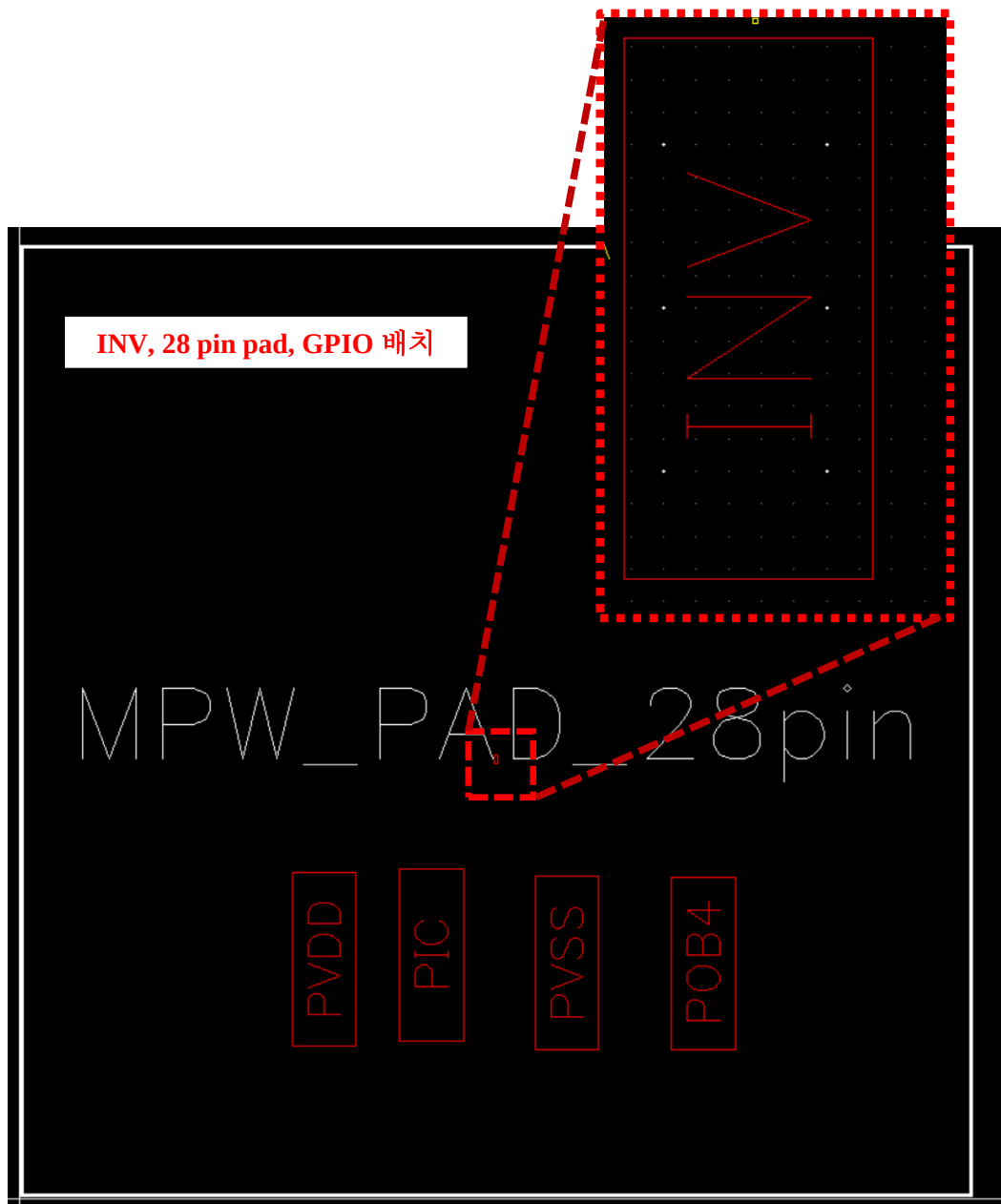


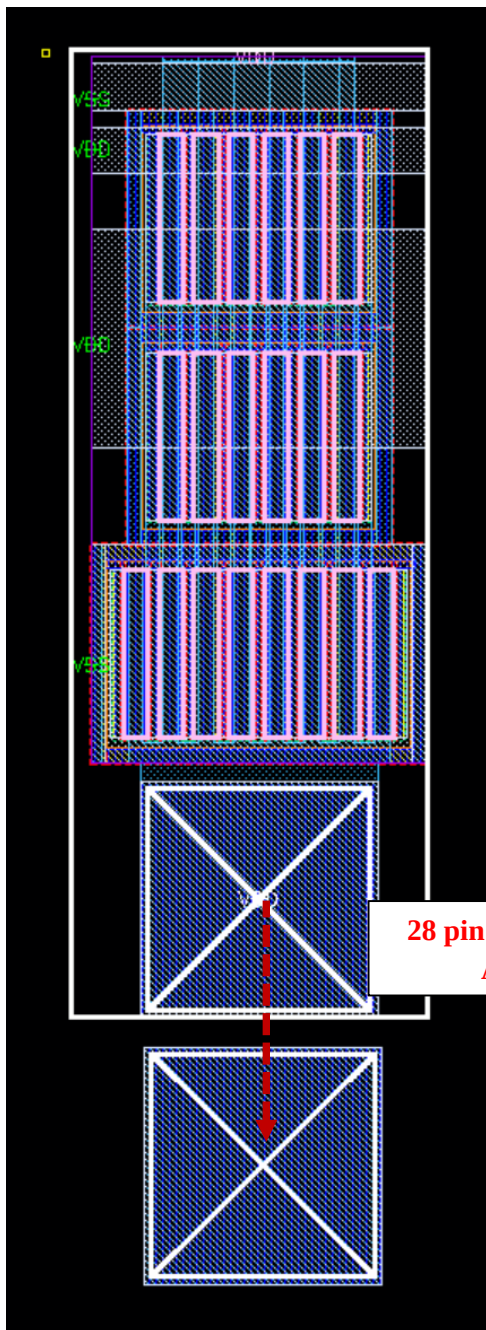
④ INV_PKG cell → Presim (3.2 시뮬레이션을 위한 회로 설계 참고)



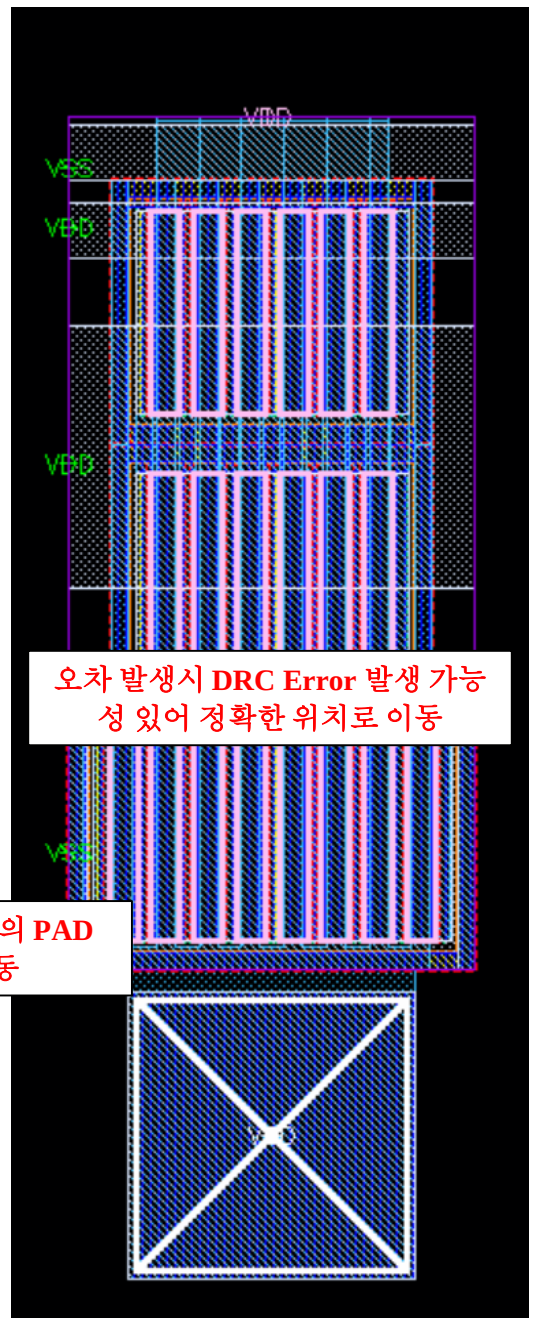
⑤ 28-pin pad 및 GPIO 포함 Layout 설계



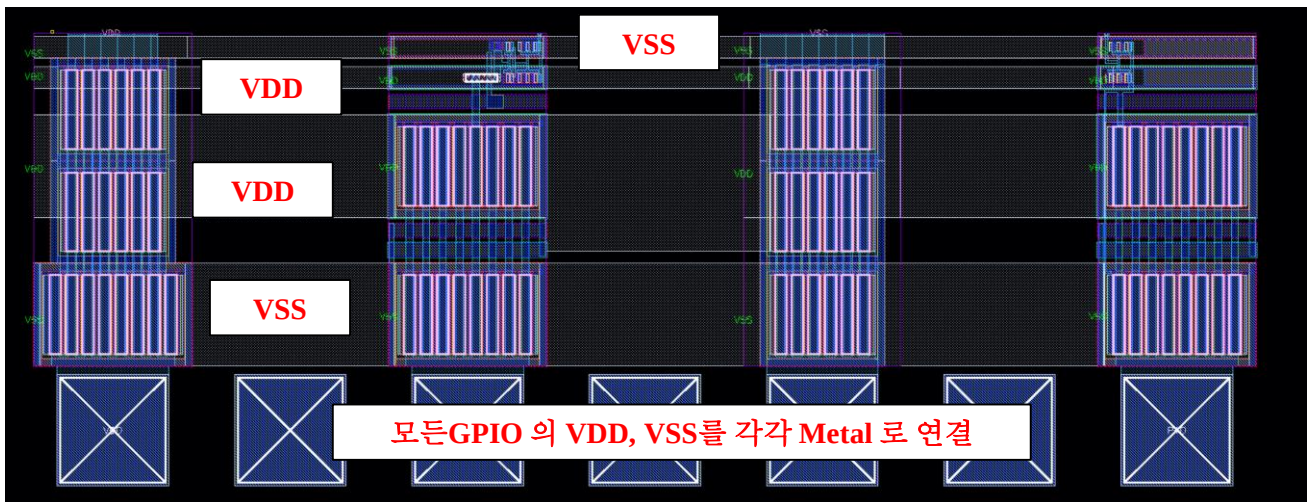
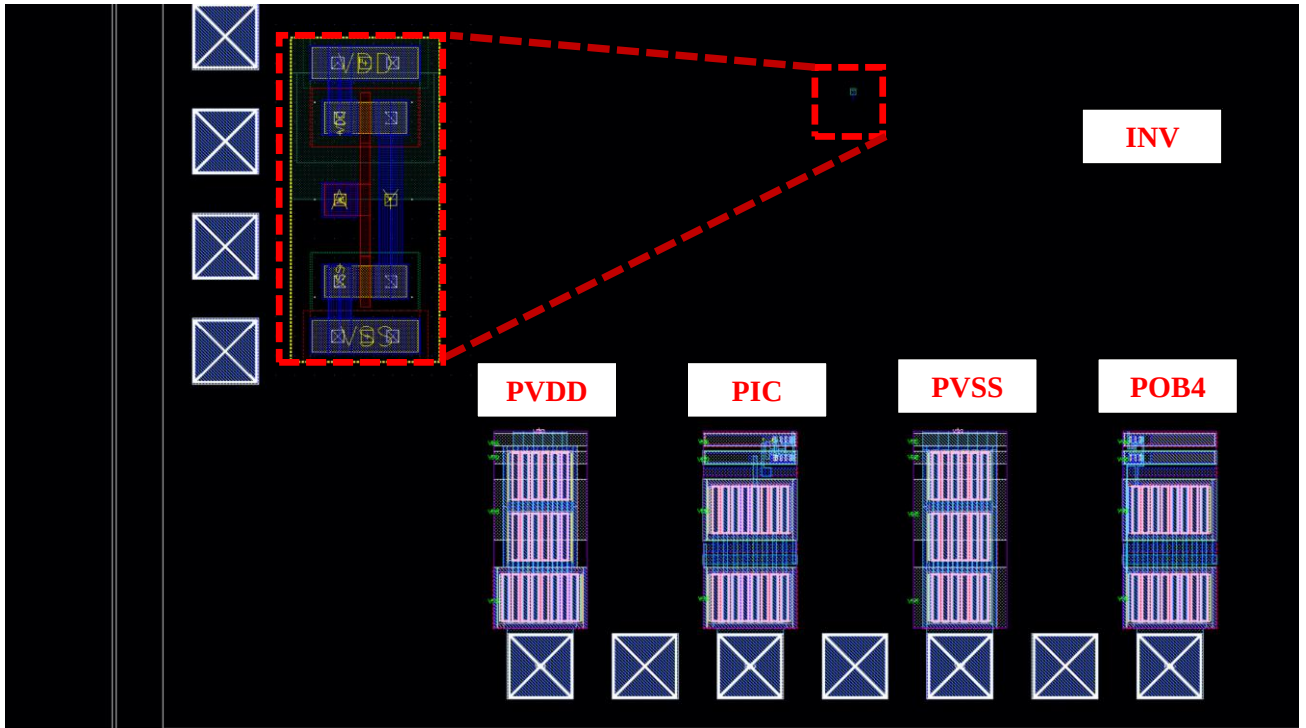


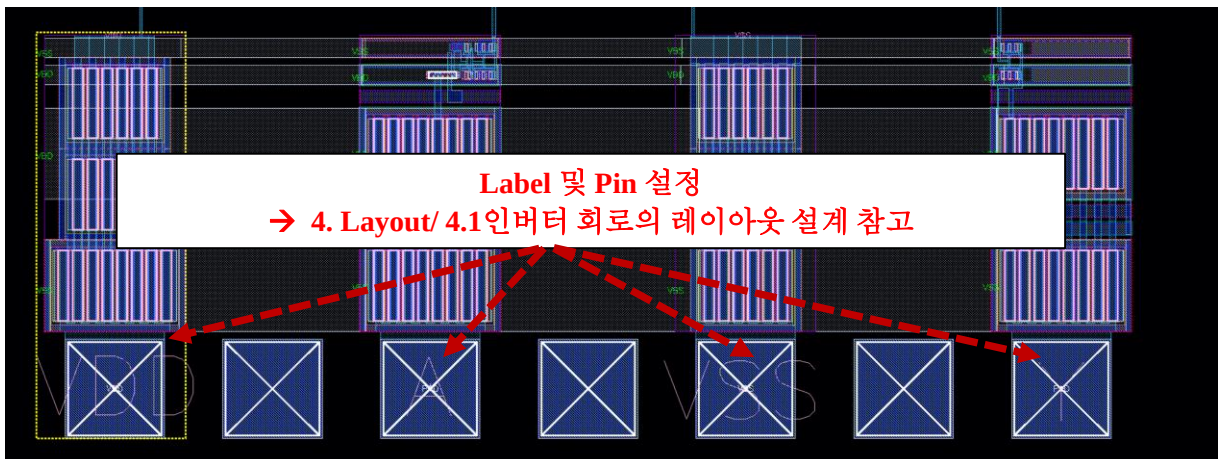
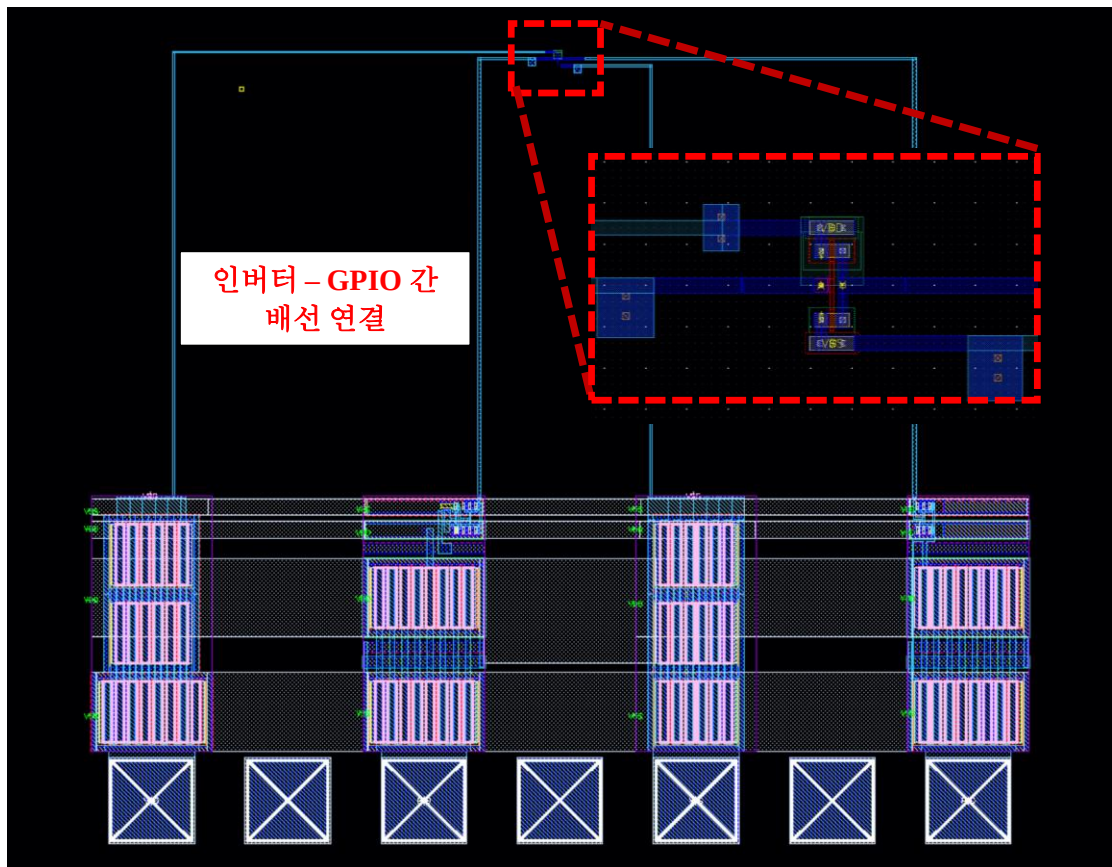


28 pin pad 와 GPIO의 PAD
Align 맞춰 이동

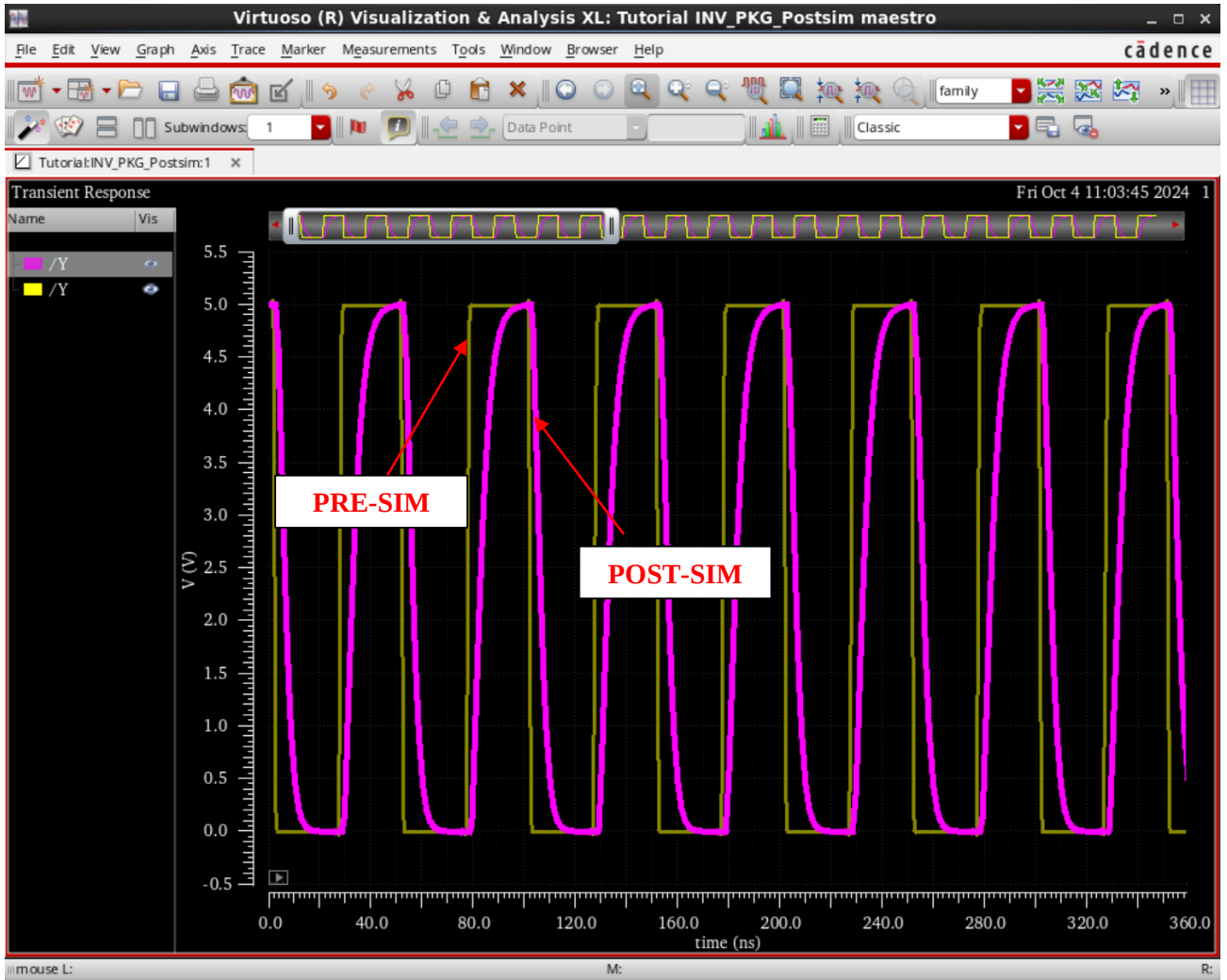


오차 발생시 DRC Error 발생 가능
성 있어 정확한 위치로 이동





DRC, LVS, PEX 및 POST-SIM 진행!



룰체크 (DRC, LVS, PEX 중 1개)를 통해 생성된 최종 xxx.gds 파일 제출